

Supplementary Material document for
Diaconis-Ylvisaker prior penalized likelihood for
 $p/n \rightarrow \kappa \in (0, 1)$ *logistic regression*

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S1 Overview

This document contains further simulation results and the proofs of all results in the main text of *Diaconis-Ylvisaker prior penalized likelihood for $p/n \rightarrow \kappa \in (0,1)$ logistic regression*. Section S2 provides another demonstration of the signal recovery, similar to Figure 1 of the main text, results on the suitability of adjusted Z-statistics for inference in support of Conjecture 6.1, and the empirical distribution of the estimator of v from the simulation study of Section 5.2. To provide context for the proofs of our results, Section S3 restates the MDYPL setting considered in the main paper. Appendices S4-S7 contain proofs of the main results in the paper. Section S8 gives a result on the boundedness of the MDYPL estimator, that is required for the AMP results of Theorem 3.1.

Key lemmas are listed in the appendices of the respective proofs and minor auxiliary lemmas are listed in Section S9.

S2 Further numerical results and simulation experiments

S2.1 Signal recovery of the rescaled MDYPL estimator

As an additional demonstration to Figure 1, we revisit the synthetic data example of Sur and Candès (2019, Figure 2). Specifically, $\mathbf{x}_j \sim N(\mathbf{0}_p, n^{-1}\mathbf{I}_p)$ and β_0 consists of $n\kappa/8$ replications of $(-10, 10, 0, 0, 0, 0, 0)^\top$ adequately scaled to achieve $\|\beta_0\|_2^2 = \gamma^2$ for various values of (κ, γ) and $n = 2000$. Figure S1 shows the MDYPL estimates and their rescaled versions $\hat{\beta}^{\text{DY}}/\mu_*$ for various values of (κ, γ) and $\alpha = 1/(1 + \kappa)$. The rescaled estimates are found to recover the signal β_0 at either side of the phase transition curve as is to be expected from (x).

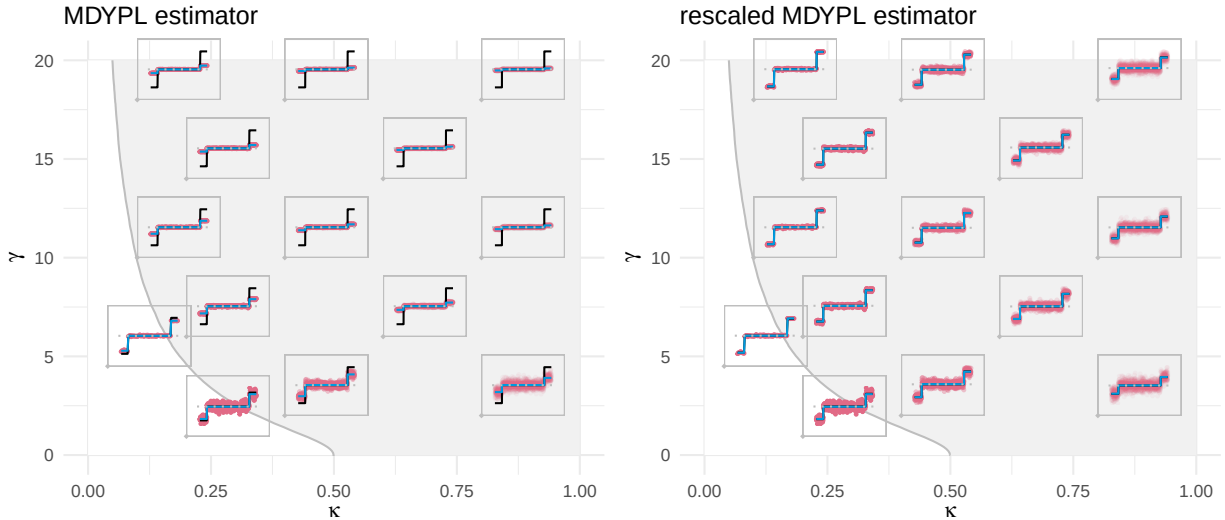


Figure S1: The MDYPL (left) and rescaled MDYPL (right) estimates for various configurations of (κ, γ) and $\alpha = 1/(1 + \kappa)$, $n = 2000$ in the setting of Figure 2 in Sur and Candès (2019). The white and grey area indicate where the ML estimate does or does not exist asymptotically, respectively. Red points show the average coefficient estimates over 10 independent replications per (κ, γ) setting. The cyan segments show the sample mean of the estimates for each value of the truth, and the black segments are the truth.

Table S1: Coordinatewise and aggregate bias and root aggregate MSE of the MDYPL estimator $\hat{\beta}^{\text{DY}}$ under the scenarios (a), (b) and (c) in the simulation setting of Section 4.1.

		$\beta_{0,j}$					Aggregate
		-3	-3/2	0	3/2	3	
$\kappa = 0.2, \gamma^2 = 0.9$	Bias	0.252	0.116	0.003	-0.126	-0.255	-0.002
	Root MSE	2.264	2.243	2.243	2.253	2.266	2.254
		-63.2	-31.6	0	31.6	63.2	
$\kappa = 0.05, \gamma^2 = 100$	Bias	31.953	15.959	0.005	-15.996	-31.958	-0.007
	Root MSE	32.091	16.217	2.826	16.252	32.093	22.781
		-3	-3/2	0	3/2	8	
$\kappa = 0.2, \gamma^2 = 3.1$	Bias	0.557	0.268	0.003	-0.281	-1.486	-0.188
	Root MSE	2.363	2.308	2.291	2.315	2.736	2.408

S2.2 Coordinate-wise and aggregate behaviour of MDYPL estimator with $\alpha = 1/(1 + \kappa)$

This section provides further evidence to support the observations in Section 4.1 that the proposal $\alpha = 1/(1 + \kappa)$ in Rigon and Aliverti (2023) is inadequate as a means of bias reduction in high-dimensional logistic regression. We simulate 5000 datasets with $n = 1000$ under the following three scenarios: (a) as in Section 1.3, we set $\kappa = 0.2$, $\gamma^2 = 0.9$, and β_0 constructed from $n\kappa/5$ replications of the vector $(-3, -3/2, 0, 3/2, 3)^\top$, (b) we rescale the signal β_0 in (a) such that $\gamma^2 = 100$ and set $\kappa = 0.05$, and (c) we replace all coordinates of β_0 with value 3 by 8, so that $\sum_{j=1}^p \beta_{0,j}/p = 1 = \mathbb{E}[\bar{\beta}]$, for $\bar{\beta}$ taking values in $\{-3, -3/2, 0, 3/2, 8\}$ with equal probability, and maintain $\kappa = 0.2$, so that $\gamma^2 = 3.1$.

The top subtable of Table S1 reports the empirical aggregate bias and root aMSE of the MDYPL estimator for scenario (a). These are found to be close to what our theory predicts, namely $\sum_{j=1}^p (\hat{\beta}_j^{\text{DY}} - \beta_{0,j})/p \xrightarrow{a.s.} \mathbb{E}[\sigma_* G] - (1 - \mu_*)\mathbb{E}[\bar{\beta}] = 0$, and $\|\hat{\beta}^{\text{DY}} - \beta_0\|_2^2/p \xrightarrow{a.s.} \mathbb{E}[(\sigma_* G - (1 - \mu_*)\bar{\beta})^2] = \sqrt{\sigma_*^2 + (1 - \mu_*)^2 \gamma^2 \kappa^{-1}} \approx 2.25$, respectively. The same holds for the empirical coordinatewise bias, which is approximately $(\mu_* - 1)\beta_{0,j}$ (see h) in Section 3.2). The mid subtable of Table S1 reports results for scenario (b). As expected from Figure 3, both coordinatewise bias and root MSE are substantially larger than they are in scenario (a). The MDYPL estimator underestimates the nonzero components of the regression coefficient by about half their absolute value, while the root aMSE is now 22.78 in contrast to 2.25 in scenario (a). Finally, the bottom subtable of Table S1 reports results for scenario (c). There is substantial aggregate bias due to the shrinkage of the MDYPL estimator towards zero, and in concordance with our theory, that aggregate bias is no longer zero but close to $(\mu_* - 1) \approx -0.187$.

S2.3 Adjusted Z-statistics for MDYPL with intercept

In this section we present further simulation evidence to support Conjecture 6.1 in the main text. We fix $n = 2000$ and let p vary so that $\kappa \in \{0.2, 0.4, 0.6, 0.8\}$, set $\alpha = 1/(1 + \kappa)$, and consider intercept values $\theta_0 \in \{0.5, 1.5, 2.5\}$. We let $\mathbf{X}$ be the n times p matrix with i.i.d $N(0, 1/n)$ entries. Hence, $\tau_j = 1/\sqrt{n}$. We also assume that β_0 has all but $p/4$ randomly chosen entries set to zero. The nonzero entries are all of the same value and scaled such that $\gamma^2 = \|\beta_0\|_2^2/n = 5$. For each combination of (κ, θ_0) , we draw 5000 independent copies of $\{\mathbf{y}, \mathbf{X}\}$ from a logistic regression model with intercept. We compute the adjusted Z-statistic of Theorem 3.3 with μ_*, σ_* from (10). Tables S2 and S3 show the estimated probability that the adjusted Z-statistics for the first zero and first non-zero coordinates of β_0 , respectively, are less than the 100q% quantile of a $N(0, 1)$ distribution, with q ranging from 0.01 to 0.99.

Table S2: Estimates of $P(Z_{*,j} \leq z_q)$ (in percentage) from the simulation setting of Section 6 ($\tau_j = 1/\sqrt{n}$), where $Z_{*,j}$ is the adjusted Z -statistic of Theorem 3.3, j is the index of the first zero coordinate of β_0 ($\beta_{0,j} = 0$), and z_q the $100q\%$ quantile of a $N(0, 1)$ distribution. The constant σ_* results from the solution of (10) for $\gamma^2 = 5$, and each combination of $\kappa \in \{0.2, 0.4, 0.6, 0.8\}$, and $\theta_0 \in \{0.5, 1.5, 2.5\}$.

κ	θ_0	100q								
		1	5	10	25	50	75	90	95	99
0.2	0.5	0.9	5.0	10.1	25.3	50.8	75.6	89.9	94.8	99.0
	1.5	1.0	5.0	10.5	25.6	50.7	76.1	90.1	94.9	99.0
	2.5	0.9	4.9	10.4	24.8	51.0	75.4	90.4	95.5	99.3
0.4	0.5	1.1	5.0	9.9	24.5	49.6	75.4	90.0	95.1	99.2
	1.5	1.0	5.0	9.7	24.7	49.6	74.2	89.9	94.7	99.1
	2.5	0.9	4.5	9.7	24.3	49.5	75.0	89.6	94.9	99.1
0.6	0.5	1.2	5.2	9.8	25.3	50.4	75.8	90.4	95.0	99.2
	1.5	0.8	5.3	11.0	26.1	52.1	76.6	90.4	94.8	99.0
	2.5	1.1	5.0	9.7	25.3	49.3	74.0	89.3	94.6	98.9
0.8	0.5	0.9	4.5	9.4	23.8	49.2	74.3	90.3	94.7	98.9
	1.5	1.0	5.1	9.9	24.7	50.0	74.9	89.9	95.1	99.2
	2.5	1.0	4.6	9.2	24.1	48.0	74.1	89.6	94.8	98.9

Table S3: Estimates of $P(Z_{*,j} \leq z_q)$ (in percentage) from the simulation setting of Section 6 ($\tau_j = 1/\sqrt{n}$), where $Z_{*,j}$ is the adjusted Z -statistic of Theorem 3.3, j is the index of the first non-zero coordinate of β_0 , and z_q the $100q\%$ quantile of a $N(0, 1)$ distribution. The constants μ_* and σ_* result from the solution of (10) for $\gamma^2 = 5$, and each combination of $\kappa \in \{0.2, 0.4, 0.6, 0.8\}$, and $\theta_0 \in \{0.5, 1.5, 2.5\}$.

κ	θ_0	100q								
		1	5	10	25	50	75	90	95	99
0.2	0.5	1.0	5.5	10.5	25.9	50.5	75.5	90.1	95.0	98.7
	1.5	1.0	5.0	10.0	25.0	50.6	75.7	90.0	95.0	99.0
	2.5	1.2	5.6	11.1	24.6	49.7	75.1	89.6	94.6	98.8
0.4	0.5	1.0	5.1	9.2	24.2	49.1	74.2	89.5	94.7	98.8
	1.5	0.9	4.6	9.6	24.4	49.9	74.8	89.8	94.9	98.9
	2.5	1.1	5.2	9.8	25.2	50.0	75.1	90.1	94.9	98.8
0.6	0.5	0.9	4.9	10.0	25.6	50.2	75.7	89.7	94.8	98.8
	1.5	1.2	4.9	10.6	25.2	49.9	75.5	90.2	94.8	99.0
	2.5	0.9	4.8	9.4	24.4	49.1	74.6	90.3	95.4	99.2
0.8	0.5	0.9	4.7	10.6	25.3	49.9	74.2	89.0	94.2	98.9
	1.5	1.0	4.6	9.7	25.6	49.7	75.1	90.0	95.1	98.9
	2.5	0.8	5.1	10.1	25.5	50.7	74.6	90.3	95.3	99.1

S2.4 Estimates of v for the simulation study of Section 5.2

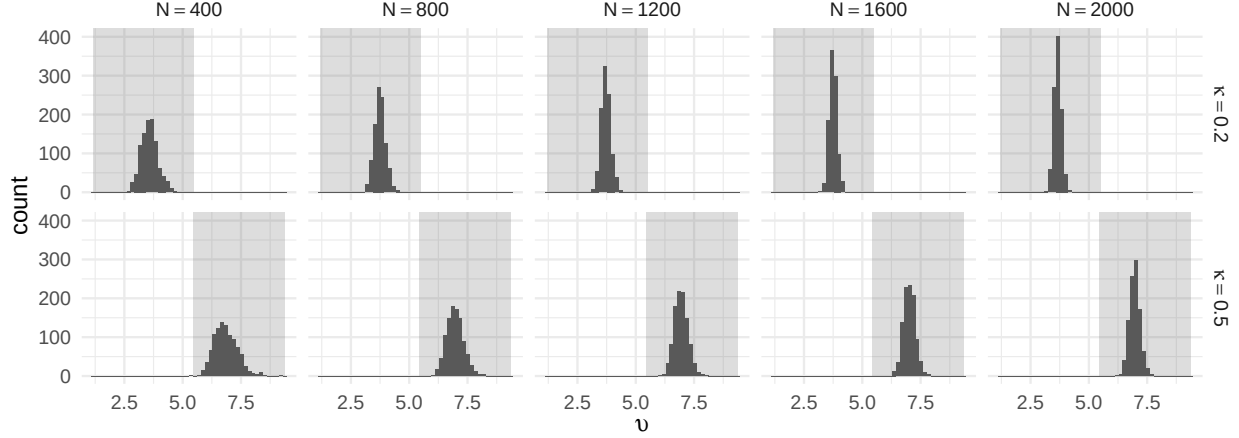


Figure S2: Empirical distribution of the SLOE estimator of v in the experiments of Section 5.2. The shaded area is the image of the function $v = h(\gamma)$ for $\kappa \in \{0.2, 0.5\}$ and $\alpha = 0.95$. See Figure 8 of the main text for $v = h(\gamma)$.

S2.5 Bradley-Terry model with competitions missing at random

Consider a binary observation y_{ij} with value 1 if player i beats player j , and value 0 if player i loses from player j for $i, j \in \{1, \dots, N\}$, in a competition with only two possible outcomes. The Bradley-Terry model assumes that the outcomes of the available competitions are realisations of independent random variables with

$$\log \frac{\Pr(y_{ij} = 1)}{1 - \Pr(y_{ij} = 1)} = \beta_i - \beta_j.$$

If $\beta_i > \beta_j$ then $\Pr(y_{ij} = 1) > 1/2$, if $\beta_i < \beta_j$ then $\Pr(y_{ij} = 1) < 1/2$, and if $\beta_i = \beta_j$ then $\Pr(y_{ij} = 1) = 1/2$. Hence, β_i can be interpreted as the ability of player i .

The Bradley-Terry model has seen a wealth of applications in the modelling of pairwise comparison data in numerous fields of enquiry, such as sport, ecology, education, risk analysis, and large language models, to name a few. The ubiquitous application of the Bradley-Terry model is partly due to the fact that it arises naturally from numerous and sometimes seemingly unrelated axiomatic frameworks and game theoretic settings; see, for example, Hamilton *et al.* (2025) for an authoritative overview. The Bradley-Terry model is a logistic regression model with a model matrix with entries 0, 1, and -1 , depending on the contrasts required to enforce the relationship S2.5 across the parameters.

Identifiability requires one constraint, such as $\beta_1 = 0$ or $\sum_{i=1}^N \beta_i = 1$. Hence, the model describes $\binom{N}{2}$ probabilities with $N-1$ parameters. Depending on the available competitions in the data and the competition outcomes, the ML estimator of the abilities can take infinite values. In contrast, as we discuss in Section 2, the MDYPL estimator is always finite for $\alpha \in (0, 1)$.

A common competition format is a double round-robin competition, where each player would compete with every other player twice, giving rise to $N(N-1)$ observed outcomes. Apart from the possibility of infinite ML estimates, such a setting can be handled well by the classical asymptotic results for likelihood-based estimation and inference, because the number of available observations grows quadratically with the number of parameters.

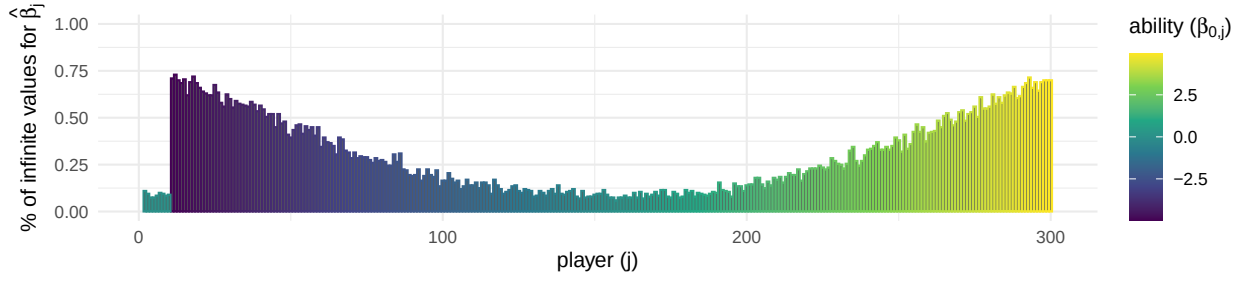


Figure S3: The percentage of samples with infinite ML estimates for the ability of each player from the simulation experiment of Section S2.5.

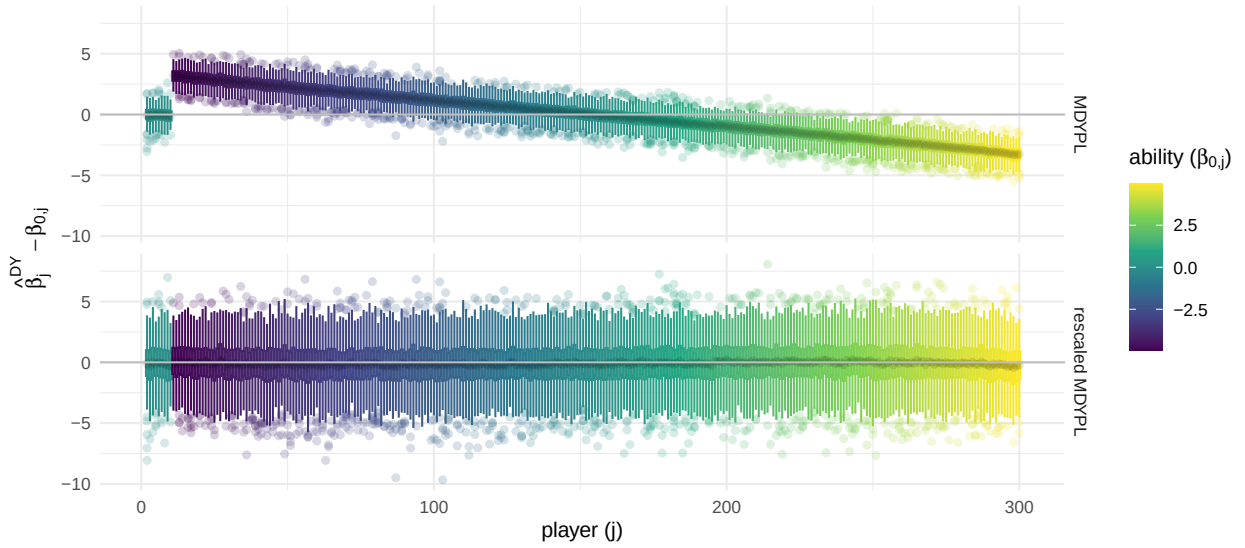


Figure S4: Boxplots of centred estimated player abilities from the simulation experiment of Section S2.5 using the MDYPL estimator and its rescaled version. The grey points are the biases in the estimation of the ability of each player.

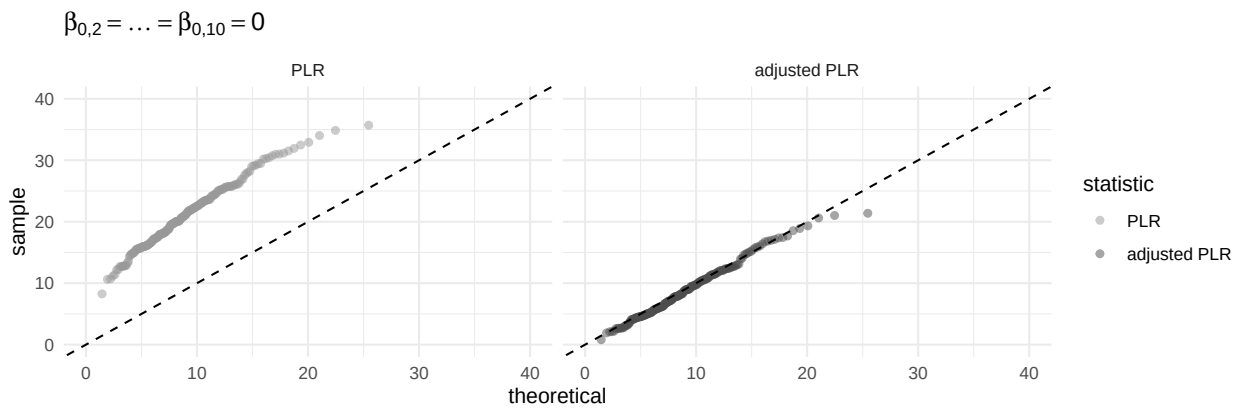


Figure S5: Q-Q plots comparing χ^2_9 quantiles to empirical quantiles of the DY prior penalized likelihood ratio statistic (light grey) and its rescaled version according to Theorem 3.4 (dark grey), for the simulation experiment of Section S2.5.

We carry out a simulation experiment to investigate the effectiveness of the high-dimensionality corrections for MDYPL estimation and inference in the main text, when competitions are missing at random in a double round-robin competition. In particular, we assume that we observe only a fraction $1/(N\kappa)$ of the $N(N-1)$ possible competitions, for $\kappa \in (0, 1)$. The high-dimensionality corrections for the estimates and the PLR statistic implied by Theorem 3.1 and Theorem 3.4 in the main text, have been developed under the sufficient condition of a random model matrix with normal rows, which is clearly not satisfied for the Bradley-Terry model.

We consider $N = 300$ players, with the abilities of the first 10 fixed at zero, and the abilities of the remainder 290 on an equi-spaced grid between -5 and 5 . We generate a double round-robin competition, which involves 89700 contests, and scale the abilities so that the variance of the implied vector of linear predictors is equal to $\gamma^2 = 16$. A sample of competitions and outcomes is then generated by simulating the outcomes of all competitions according to the Bradley-Terry model at the scaled abilities, and randomly selecting only $1/60$ of those, that is taking $\kappa = 0.2$. This results in 1495 competitions and their outcomes. We generate 200 independent samples in this way, and for each sample we estimate the parameters of Bradley-Terry model using MDYPL with $\alpha = 0.7$, after fixing $\beta_1 = 0$ for identifiability.

Figure S3 shows the simulation-based estimate of the probability of the ML estimate for the ability of each player. As is apparent, there is at least one infinite estimated ability in all samples. As expected, the probability of infinite estimates increases with the absolute value of the ability.

The MDYPL estimates and the simulation based estimates of the finite sample bias in the estimation of each ability are shown in Figure S4. MDYPL appears to perform well in estimating zero abilities, but tends to underestimate non-zero abilities in absolute value. Figure S4 also shows the rescaled MDYPL ability estimates, where the quantities necessary for the rescaling are computed using the data-driven procedures of Section 7 of the main text for a model without intercept. Evidently, rescaling the estimates almost eliminates the persistent bias in the estimation of the abilities. Figure S5 shows that the distribution of the PLR statistic for testing that $\beta_{0,2} = \dots \beta_{0,10} = 0$ is far from χ_9^2 , which is what is expected by standard asymptotic theory. In contrast, the rescaled version of the PLR statistic according to Theorem 3.4 closely agrees to the χ_9^2 distribution.

S3 Maximum Diaconis-Ylvisaker prior penalized likelihood

Consider data $\{\mathbf{y}, \mathbf{X}\}$ with responses $\mathbf{y} = (y_1, \dots, y_n)^\top$, $y_j \in \{0, 1\}$, and covariate matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ with rows $\mathbf{x}_j \in \mathbb{R}^p$ ($j = 1, \dots, n$). A logistic regression model assumes that conditional on $\mathbf{x}_j$, the responses y_j are independent realizations of Bernoulli random variables with mean $\zeta'(\mathbf{x}_j^\top \boldsymbol{\beta}_0)$ where $\zeta(x) = \log(1 + e^x)$ and $\boldsymbol{\beta}_0 \in \mathbb{R}^p$ is the unknown p -dimensional vector of regression coefficients.

Given hyperparameters $\alpha \in (0, 1]$ and $\boldsymbol{\beta}_P \in \mathbb{R}^p$, the logarithm of the Diaconis and Ylvisaker (1979) prior, bar the normalizing constant, for the logistic regression model is given by

$$\log p(\boldsymbol{\beta}; \mathbf{X}) = \frac{1 - \alpha}{\alpha} \sum_{j=1}^n \left\{ \zeta'(\mathbf{x}_j^\top \boldsymbol{\beta}_P) \mathbf{x}_j^\top \boldsymbol{\beta} - \zeta(\mathbf{x}_j^\top \boldsymbol{\beta}) \right\},$$

where $\boldsymbol{\beta}_P$ is the mode of the prior, while α controls the variability of the prior distribution. The DY-prior penalized log-likelihood, without the normalizing constant is given by

$$\ell^*(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \frac{1}{\alpha} \sum_{j=1}^n \left\{ (\alpha y_j + (1 - \alpha) \zeta'(\mathbf{x}_j^\top \boldsymbol{\beta}_P)) \mathbf{x}_j^\top \boldsymbol{\beta} - \zeta(\mathbf{x}_j^\top \boldsymbol{\beta}) \right\}. \quad (\text{S1})$$

The maximiser of (S1) is denoted by $\hat{\boldsymbol{\beta}}^{\text{DY}}$. Hence, $\hat{\boldsymbol{\beta}}^{\text{DY}}$ is the maximiser of a logistic regression model log-likelihood with pseudo-responses $\mathbf{y}^* = (y_1^*, \dots, y_n^*)^\top$, $y_j^* = \alpha y_j + (1 - \alpha) \zeta'(\mathbf{x}_j^\top \boldsymbol{\beta}_P)$, $\alpha \in (0, 1]$. For the remainder of this work, $\boldsymbol{\beta}_P = \mathbf{0}_p$ and $\hat{\boldsymbol{\beta}}^{\text{DY}}$ is understood as the maximiser of $\ell(\boldsymbol{\beta}; \mathbf{y}^*, \mathbf{X})$.

S4 Proof of Theorem 3.1

As in Sur and Candès (2019), we assume that the model matrix $\mathbf{X}$ has independent rows $\mathbf{x}_i \sim \text{N}(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$, where $\mathbf{I}_p$ is the $p \times p$ identity matrix, and that the linear predictors $\mathbf{x}_i^\top \boldsymbol{\beta}_0$ have asymptotic variance

$\lim_{n \rightarrow \infty} \text{var}(\mathbf{x}_i^\top \boldsymbol{\beta}_0) = \lim_{n \rightarrow \infty} \|\boldsymbol{\beta}_0\|_2^2/n = \gamma^2$. It is further assumed that there exists a random variable $\bar{\beta} \sim \pi_{\bar{\beta}}$ with finite second moment, such that the empirical distribution of the entries of $\boldsymbol{\beta}_0$ converges weakly in distribution to $\pi_{\bar{\beta}}$ and $\lim_{n \rightarrow \infty} \frac{1}{p} \sum_{j=1}^n \beta_{0,j}^2 = \mathbb{E}(\bar{\beta}^2)$. Finally, let $\lim_{n \rightarrow \infty} p/n = \kappa$ for $\kappa \in (0, 1)$.

S4.1 Asymptotic characterisation via stationary state evolution

Recall the state evolution equations (S2), (S3), (S4) below.

$$\mathbb{E} \left[2\zeta'(Z)Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b\zeta} \left(Z_* + \frac{1+\alpha}{2}b \right) \right) \right\} \right] = 0 \quad (\text{S2})$$

$$1 - \kappa - \mathbb{E} \left[\frac{2\zeta'(Z)}{1 + b\zeta''(\text{prox}_{b\zeta}(Z_* + \frac{1+\alpha}{2}b))} \right] = 0 \quad (\text{S3})$$

$$\sigma^2 - \frac{b^2}{\kappa^2} \mathbb{E} \left[2\zeta'(Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b\zeta} \left(Z_* + \frac{1+\alpha}{2}b \right) \right) \right\}^2 \right] = 0. \quad (\text{S4})$$

Here, $Z \sim \mathcal{N}(0, \gamma^2)$, $Z_* = \mu Z + \kappa^{1/2} \sigma G$ for $G \sim \mathcal{N}(0, 1)$ which is independent of Z , $\zeta'(\cdot), \zeta''(\cdot)$ are the first and second derivatives of $\zeta(\cdot)$ and $\text{prox}_{b\zeta}(x) = \arg \min_u \{b\zeta(u) + \frac{1}{2}(x-u)^2\}$ denotes the proximal operator. Equations (S2)-(S4) define the stationary points to the state evolution iterates (S15)-(S17) of the AMP recursion (S11)-(S12). A discussion about how these equations are derived is thus postponed to S4.4.2.

Denote by $\mathbf{J}(\mu, b, \sigma)$ the Jacobian matrix of the LHS of (S2)-(S4) with respect to (μ, b, σ) .

Theorem 3.1. *Let $\hat{\boldsymbol{\beta}}^{DY}$ be the MDYPL estimator for a logistic regression model with i.i.d. $\mathcal{N}(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$ covariates. Assume that (α, κ, γ) are such that (4) admits a solution (μ_*, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is nonsingular. Then for any function $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}$ that is pseudo-Lipschitz of order 2,*

$$\frac{1}{p} \sum_{j=1}^p \psi(\hat{\boldsymbol{\beta}}_j^{DY} - \mu_* \boldsymbol{\beta}_{0,j}, \boldsymbol{\beta}_{0,j}) \xrightarrow{a.s.} \mathbb{E}[\psi(\sigma_* G, \bar{\beta})], \quad \text{as } n \rightarrow \infty, \quad (\text{S5})$$

where $G \sim \mathcal{N}(0, 1)$ is independent of $\bar{\beta} \sim \pi_{\bar{\beta}}$.

The proof of Theorem 3.1 closely follows the proof of Theorem 1 in Sur and Candès (2019). We replace the logistic indicator function $h(u, v) = \{1 - \text{sign}(v \leq \zeta'(u))\}/2$ is simply replaced by $h(u, v) = \alpha \{1 - \text{sign}(v \leq \zeta'(u))\}/2 + (1 - \alpha)/2$ and the derivations are adjusted accordingly. Further, we provide a smoothing argument that formally justifies the asymptotic results of the relevant AMP recursion, and which has been missing hitherto.

S4.2 Proof outline for Theorem 3.1

The following steps constitute the proof of Theorem 3.1:

- A.2.1) *Target AMP recursion:* As a starting point, we formulate an AMP recursion, whose stationary point coincides with $\hat{\boldsymbol{\beta}}^{DY}$.
- A.2.2) *Centred target AMP recursion:* Based on the target AMP recursion, we devise a centred AMP recursion, for which the aggregate asymptotic behaviour can be established via existing AMP results.
- A.2.3) *Matrix-valued AMP embedding:* The centred AMP recursion is embedded into a symmetric, matrix valued AMP recursion of Javanmard and Montanari (2013) for which the topical AMP asymptotics hold. A reduction argument, which is also due to Javanmard and Montanari (2013) and which was subsequently used in Donoho and Montanari (2016) and Sur and Candès (2019), identifies the correct asymptotic behaviour for the centred AMP recursion.

- A.2.4) *Smoothing argument*: As the logistic indicator function is not Lipschitz, which is a prerequisite for the application of the asymptotic result of Javanmard and Montanari 2013, Theorem 1, we instead establish a Lipschitz approximation to the indicator function that is parametrised by $\epsilon > 0$, such that $\epsilon = 0$ recovers the original recursions of A.2.1 and A.2.2.
- A.2.5) *Asymptotic equivalence of target and centred AMP*: Next, we show that the target recursion and the centred AMP recursion, whose asymptotic behaviour is characterised through A.2.3, are asymptotically equivalent for sufficiently large iterations.
- A.2.6) *Asymptotic equivalence of AMP and MDYPL*: Finally, we shown that for $\epsilon \rightarrow 0$, the approximate AMP iterates are arbitrarily close to $\hat{\beta}^{\text{DY}}$, asymptotically, if iterated for sufficiently long.

S4.3 Matrix-valued AMP

For the readers' convenience, this subsection introduces the matrix valued AMP recursion of Javanmard and Montanari (2013), which is the theoretical workhorse to derive the asymptotic behaviour of the AMP recursion of interest. It is more or less a verbatim replication of Javanmard and Montanari 2013, Section 2, tailored to the specific requirements of the problem at hand.

Given integers q, N , let $\mathcal{V}_{q,N} \equiv (\mathbb{R}^q)^N$, so that a vector $x \in \mathcal{V}_{q,N}$ is regarded as a N -vector with entries in $\mathbb{R}^q$, i.e. $x = (\mathbf{x}_1, \dots, \mathbf{x}_N)$ where $\mathbf{x}_i \in \mathbb{R}^q$ with components denoted by $\mathbf{x}_i = (\mathbf{x}_i(1), \dots, \mathbf{x}_i(q))$ ($i = 1, \dots, N$). Given $x \in \mathcal{V}_{q,N}$, define its ℓ_2 -norm as $\|x\|_2 = \left(\sum_{i=1}^N \|\mathbf{x}_i\|_2^2 \right)^{1/2}$. Given a matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, let it act on $\mathcal{V}_{q,N}$ so that for $v, v' \in \mathcal{V}_{q,N}$, $v' = \mathbf{A}v$ for $\mathbf{v}'_i = \sum_{j=1}^N A_{ij} \mathbf{v}_j$ for $i \in [N]$, where $[N]$ denotes the integers $1, \dots, N$. That is, $\mathbf{A}$ is identified with the Kronecker product $\mathbf{A} \otimes \mathbf{I}_q$ when acting on elements of $\mathcal{V}_{N,q}$.

Definition S1 (Symmetric AMP). A symmetric AMP instance is a triple $(\mathbf{A}, \mathcal{F}, x^0)$, where

- 1) $\mathbf{A} = \mathbf{G} + \mathbf{G}^\top$, where $\mathbf{G} \in \mathbb{R}^{N \times N}$ has i.i.d. $N(0, (2N)^{-1})$ entries
- 2) $\mathcal{F} = \{f^k : k \in [N]\}$ is a collection of mappings $f^k : \mathbb{R}^q \times \mathbb{N} \rightarrow \mathbb{R}^q$, $(\mathbf{x}, t) \mapsto f^k(\mathbf{x}, t)$ is locally Lipschitz in its first argument
- 3) $x^0 \in \mathcal{V}_{q,N}$ is an initial condition.

Given $\mathcal{F} = \{f^k : k \in [N]\}$, denote by $f(\cdot, t)$ the mapping from $\mathcal{V}_{q,N} \rightarrow \mathcal{V}_{q,N}$ such that $v' = f(v, t)$ for $\mathbf{v}'_i = f^i(\mathbf{v}_i, t)$ and all $i \in [N]$.

Definition S2 (AMP orbit). The AMP orbit corresponding to an AMP instance $(\mathbf{A}, \mathcal{F}, x^0)$ is the sequence of vectors $\{x^t\}_{t \in \mathbb{N}_0}$, $x^t \in \mathcal{V}_{q,N}$ such that for $t \geq 0$,

$$x^{t+1} = \mathbf{A}f(x^t, t) - \mathbf{B}_t f(x^{t-1}, t-1), \quad (\text{S6})$$

where $\mathbf{B}_t : \mathcal{V}_{q,N} \rightarrow \mathcal{V}_{q,N}$ is the linear operator such that for $v' = \mathbf{B}_t v$,

$$\mathbf{v}'_i = \left(\frac{1}{N} \sum_{j=1}^N \frac{\partial f^j}{\partial \mathbf{x}}(\mathbf{x}_j^t, t) \right) \mathbf{v}_i,$$

and where $\frac{\partial f^j}{\partial \mathbf{x}}$ denotes the Jacobian matrix of $f^j(\cdot, t) : \mathbb{R}^q \rightarrow \mathbb{R}^q$ given t .

Any expression with negative t -index is considered zero in the above definition.

Definition S3 (Converging AMP). A sequence of AMP instances $\{(\mathbf{A}(N), \mathcal{F}_N, x^{0,N})\}_{N \in \mathbb{N}_0}$ is said to be converging if

- i) There exist integers q, q' , and
- ii) a function $g : \mathbb{R}^q \times \mathbb{R}^q \times [q'] \times \mathbb{N} \rightarrow \mathbb{R}^q$ with $g(\mathbf{x}, \mathbf{y}, a, t) = (g_1(\mathbf{x}, \mathbf{y}, a, t), \dots, g_q(\mathbf{x}, \mathbf{y}, a, t))$, such that for each $r \in [q], a \in [q'], t \in \mathbb{N}$, $g_r(\cdot, \cdot, a, t)$ is Lipschitz,
- iii) q' probability measures $P_1, \dots, P_{q'}$ on $\mathbb{R}^q$,
- iv) for each N , a finite partition $C_1^N \cup \dots \cup C_{q'}^N = [N]$,
- v) q' positive definite matrices $\widehat{\Sigma}_1^0, \dots, \widehat{\Sigma}_{q'}^0 \in \mathbb{R}^{q \times q}$,

such that the following happens:

- 1) For each $a \in [q']$, one has $\lim_{N \rightarrow \infty} \frac{|C_a^N|}{N} = c_a \in (0, 1)$,
- 2) There exists $y \in \mathcal{V}_{q,N}$ with $y = (\mathbf{y}_1, \dots, \mathbf{y}_N), \mathbf{y}_i \in \mathbb{R}^q (i = 1, \dots, N)$, such that for each $N \in \mathbb{N}_0$, $a \in [q'], i \in C_a^N$, $f^i(\mathbf{x}, t) = g(\mathbf{x}, \mathbf{y}_i, a, t)$. Further, the empirical distribution of $\{\mathbf{y}_i\}_{i \in C_a^N}$, denote it by $\widehat{P}_a$, converges weakly to P_a , i.e.

$$\frac{1}{|C_a^N|} \sum_{i \in C_a^N} \delta_{\mathbf{y}_i} \xrightarrow{d} P_a,$$

- 3) For each $a \in [q']$, in probability,

$$\lim_{N \rightarrow \infty} \frac{1}{|C_a^N|} \sum_{i \in C_a^N} g(\mathbf{x}_i^0, \mathbf{y}_i, a, 0) g(\mathbf{x}_i^0, \mathbf{y}_i, a, 0)^\top = \widehat{\Sigma}_a^0.$$

This definition follows Remark 1 of Javanmard and Montanari (2013) and allows for a partition size q' that is distinct from q . Further note that the functions $f^i(\cdot, \cdot)$ of Definition S3, 2), depends on $\mathbf{y}_i$, but as these do not change between iterations, this dependence is suppressed for notational convenience.

Definition S4 (State evolution). Given $q, q' \in \mathbb{N}_0$ and starting points $\{\widehat{\Sigma}_a^0\}_{a \in [q']}, \widehat{\Sigma}_a^0 \in \mathbb{R}^{q \times q}$ positive definite, and weights $\{c_a\}_{a \in [q']}, c_a \in \mathbb{R}$, the state evolution defines a sequence of positive definite matrices $\{\Sigma^t\}_{t \in \mathbb{N}}$, such that for each $t \in \mathbb{N}$,

$$\Sigma^t = \sum_{b=1}^{q'} c_b \widehat{\Sigma}_b^{t-1} \tag{S7}$$

$$\widehat{\Sigma}_a^t = \mathbb{E} [g(Z_a^t, Y_a, a, t) g(Z_a^t, Y_a, a, t)^\top] \quad (a = 1, \dots, q'), \tag{S8}$$

where $Y_a \sim P_a$, $Z_a^t \sim \mathcal{N}(\mathbf{0}_q, \Sigma^t)$ and where $Y_a \perp Z_a^t$ for $a \in [q']$.

It is now possible to state Theorem 1 of Javanmard and Montanari (2013) which will serve as the blackbox result to establish the asymptotic behaviour of the MDYPL estimator. This result makes use of pseudo-Lipschitz functions, a definition of which is given for completeness.

Definition S5 (Pseudo-Lipschitz continuity). For $k \geq 1$, a function $\psi : \mathbb{R}^m \rightarrow \mathbb{R}$ is said to be pseudo-Lipschitz of order k , denote the corresponding class of functions by $\text{PL}(k)$, if there exists a constant $L > 0$ such that for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$,

$$|\psi(\mathbf{u}) - \psi(\mathbf{v})| \leq L \left(1 + \|\mathbf{u}\|_2^{k-1} + \|\mathbf{v}\|_2^{k-1} \right) \|\mathbf{u} - \mathbf{v}\|_2.$$

Hence, if $\psi \in \text{PL}(k)$, there exists a constant L' such that for all $\mathbf{u} \in \mathbb{R}^m$, $|\psi(\mathbf{u})| \leq L'(1 + \|\mathbf{u}\|_2^k)$. Further, note that every Lipschitz function is also pseudo Lipschitz for any $k \geq 1$.

Theorem S1 (Javanmard and Montanari (2013) Theorem 1). *Let $\{(\mathbf{A}(N), \mathcal{F}_N, x^{0,N})\}_{N \in \mathbb{N}_0}$ be a converging sequence of AMP instances and denote by $\{x^t\}_{t \in \mathbb{N}_0}$ the corresponding AMP orbit. Suppose further that for some $k \geq 2$ and all $a \in [q']$, $E_{P_a}[\|Y_a\|^{2k-2}]$ is bounded and that $E_{\hat{P}_a}[\|Y_a\|^{2k-2}] \rightarrow E_{P_a}[\|Y_a\|^{2k-2}]$ as $N \rightarrow \infty$. Then for all $t \in \mathbb{N}$, $a \in [q']$ and any pseudo-Lipschitz function $\psi : \mathbb{R}^q \times \mathbb{R}^q \rightarrow \mathbb{R}$ of order k ,*

$$\lim_{N \rightarrow \infty} \frac{1}{|C_a^N|} \sum_{i \in C_a^N} \psi(\mathbf{x}_i^t, \mathbf{y}_i) = E[\psi(Z_a^t, Y_a)],$$

where $Y_a \sim P_a$, $Z_a^t \sim N(\mathbf{0}_q, \Sigma^t)$ and $Y_a \perp Z_a^t$.

S4.4 Proof of Theorem 3.1

S4.4.1 Target AMP recursion

Consider the following recursion. Given initial conditions $\hat{\beta}^0$, $\mathbf{S}^0 = \mathbf{X}\hat{\beta}^0$, and a sequence of constants $\{\mu_t, b_t, \sigma_t\}_{t \in \mathbb{N}_0}$, for $t \in \mathbb{N}$, set

$$\hat{\beta}^t = \hat{\beta}^{t-1} + \kappa^{-1} \mathbf{X}^\top \Psi^{t-1}(\mathbf{y}^*, \mathbf{S}^{t-1}) \quad (\text{S9})$$

$$\mathbf{S}^t = \mathbf{X}\hat{\beta}^t - \Psi^{t-1}(\mathbf{y}^*, \mathbf{S}^{t-1}), \quad (\text{S10})$$

where $\Psi^t(\mathbf{y}^*, \mathbf{S}^t)$ acts elementwise on $\mathbf{y}^*, \mathbf{S}^t$ as

$$\Psi^t(y_i^*, \mathbf{S}_i^t) = b_t \{y_i^* - \zeta'(\text{prox}_{b_t \zeta}(b_t y_i^* + \mathbf{S}_i^t))\},$$

for $i = 1, \dots, n$ and where $y_i^* = \alpha y_i + (1 - \alpha)/2$. For $\alpha = 1$, this recursion corresponds to equations (49) and (50) of the supplementary material of Sur and Candès (2019).

The relation of this recursion to the maximiser of $\ell(\beta; \mathbf{y}^*, \mathbf{X})$ is shown in Sur and Candès 2019, Appendix B: If there exists a fixed points $\hat{\beta}_*, \mathbf{S}_*$ along with (μ_*, b_*, σ_*) , then by construction

$$\begin{aligned} \mathbf{X}^\top b_* (\mathbf{y}^* - \zeta'(\text{prox}_{b_* \zeta}(b_* \mathbf{y}^* + \mathbf{S}_*))) &= \mathbf{0}_p \\ (b_* \mathbf{y}^* + \mathbf{S}_*) - b_* \zeta'(\text{prox}_{b_* \zeta}(b_* \mathbf{y}^* + \mathbf{S}_*)) &= \mathbf{X}\hat{\beta}_*. \end{aligned}$$

Now using that $x - b\zeta'(\text{prox}_{b\zeta}(x)) = \text{prox}_{b\zeta}(x)$, it follows immediately that

$$\mathbf{X}^\top (\mathbf{y}^* - \zeta'(\mathbf{X}\hat{\beta}_*)) = \mathbf{0}_p,$$

the left hand side of which is proportional to the gradient of (S1), and by strict concavity thereof, $\hat{\beta}_*$ is equal to $\hat{\beta}^{\text{DY}}$.

S4.4.2 Centred target AMP recursion

Instead of analysing the target AMP recursion directly, a centred variant thereof is analysed. This centring ensures that the iterates of the resulting recursion have well-defined limits that are understood by AMP theory (See remarks in Donoho and Montanari 2016, Appendix B). In particular, as in Sur and Candès (2019) and Donoho and Montanari (2016), consider the following centred recursion with adjusted initial conditions $\boldsymbol{\nu}^0 = \hat{\beta}^0 - \mu_0 \beta_0$, $\mathbf{R}^0 = \mathbf{S}^0$ for $\hat{\beta}^0, \beta_0 \in \mathbb{R}^p$, $\mu_0 \in \mathbb{R}$ and $\mathbf{S}^0 \in \mathbb{R}^n$, for $t \in \mathbb{N}$, let

$$\boldsymbol{\nu}^t = q_{t-1}(\boldsymbol{\nu}^{t-1} + \mu_{t-1} \beta_0) - a_t \beta_0 + \kappa^{-1} \mathbf{X}^\top \Psi^{t-1}(\mathbf{y}^*, \mathbf{R}^{t-1}) \quad (\text{S11})$$

$$\mathbf{R}^t = \mathbf{X}(\boldsymbol{\nu}^t + \mu_t \beta_0) - \Psi^{t-1}(\mathbf{y}^*, \mathbf{R}^{t-1}), \quad (\text{S12})$$

where $a_0 = \mu_0$ and for $t \in \mathbb{N}$,

$$q_t = -\frac{1}{\kappa n} \sum_{i=1}^n \frac{\partial \Psi^t}{\partial R}(y_i^*, R) \Big|_{R=\mathbf{R}_i^t} \quad (\text{S13})$$

$$a_t = \frac{1}{\kappa n} \sum_{i=1}^n \frac{\partial \Psi^{t-1}}{\partial Z}(h(Z, \bar{\varepsilon}_i), \mathbf{R}_i^{t-1}) \Big|_{Z=\mathbf{x}_i^\top \beta_0}, \quad (\text{S14})$$

where $\bar{\varepsilon}_i \sim U(0, 1)$ ($i = 1, \dots, n$) are i.i.d. draws independent of all other random variables.

Given initial conditions (μ_0, b_0, σ_0) , let the state evolution of (μ_t, b_t, σ_t) be characterised by the solution to the following system of nonlinear equations

$$1 - \kappa = \mathbb{E} \left[\frac{1}{1 + b_t \zeta''(\text{prox}_{b_t \zeta}(b_t Y^* + Z_t))} \right] \quad (\text{S15})$$

$$\mu_{t+1} = \frac{b_t}{\kappa \gamma^2} \mathbb{E} [Z \{Y^* - \zeta'(\text{prox}_{b_t \zeta}(b_t Y^* + Z_t))\}] + \mu_t \quad (\text{S16})$$

$$\sigma_{t+1}^2 = \frac{b_t^2}{\kappa^2} \mathbb{E} [\{Y^* - \zeta'(\text{prox}_{b_t \zeta}(b_t Y^* + Z_t))\}^2], \quad (\text{S17})$$

for independent $Z \sim \mathcal{N}(0, \gamma^2)$, $G \sim \mathcal{N}(0, 1)$, and $\bar{\varepsilon} \sim U(0, 1)$, and letting

$$Y^* = \frac{\alpha}{2} \{1 - \text{sign}(\bar{\varepsilon} - \zeta'(Z))\} + \frac{1 - \alpha}{2}, \quad Z_t = \mu_t Z + \kappa^{1/2} \sigma_t G.$$

The iterate of σ_{t+1} in (S17) arises from the state evolution Σ^t (cf. Definition S4) that characterises the variance of the AMP iterates (S9) and (S10) as is shown in (S49). On the other hand, b_t, μ_t come from the nonlinear corrections a_t, q_t in (S13) and (S13) to the recursion (S9) and (S10). (S52) and (S54)-(S58), establish the relation between b_t, μ_t and the corrections a_t, q_t . These corrections ensure that the AMP iterates have a tractable asymptotic distribution. It is customary in the AMP literature to refer to the iterates μ_t, b_t, σ_t as state evolutions and equations (S35)-(S37) as state evolution equations – a convention which is henceforth adhered to in this text. Feng *et al.* (2022), Section 2.1, and Section 4 provide a thorough discussion of the state evolutions and their role in AMP.

The relation to equations (22)-(24) of the supplementary material of Sur and Candès (2019) and equation (118) in Feng *et al.* (2022) is evident.

S4.4.3 Matrix-valued AMP-embedding

This section embeds the centred AMP recursion of (S4.4.2) in a symmetric AMP-instance, which is shown to satisfy all properties of Definitions S1-S4, apart from a Lipschitz continuity constraint on $g(\cdot, \cdot, a, t)$ in Definition S3, which is dealt with in Section S4.4.4. Theorem S1 then characterises the asymptotic behaviour of said recursion. In particular, the centred recursion of (S11)-(S14), which is not symmetric, is embedded in a symmetric AMP recursion as defined in S4.4.1. A subset of the resulting orbit then corresponds to the orbit of interest, namely (S11)-(S14). This approach was proposed in Javanmard and Montanari (2013, Lemma 1) and also followed in Sur and Candès (2019, Lemma 3) and Donoho and Montanari (2016, Lemma C.1). The symmetric AMP instance is built up for even and odd iterates and the partition $[n] \cup [N] \setminus [n]$ of $[N]$ separately, which is then extended to a complete AMP recursion.

In what follows, iterates are indicated by the letters $t, t-1, t-2$ and 0, which for scalar values are used as subscripts, e.g. μ_0, σ_{t-1} and are used as superscripts for functions and nonscalar elements, e.g. $\Psi^{t-1}(\mathbf{y}, \mathbf{R}^{t-1})$.

Relating a recursion that is parametrised in terms of model parameters $\mathbf{y}, y_i, \mathbf{X}, \mathbf{x}_i$ etc. and a general AMP recursion, which is parametrised by elements such as $x, y, \mathbf{x}_i, \mathbf{y}_i, Y_a$ may lead to confusion. For this reason, an effort is made to point out the disparity. It is specifically pointed out that boldface elements pertaining to the original model use mathematical boldface (e.g. $\mathbf{x}_i$), while boldface elements of a AMP recursion use standard boldface (e.g. $\mathbf{x}_i$). Moreover, note that Javanmard and Montanari (2013) chose symbols x, y to

denote elements in $\mathcal{V}_{q,N}$, and in particular, $y = (\mathbf{y}_1, \dots, \mathbf{y}_N)$, a convention which is kept for comparability. This is not the same as $\mathbf{y} = (y_1, \dots, y_n)$, which is used to denote the response variables in the logistic regression model. Moreover, the response variables y_i are assumed to be realizations of random variables Y_i for $i = 1, \dots, n$. This may stand in conflict with Y_a , the random variables in $\mathfrak{R}^q$ as defined in the AMP setup, which are indexed by partition cell labels $a \in [q']$. To avoid ill-definedness, the index i is exclusively used when referring to the Bernoulli random variables Y_i that give rise to observations y_i , while the subscript a is used in the context of the random variables for the AMP recursion. Finally, recall that $\beta_0 \in \mathfrak{R}^p$ is the parameter vector that is to be estimated, not an initial condition.

Let q be an arbitrarily large integer, $q' = 2$, $N = n + p$, where n, p denote the rows and columns of $\mathbf{X}$, respectively, and $\kappa = \kappa(N) = p/n$. Following Definition S1, 1), let $\mathbf{A}$ be the symmetric $N \times N$ matrix whose upper triangular entries are given by $\mathbf{A}_{ij} = \sqrt{n/N} \mathbf{X}_{i,j-n}$ for $i \in [n]$ and $j \in [N] \setminus [n]$, and all other entries $\mathbf{A}_{ij}$ $i < j$ are i.i.d. $N(0, N^{-1})$ independent of all other entries in $\mathbf{A}$. That is, letting $\mathbf{B}_1, \mathbf{B}_2$ be $n \times n$ matrices with i.i.d. $N(0, (2N)^{-1})$ entries and defining $\mathbf{C}_k = \mathbf{B}_k + \mathbf{B}_k^\top$ for $k = 1, 2$, then

$$\mathbf{A} = \begin{bmatrix} \mathbf{C}_1 & \sqrt{\frac{n}{N}} \mathbf{X} \\ \sqrt{\frac{n}{N}} \mathbf{X}^\top & \mathbf{C}_2 \end{bmatrix}. \quad (\text{S18})$$

Next, the collection of functions $\mathcal{F}$ is constructed. For this, recall from Definition S3, 2), that the collections $\mathcal{F}_N$ of a converging sequence of AMP-instances with functions $f^i(\cdot, \cdot)$ may depend on $\mathbf{y}_i$ for some $y \in \mathcal{V}_{q,N}$. Hence, given $\mathbf{x}, \mathbf{y}_i \in \mathfrak{R}^q$, which should be thought of as entries of some $x, y \in \mathcal{V}_{q,N}$ that are concretely specified in (S21) and (S22), for $i \in [n]$, and odd t , define $f^i(\mathbf{x}, t)$ as

$$f^i(\mathbf{x}, t) = \kappa^{-1} \sqrt{\frac{N}{n}} \left[0, \Psi^0(h(\mathbf{x}(1), \mathbf{y}_i(1)), \mathbf{x}(3)), 0, \Psi^1(h(\mathbf{x}(1), \mathbf{y}_i(1)), \mathbf{x}(5)), \dots, \Psi^{\frac{t-1}{2}}(h(\mathbf{x}(1), \mathbf{y}_i(1)), \mathbf{x}(t+2)), 0, 0, \dots \right]^\top. \quad (\text{S19})$$

whereas for $i \in [N] \setminus [n]$ and odd t , let $f^i(\mathbf{x}, t) = \mathbf{0}_q$. Further, for $i \in [n]$ and even t , $f^i(\mathbf{x}, t) = \mathbf{0}_q$ and for $i \in [N] \setminus [n]$, even t , let

$$f^i(\mathbf{x}, t) = \sqrt{\frac{N}{n}} \left[\mathbf{y}_i(1), 0, \mathbf{y}_i(2) + \mu_0 \mathbf{y}_i(1), 0, \mathbf{x}(2) + \mu_1 \mathbf{y}_i(1), 0, \mathbf{x}(4) + \mu_2 \mathbf{y}_i(1), 0, \dots, \mathbf{x}(t) + \mu_{\frac{t}{2}} \mathbf{y}_i(1), 0, 0, \dots \right]^\top. \quad (\text{S20})$$

For well-definedness of (S19), (S20), which map to $\mathfrak{R}^q$, it must be the case that $t \leq q - 3$, which is of no consequence as q can always be chosen large enough to accommodate any $t \in \mathfrak{R}$. Hence, without loss of generality, all results are stated for $t \in [q]$, rather than $t \in [\tilde{q}]$ for $\tilde{q} = q + 3$ to simplify presentation.

It is straightforward to establish that $f^i(\cdot, t)$ is locally Lipschitz in its first argument, which is shown formally for the smoothed version of S4.4.4, to which Theorem S1 applies ultimately. To conclude the symmetric AMP-instance, let $x^0 = \mathbf{0}_{N,q}$ be the N -vector with elements $\mathbf{0}_q$.

With these definitions in place, it is possible to define the iterates x^t and y that give rise to the target AMP recursion. In particular, let $\eta_0 = \mathbf{X} \beta_0$ and for some integer $k, t \in [q]$, define $\mathbf{x}_i^t$ as

$$\mathbf{x}_i^t = \begin{cases} \left[\eta_{0,i}, 0, \mathbf{R}_i^0, 0, \mathbf{R}_i^1, 0, \dots, \mathbf{R}_i^{\frac{t-1}{2}}, 0, 0, \dots \right]^\top & \text{if } t = 2k + 1, i \in [n] \\ \mathbf{0}_q & \text{if } t = 2k + 1, i \in [N] \setminus [n] \\ \mathbf{0}_q & \text{if } t = 2k, i \in [n] \\ \left[0, \nu_{i-n}^1, 0, \nu_{i-n}^2, \dots, \nu_{i-n}^{\frac{t}{2}}, 0, 0, \dots \right]^\top & \text{if } t = 2k, i \in [N] \setminus [n] \end{cases} \quad (\text{S21})$$

Finally, let $y \in \mathcal{V}_{N,q}$ have vectors $\mathbf{y}_i \in \mathfrak{R}^q$, $i = 1, \dots, N$, such that

$$\mathbf{y}_i(k) = \begin{cases} \bar{\varepsilon}_i & \text{if } i \in [n], k = 1 \\ \beta_{0,i-n} & \text{if } i \in [N] \setminus [n], k = 1 \\ \nu_{i-n}^0 & \text{if } i \in [N] \setminus [n], k = 2 \\ 0 & \text{else} \end{cases} \quad (\text{S22})$$

That is, y can be seen as the $q \times N$ matrix

$$y = \begin{bmatrix} \bar{\varepsilon}_1 & \bar{\varepsilon}_2 & \dots & \bar{\varepsilon}_n & \beta_{0,1} & \beta_{0,2} & \dots & \beta_{0,p} \\ 0 & 0 & \dots & 0 & \nu_1^0 & \nu_2^0 & \dots & \nu_p^0 \\ 0 & & & & \dots & & & 0 \\ \vdots & & & & \ddots & & & \vdots \\ 0 & & & & \dots & & & 0 \end{bmatrix}$$

It is now possible to define an AMP-orbit corresponding to the above AMP instance, which is subject of the next two Lemmas. For this, note from Definition S2, equation (S6), that the i th component of an AMP orbit $\{x^t\}_{t \in \mathbb{N}}$ for an AMP instance $(\mathbf{A}, \mathcal{F}, x^0)$ is given by

$$\mathbf{x}_i^t = \sum_{j=1}^N \mathbf{A}_{ij} f^j(\mathbf{x}_j^{t-1}, t-1) - \frac{1}{N} \left\{ \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, t-1) \Big|_{\mathbf{x}=\mathbf{x}_j^{t-1}} \right\} f^i(\mathbf{x}_i^{t-2}, t-2). \quad (\text{S23})$$

Lemma S1. *Consider the symmetric AMP instance $(\mathbf{A}, \mathcal{F}, x^0)$ defined by (S18), (S19), (S20) and $x^0 = 0_{N,q}$ and the iterates $\{x^t\}_{t \in [q]}$, the entries of which are defined in (S21) and (S11)-(S14). Then, for $t \in [q]$ being even, and $i \in [N] \setminus [n]$ and $t \in [q]$ odd with $i \in [n]$, these iterates satisfy the recursion of (S23).*

Proof. The proof follows from strong induction on t for even and odd iterates and straightforward algebra.

Base case (i): $t = 1, i \in [n]$. Note that by construction

$$\begin{aligned} \sum_{j=1}^N \mathbf{A}_{ij} f^j(\mathbf{x}_j^0, 0) &= \sum_{j=n+1}^{n+p} \mathbf{A}_{ij} f^j(\mathbf{x}_j^0, 0) \\ &= \sqrt{\frac{n}{N}} \sum_{j=1}^p \mathbf{X}_{ij} f^{j+n}(\mathbf{x}_{j+n}^0, 0) \\ &= \sum_{j=1}^p \mathbf{X}_{ij} [\mathbf{y}_{j+n}(1), 0, \mathbf{y}_{j+n}(2) + \mu_0 \mathbf{y}_{j+n}(1), 0, 0, \dots]^\top \\ &= \sum_{j=1}^p \mathbf{X}_{ij} [\beta_{0,j}, 0, \nu_j^0 + \mu_0 \beta_{0,j}, 0, 0, \dots]^\top \\ &= \sum_{j=1}^p \mathbf{X}_{ij} [\beta_{0,j}, 0, \hat{\beta}_j^0 - \mu_0 \beta_{0,j} + \mu_0 \beta_{0,j}, 0, 0, \dots]^\top \\ &= \left[\sum_{j=1}^p \mathbf{X}_{ij} \beta_{0,j}, 0, \sum_{j=1}^p \mathbf{X}_{ij} \hat{\beta}_j^0, 0, 0, \dots \right]^\top \\ &= [\boldsymbol{\eta}_{0,i}, 0, \mathbf{R}_i^0, 0, 0, \dots]^\top \\ &= \mathbf{x}_i^1, \end{aligned} \quad (\text{S24})$$

The first equality follows from the definition of (S20), the second from the definition of (S18), the third again from (S20), the fourth from (S22), the fifth and seventh from the definition of $\nu^0, \mathbf{R}^0$ above (S11) and since $\boldsymbol{\eta}_0 = \mathbf{X} \beta_0$ and the last from (S21). Concordance with the full recursion of (S23) follows since $f^i(\mathbf{x}_i^{-1}, -1) = \mathbf{0}_q$ by definition.

Base case (ii): $t = 2, i \in [N] \setminus [n]$. Again it is easily checked that

$$\begin{aligned}
\sum_{j=1}^N \mathbf{A}_{ij} f^j(\mathbf{x}_j^1, 1) &= \sum_{j=1}^n \mathbf{A}_{ij} f^j(\mathbf{x}_j^1, 1) \\
&= \sqrt{\frac{N}{n}} \sum_{j=1}^n \mathbf{X}_{ji} f^j(\mathbf{x}_j^1, 1) \\
&= \kappa^{-1} \sum_{j=1}^n \mathbf{X}_{ji} [0, \Psi^0(h(\mathbf{x}_j^1(1), y_j(1)), \mathbf{x}_j^1(3)), 0, 0 \dots]^\top \\
&= \kappa^{-1} \sum_{j=1}^n \mathbf{X}_{ji} [0, \Psi^0(h(\boldsymbol{\eta}_{0,j}, \bar{\varepsilon}_j), \mathbf{R}_j^0), 0, 0 \dots]^\top,
\end{aligned} \tag{S25}$$

where the first equality comes from (S19), the second and third from (S18), the fourth again from (S19) and the last from (S24). Now recalling that $y_i^* = h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i)$ for $h(Z, \bar{\varepsilon}) = \alpha \{1 - \text{sign}(\bar{\varepsilon} - \zeta'(Z))\} / 2 + (1 - \alpha)/2$, it immediately follows that the nonzero element in the last line of (S25) corresponds to the i th entry of $\kappa^{-1} \mathbf{X}^\top \Psi^0(\mathbf{y}^*, \mathbf{R}^0)$.

Next consider

$$\begin{aligned}
\sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, 1) \Big|_{\mathbf{x}=\mathbf{x}_j^1} &= \sum_{j=1}^n \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, 1) \Big|_{\mathbf{x}=\mathbf{x}_j^1} \\
&= \kappa^{-1} \sqrt{\frac{N}{n}} \sum_{j=1}^n [0, \frac{\partial}{\partial \mathbf{x}} \Psi^0(h(\mathbf{x}(1), \mathbf{y}(1)), \mathbf{x}(3)), 0, 0, \dots]^\top \Big|_{\mathbf{x}=\mathbf{x}_j^1} \\
&= \kappa^{-1} \sqrt{\frac{N}{n}} \sum_{j=1}^n \begin{bmatrix} 0 & 0 & \dots & 0 & 0 & \dots \\ \frac{\partial}{\partial Z} \Psi^0(h(Z, \bar{\varepsilon}_j), \mathbf{R}_j^0) \Big|_{Z=\boldsymbol{\eta}_{0,j}} & 0 & \frac{\partial}{\partial R} \Psi^0(y_j^*, R) \Big|_{R=\mathbf{R}_j^0} & 0 & 0 & \dots \\ 0 & 0 & \dots & & & \\ \vdots & \ddots & & & & \end{bmatrix},
\end{aligned} \tag{S26}$$

where the first two equalities follow from (S19) and the last from straightforward differentiation and $\mathbf{x}^1$ from (S24).

Finally, it immediately follows from (S20) that

$$\begin{aligned}
f^i(\mathbf{x}_i^0, 0) &= \sqrt{\frac{N}{n}} [\mathbf{y}_i(1), 0, \mathbf{y}_i(2) + \mu_0 \mathbf{y}_i(1), 0, 0 \dots]^\top \\
&= \sqrt{\frac{N}{n}} [\boldsymbol{\beta}_{0,i-n}, 0, \boldsymbol{\nu}_{i-n}^0 + \mu_0 \boldsymbol{\beta}_{0,i-n}, 0, 0 \dots]^\top
\end{aligned} \tag{S27}$$

so that (S26) and (S27) yield

$$\begin{aligned}
& -\frac{1}{N} \left\{ \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, 1) \Big|_{\mathbf{x}=\mathbf{x}_j^1} \right\} f^i(\mathbf{x}_i^0, 0) \\
&= -\frac{1}{n\kappa} \begin{bmatrix} 0 & 0 & \dots & 0 & 0 & \dots \\ \sum_{j=1}^n \frac{\partial}{\partial Z} \Psi^0(h(Z, \bar{\varepsilon}_j), \mathbf{R}_j^0) \Big|_{Z=\eta_{0,j}} & 0 & \sum_{j=1}^n \frac{\partial}{\partial R} \Psi^0(y_j^*, R) \Big|_{R=\mathbf{R}_j^0} & 0 & 0 & \dots \\ 0 & 0 & \dots & 0 & 0 & \dots \\ \vdots & \ddots & & \vdots & \vdots & \end{bmatrix} \begin{bmatrix} \beta_{0,i-n} \\ 0 \\ \nu_{i-n}^0 + \mu_0 \beta_{0,i-n} \\ 0 \\ 0 \\ \vdots \end{bmatrix} \\
&= \left[0, \left\{ -\frac{1}{n\kappa} \sum_{j=1}^n \frac{\partial}{\partial Z} \Psi^0(h(Z, \bar{\varepsilon}_j), \mathbf{R}_j^0) \Big|_{Z=\eta_{0,j}} \right\} \beta_{0,i-n} + \left\{ -\frac{1}{n\kappa} \sum_{j=1}^n \frac{\partial}{\partial R} \Psi^0(y_j^*, R) \Big|_{R=\mathbf{R}_j^0} \right\} (\nu_{i-n}^0 + \mu_0 \beta_{0,i-n}), 0, 0, \dots \right] \\
&= [0, -a_1 \beta_{0,i-n} + q_0 (\nu_{i-n}^0 + \mu_0 \beta_{0,i-n}), 0, 0, \dots]^\top, \quad (\text{S28})
\end{aligned}$$

Hence, from (S25), (S28) and (S11), (S13), (S14), it follows that the second entry of (S23) equals

$$\kappa^{-1} \sum_{j=1}^n \mathbf{X}_{ji} \Psi^0(h(\eta_{0,j}, \bar{\varepsilon}_j), \mathbf{R}_j^0) - a_1 \beta_{0,i-n} + q_0 (\nu_{i-n}^0 + \mu_0 \beta_{0,i-n}) = \nu_{i-n}^1,$$

while all other entries are zero. Hence, by construction of (S21) the second base case holds.

Induction hypothesis: Assume that there exists some $t < q$, $t \geq 2$ such that for all $t' \leq t$ if t' is even, and $i \in [N] \setminus [n]$ or if t' is odd with $i \in [n]$, the iterates of (S21) satisfy the recursion of (S23).

Induction step (i): $t+1$ odd, $i \in [n]$. Most derivations are analogous to Base case (i), so that details are skipped on repetitive steps. First note that due to the nested nature of the iterates $\mathbf{x}_i^t$ in (S21) and the induction hypothesis, the induction step must only be established for the last nonzero entry in $\mathbf{x}_j^{t+1}$.

For this, note that

$$\sum_{j=1}^N \mathbf{A}_{ij} f^j(\mathbf{x}_j^t, t) = \sum_{j=1}^p \mathbf{X}_{ij} \left[\beta_{0,j}, 0, \nu_j^0 + \mu_0 \beta_{0,j}, 0, \nu_j^1 + \mu_1 \beta_{0,j}, 0, \dots, \nu_j^{\frac{t}{2}} + \mu_{\frac{t}{2}} \beta_{0,j}, 0, 0, \dots \right]^\top. \quad (\text{S29})$$

Next, it is easily seen that

$$\begin{aligned}
& \sqrt{\frac{n}{N}} \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, t) \Big|_{\mathbf{x}=\mathbf{x}_j^t} = \sqrt{\frac{n}{N}} \sum_{j=1}^p \frac{\partial}{\partial \mathbf{x}} f^{n+j}(\mathbf{x}, t) \Big|_{\mathbf{x}=\mathbf{x}_{n+j}^t} \\
&= \sum_{j=1}^p \frac{\partial}{\partial \mathbf{x}} \left[\mathbf{y}_i(1), 0, \mathbf{y}_i(2) + \mu_0 \mathbf{y}_i(1), 0, \mathbf{x}(2) + \mu_1 \mathbf{y}_i(1), 0, \mathbf{x}(4) + \mu_2 \mathbf{y}_i(1), 0, \dots, \mathbf{x}(t) + \mu_{\frac{t}{2}} \mathbf{y}_i(1), 0, \dots \right]^\top \Big|_{\mathbf{x}=\mathbf{x}_{n+j}^t} \\
&= \sum_{j=1}^p \tilde{I}_{q \times q} = p \tilde{I}_{q \times q}, \quad (\text{S30})
\end{aligned}$$

where $\tilde{I}_{q \times q}$ whose odd rows with row index $5 \leq i \leq t+3$ have the $(i-3)$ rd element equal to one, while all other elements of this matrix are equal to zero.

Further, note that for $t + 1$ odd, $i \in [n]$,

$$\kappa \sqrt{\frac{n}{N}} f^i(\mathbf{x}_i^{t-1}, t-1) = \begin{bmatrix} 0 \\ \Psi^0(h(\mathbf{x}_i^{t-1}(1), \mathbf{y}_i(1)), \mathbf{x}_i^{t-1}(3)) \\ 0 \\ \Psi^1(h(\mathbf{x}_i^{t-1}(1), \mathbf{y}_i(1)), \mathbf{x}_i^{t-1}(5)) \\ 0 \\ \vdots \\ \Psi^{\frac{t-2}{2}}(h(\mathbf{x}_i^{t-1}(1), \mathbf{y}_i(1)), \mathbf{x}_i^{t-1}(t+1)) \\ 0 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ \Psi^0(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^0) \\ 0 \\ \Psi^1(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^1) \\ 0 \\ \vdots \\ \Psi^{\frac{t}{2}-1}(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^{\frac{t}{2}-1}) \\ 0 \\ \vdots \end{bmatrix}, \quad (\text{S31})$$

by definition of (S20), (S21) and (S22). Hence, from (S30) and (S31), it follows that

$$-\frac{1}{N} \left\{ \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, t) \Big|_{\mathbf{x}=\mathbf{x}_j^t} \right\} f^i(\mathbf{x}_i^{t-1}, t-1) = -\kappa^{-1} \frac{p}{n} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \Psi^0(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^0) \\ 0 \\ \Psi^1(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^1) \\ 0 \\ \vdots \\ \Psi^{\frac{t}{2}-1}(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^{\frac{t}{2}-1}) \\ 0 \\ \vdots \end{bmatrix},$$

Hence, it follows from equations (S29) and (S29), that the last nonzero entry of

$$\sum_{j=1}^N \mathbf{A}_{ij} f^j(\mathbf{x}_j^t, t) - \frac{1}{N} \left\{ \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\mathbf{x}, t) \Big|_{\mathbf{x}=\mathbf{x}_j^t} \right\} f^i(\mathbf{x}_i^{t-1}, t-1)$$

is given by

$$\sum_{j=1}^p \mathbf{X}_{ij} \left(\nu_j^{\frac{t}{2}} + \mu_{\frac{t}{2}} \beta_{0,j} \right) - \Psi^{\frac{t}{2}-1} \left(h(\boldsymbol{\eta}_{0,i}, \bar{\varepsilon}_i), \mathbf{R}_i^{\frac{t}{2}-1} \right),$$

which by (S12) corresponds to the i th entry of $\mathbf{R}_i^{\frac{t}{2}}$ and that corresponds precisely to the last nonzero entry of $\mathbf{x}_i^{t+1}$ as defined in (S21). This establishes the first part of the induction step.

Induction step (ii): $t + 1$ even, $i \in [N] \setminus [n]$. This step is completely analogous to base case (ii) and therefore omitted.

By strong induction, the claim of the Lemma holds for any $t \in [q]$. $\square$

With this result it is possible to fill up the missing parts the incomplete recursion of Lemma S1. Hence, define iterate $\tilde{x}^t$ as follows. Let $\tilde{x}^0 = 0_{N,q}$ and for $t \in [q]$ odd and $i \in [n]$ or if t is even and $i \in [N] \setminus [n]$, set $\tilde{\mathbf{x}}_i^t = \mathbf{x}_i^t$. Otherwise, let

$$\tilde{\mathbf{x}}_i^t = \sum_{j=1}^N \mathbf{A}_{ij} f^j(\tilde{\mathbf{x}}_j^{t-1}, t-1) - \frac{1}{N} \left\{ \sum_{j=1}^N \frac{\partial}{\partial \mathbf{x}} f^j(\tilde{\mathbf{x}}, t-1) \Big|_{\tilde{\mathbf{x}}=\tilde{\mathbf{x}}_j^{t-1}} \right\} f^i(\tilde{\mathbf{x}}_i^{t-2}, t-2), \quad (\text{S32})$$

again with the convention that terms involving negative t -indices are zero. The Lemma below establishes

that this recursion is indeed a AMP orbit corresponding to instance $(\mathbf{A}, \mathcal{F}, \tilde{x}^0)$ and that the orbit of interest, i.e. the iterates of Lemma S1, are preserved.

Lemma S2. *Given the AMP instance $(\mathbf{A}, \mathcal{F}, \tilde{x}^0)$ of Lemma S1, the recursion of (S32) is an orbit according to Definition S2. Further, for $t \in [q]$ odd and $i \in [n]$ or if t is even and $i \in [N] \setminus [n]$, the iterates $\tilde{\mathbf{x}}_i^t$ correspond to those in Lemma S1.*

Proof. The statement of the Lemma is shown by strong induction on t .

Base case (i): $t = 1$. Since $\tilde{x}^0 = x^0 = 0_{n,q}$, by base case (i) of Lemma S1, the second part of the Lemma holds for $t = 1$. For $i \in [n]$, the first part of the claim follows from base case (i) of Lemma S1 too, while for $i \in [N] \setminus [n]$ it follows trivially from (S32).

Base case (ii): $t = 2$. Again, note that by the base case (ii) in Lemma S1, the claim of the Lemma immediately follows for $i \in [N] \setminus [n]$. For $i \in [n]$, the second part of the Lemma follows from (S32).

Induction hypothesis: The claim of the Lemma holds for all t' up to some $t < q$, $t \geq 2$.

Induction step (i): $t + 1$ odd. Consider $i \in [n]$. From the induction hypothesis, one has that $\tilde{\mathbf{x}}_j^t = \mathbf{x}_j^t$ for $i \in [N] \setminus [n]$ and similarly $\tilde{\mathbf{x}}_j^{t-1} = \mathbf{x}_j^{t-1}$ for $i \in [n]$. Hence again, the corresponding induction step of Lemma S1 shows that the recursion is an orbit according to Definition S2 and that the iterates $\tilde{\mathbf{x}}_i^t$ correspond to those in Lemma S1. Now consider $i \in [N] \setminus [n]$. The induction step holds trivially by construction of (S32).

Induction step (ii): $t + 1$ even. This case is handled similarly to induction step (i) and is therefore omitted.

By strong induction, the Lemma is proven for all $t \in [q]$. $\square$

Hence, if $(\mathbf{A}, \mathcal{F}, \tilde{x}^0)$ were shown to be a converging AMP instance according to Definition S3, then Theorem S1 along with the state evolution of Definition S4, would give the large sample behaviour of the corresponding orbit of interest. However, as remarked in Feng *et al.* (2022), Section 4.7, the function g in Definition S3 cannot be Lipschitz, which is owed to the indicator function $h(Z, \bar{\varepsilon})$, so that Theorem S1 does not apply. The next section deals with this technicality.

S4.4.4 A smooth approximation

This section introduces a smooth approximation to the original recursions (S9)-(S10) and to (S11)-(S14), the latter of which can be used in a matrix-valued AMP embedding for which Theorem S1 applies. A limiting argument then shows that the theorem still applies in the limit where the smooth approximation approaches the nonsmooth original recursion.

In particular, the step function $\{1 - \text{sign}(x)\}/2$ can, amongst others, be approximated by $\left\{1 - \frac{x}{|x| + \epsilon}\right\}/2$ for some positive constant ϵ . Note that defining the Bernoulli response $Y = \{1 - \text{sign}(\bar{\varepsilon} - \zeta'(Z))\}/2$ rather than the more conventional $Y = \mathbb{1}\{\bar{\varepsilon} < \zeta'(Z)\}$ comes without loss of generality, as per the modelling assumption, $\boldsymbol{\eta}_{0,i}$ are independent $N(0, \gamma^2)$ normal random variables and $\bar{\varepsilon}_i$ are independent of $\boldsymbol{\eta}_0$ and uniformly distributed on the unit interval so that the event $\{\exists i \in [n] : \bar{\varepsilon}_i = \zeta'(\boldsymbol{\eta}_{0,i})\}$ has measure zero.

Hence, in what follows, denote by $h_\epsilon(Z, \bar{\varepsilon})$ the function

$$h_\epsilon(Z, \bar{\varepsilon}) = \frac{\alpha}{2} \left\{ 1 - \frac{\bar{\varepsilon} - \zeta'(Z)}{|\bar{\varepsilon} - \zeta'(Z)| + \epsilon} \right\} + \frac{1 - \alpha}{2}.$$

Consequently, define new sets of recursions analogous to (S9)-(S10) and (S11)-(S14) where all appearances of $h(\cdot, \cdot)$ are replaced by $h_\epsilon(\cdot, \cdot)$. In particular, define the approximate target AMP recursion with initial conditions $\hat{\boldsymbol{\beta}}^0$, $\mathbf{S}^0 = \mathbf{X}\hat{\boldsymbol{\beta}}^0$, and a sequence of constants $\{\epsilon\mu_t, \epsilon b_t, \epsilon\sigma_t\}_{t \in \mathbb{N}_0}$, for $t \in \mathbb{N}$, as

$$\epsilon\hat{\boldsymbol{\beta}}^t = \epsilon\hat{\boldsymbol{\beta}}^{t-1} + \kappa^{-1} \mathbf{X}^\top \Psi^{t-1}(\epsilon \mathbf{y}^*, \epsilon \mathbf{S}^{t-1}) \quad (\text{S33})$$

$$\epsilon \mathbf{S}^t = \mathbf{X}_\epsilon \hat{\boldsymbol{\beta}}^t - \Psi^{t-1}(\epsilon \mathbf{y}^*, \epsilon \mathbf{S}^{t-1}), \quad (\text{S34})$$

where $\Psi^t(\epsilon \mathbf{y}^*, \epsilon \mathbf{S}^t)$ acts elementwise on $\epsilon \mathbf{y}^*, \epsilon \mathbf{S}^t$ as

$$\Psi^t(\epsilon y_i^*, \epsilon \mathbf{S}_i^t) = \epsilon b_t \left\{ \epsilon y_i^* - \zeta' \left(\text{prox}_{\epsilon b_t \zeta} (\epsilon b_t \epsilon y_i^* + \epsilon \mathbf{S}_i^t) \right) \right\},$$

for $i = 1, \dots, n$ and where $\epsilon y_i^* = \alpha h_\epsilon(\boldsymbol{\eta}_{0,i}, \bar{\epsilon}_i) + (1 - \alpha)/2$. Further, the state evolution is determined by the system of nonlinear equations

$$1 - \kappa = \mathbb{E} \left[\frac{1}{1 + \epsilon b_t \zeta'' (\text{prox}_{\epsilon b_t \zeta} (\epsilon b_t \epsilon Y^* + \epsilon Z_t))} \right] \quad (\text{S35})$$

$$\epsilon \mu_{t+1} = \frac{\epsilon b_t}{\kappa \gamma^2} \mathbb{E} [Z \{ \epsilon Y^* - \zeta' (\text{prox}_{\epsilon b_t \zeta} (\epsilon b_t \epsilon Y^* + \epsilon Z_t)) \}] + \epsilon \mu_t \quad (\text{S36})$$

$$\epsilon \sigma_{t+1}^2 = \frac{\epsilon b_t^2}{\kappa^2} \mathbb{E} [\{ \epsilon Y^* - \zeta' (\text{prox}_{\epsilon b_t \zeta} (\epsilon b_t \epsilon Y^* + \epsilon Z_t)) \}^2], \quad (\text{S37})$$

for independent $Z, \tilde{Z} \sim N(0, \gamma^2)$, $G \sim N(0, 1)$, and $\bar{\epsilon} \sim U(0, 1)$, and letting

$$\epsilon Y^* = h_\epsilon(Z, \bar{\epsilon}), \quad \epsilon Z_t = \epsilon \mu_t Z + \kappa^{1/2} \epsilon \sigma_t G.$$

Given initial conditions $\boldsymbol{\nu}^0 = \hat{\boldsymbol{\beta}}^0 - \mu_0 \boldsymbol{\beta}_0$, $\mathbf{R}^0 = \mathbf{S}^0$ for $\hat{\boldsymbol{\beta}}^0, \boldsymbol{\beta}_0 \in \Re^p$, $\mu_0 \in \Re$ and $\mathbf{S}^0 \in \Re^n$, for $t \in \mathbb{N}$, the approximate centred recursion of (S11)-(S14) is defined as

$$\epsilon \boldsymbol{\nu}^t = \epsilon q_{t-1}(\epsilon \boldsymbol{\nu}^{t-1} + \epsilon \mu_{t-1} \boldsymbol{\beta}_0) - \epsilon a_t \boldsymbol{\beta}_0 + \kappa^{-1} \mathbf{X}^\top \Psi^{t-1}(\epsilon \mathbf{y}^*, \epsilon \mathbf{R}^{t-1}) \quad (\text{S38})$$

$$\epsilon \mathbf{R}^t = \mathbf{X}(\epsilon \boldsymbol{\nu}^t + \epsilon \mu_t \boldsymbol{\beta}_0) - \Psi^{t-1}(\epsilon \mathbf{y}^*, \epsilon \mathbf{R}^{t-1}), \quad (\text{S39})$$

where $a_0 = \mu_0$ and for $t \in \mathbb{N}$,

$$\begin{aligned} \epsilon q_t &= -\frac{1}{\kappa n} \sum_{i=1}^n \frac{\partial \Psi^t}{\partial R}(\epsilon y_i^*, R) \Big|_{R=\epsilon \mathbf{R}_i^t} \\ \epsilon a_t &= \frac{1}{\kappa n} \sum_{i=1}^n \frac{\partial \Psi^{t-1}}{\partial Z}(h_\epsilon(Z, \bar{\epsilon}_i), \epsilon \mathbf{R}_i^{t-1}) \Big|_{Z=\mathbf{x}_i^\top \boldsymbol{\beta}_0}, \end{aligned} \quad (\text{S40})$$

where $\bar{\epsilon}_i \sim U(0, 1)$ ($i = 1, \dots, n$) are i.i.d. draws independent of all other random variables.

Similarly, define $f_\epsilon^i(\mathbf{x}, t)$ as functions (S19) and (S20) with $h_\epsilon(Z, \bar{\epsilon})$ replaced by $h_\epsilon(Z, \bar{\epsilon})$ and denote by $\epsilon \mathbf{x}_i^t$ the orbit of (S21) whose definition is unchanged but whose entries now depend on ϵ through the replacement of $h(\cdot, \cdot)$ by $h_\epsilon(\cdot, \cdot)$ in (S11)-(S14). Finally, denote by $\epsilon \tilde{\mathbf{x}}_i^t$ the corresponding full AMP recursion as defined in (S32) with $\mathbf{x}_i^t$ replaced by $\epsilon \mathbf{x}_i^t$. Completely analogous to Lemmas S1 and S2, it can be shown that $\epsilon \mathbf{x}_i^t$ is a AMP orbit to the AMP instance $(\mathbf{A}, \mathcal{F}_\epsilon, \tilde{\mathbf{x}}^0)$, where the definition of $\mathbf{A}$ remains unchanged, $\mathcal{F}_\epsilon$ is the collection of functions f_ϵ^i and $\tilde{\mathbf{x}}^0 = \mathbf{0}_{q,N}$ as before. This is easily verified by noting that the proofs of these Lemmas do not depend on the form of the indicator function but are concerned with the general structure of the recursion.

It remains to construct a converging AMP instance following Definition S3 to motivate Theorem S1. Following the structure of Definition S3,

- i) Let q be some arbitrarily large integer, and $q' = 2$
- ii) Define $g_\epsilon : \Re^q \times \Re^q \times [q'] \times \mathbb{N} \rightarrow \Re^q$ as follows. Given $k \in \mathbb{N}_0$,

$$\begin{aligned} &g_\epsilon(\mathbf{x}, \mathbf{y}, 1, 2k+1) \\ &= \kappa^{-1} \sqrt{\frac{N}{n}} \left[0, \Psi^0(h_\epsilon(\mathbf{x}(1), \mathbf{y}(1)), \mathbf{x}(3)), 0, \Psi^1(h_\epsilon(\mathbf{x}(1), \mathbf{y}(1)), \mathbf{x}(5)), \dots, \Psi^{\frac{t-1}{2}}(h_\epsilon(\mathbf{x}(1), \mathbf{y}(1)), \mathbf{x}(t+2)), 0, 0, \dots \right]^\top, \end{aligned} \quad (\text{S41})$$

and

$$\begin{aligned}
& g_\epsilon(\mathbf{x}, \mathbf{y}, 2, 2k) \\
&= \sqrt{\frac{N}{n}} \left[\mathbf{y}(1), 0, \mathbf{y}(2) + \mu_0 \mathbf{y}(1), 0, \mathbf{x}(2) + \mu_1 \mathbf{y}(1), 0, \mathbf{x}(4) + \mu_2 \mathbf{y}(1), 0, \dots, \mathbf{x}(t) + \mu_{\frac{t}{2}} \mathbf{y}(1), 0, 0, \dots \right]^\top,
\end{aligned} \tag{S42}$$

while for all other configurations of t, a , $g_\epsilon(\mathbf{x}, \mathbf{y}, a, t) = \mathbf{0}_q$.

iii) Let $\pi_{\bar{\beta}}$ be the probability distribution such that the empirical distribution of β_0 converges weakly to $\pi_{\bar{\beta}}$, which is assumed to exist. Further, given initial conditions $\hat{\beta}^0, \mu_0$, let $\pi_{\bar{\nu}}$ be the probability distribution such that the empirical distribution of $\nu^0 = \hat{\beta}^0 - \mu_0 \beta_0$ converges weakly to, which is also assumed to exist, such as the joint distribution $\pi_{\bar{\beta}, \bar{\nu}}$ which is the limit of the joint distribution of β_0, ν^0 . Finally, let π_ε be the distribution of a $U(0, 1)$ random variable. Then let P_1 be the distribution of q independent random variables whose marginals are π_ε for the first coordinate and point masses at zero for all others. Further, let P_2 be the distribution of q random variables whose first two coordinates are distributed according to $\pi_{\bar{\beta}, \bar{\nu}}$ and are independent of all other coordinates, whose distributions are independent of another and point masses at zero.

iv) For $N = n + p$, partition $[N]$ into $C_1^N = [n]$ and $C_2^N = [N] \setminus [n]$

v) For $a = 1, 2$, let

$$\widehat{\Sigma}_a^0 = \text{plim}_{N \rightarrow \infty} \frac{1}{|C_a^N|} \sum_{i \in C_a^N} g_\epsilon(\mathbf{0}_q, \mathbf{y}_i, a, 0) g_\epsilon(\mathbf{0}_q, \mathbf{y}_i, a, 0)^\top,$$

which is assumed to exist. Here, $c_1 = \lim_{N \rightarrow \infty} n/(n+p) = 1/(1+\kappa)$ and $c_2 = \lim_{N \rightarrow \infty} p/(n+p) = \kappa/(1+\kappa)$ in concordance with Definition S3 1).

Condition iii) appears a bit abstract in its stated generality. For initial condition of relevance in this paper, namely the oracle initialiser $\mu_* \beta_0 + \sigma_* \xi$, where $\xi \sim N(\mathbf{I}_p)$ independent of everything else, it is readily seen that the condition holds whenever the empirical distribution of β_0 converges to some distribution $\pi_{\bar{\beta}}$.

To conclude the definition, it must now be shown that for all $r \in [q]$, the r th component of $g_\epsilon(\cdot, \cdot, a, t)$ is Lipschitz in its first two arguments and that points 1)-3) of Definition S3 hold. This is subject of the Lemma below.

Lemma S3. *The AMP instance defined above is a converging AMP instance according to Definition S3.*

Proof. It must be shown that 0) each component of $g_\epsilon(\cdot, \cdot, a, t)$ is Lipschitz in its first two arguments and that points 1)-3) of Definition S3 hold. Each point is shown separately.

0) *Lipschitz continuity of $g_\epsilon(\cdot, \cdot, a, t)$:* It is shown in Lemma S12 ii) that $\Psi^{\frac{t-1}{2}}(h_\epsilon(\mathbf{x}(1), \mathbf{y}(1)), \mathbf{x}(t+2))$ is Lipschitz in $[\mathbf{x}(1), \mathbf{x}(t+2), \mathbf{y}(1)]^\top$. Hence by (S41), for any $k \in \mathbb{N}_0$, $a = 1$, $r \in [q]$, the r th component of $g_\epsilon(\cdot, \cdot, 1, 2k+1)$ is Lipschitz. On the other hand, by (S42), Lipschitz continuity of $g_\epsilon(\cdot, \cdot, 2, 2k)$ is immediate since it is an affine transformation of its arguments. For all other combinations of t, a , $g_\epsilon(\cdot, \cdot, a, t) = \mathbf{0}_q$ and thus trivially Lipschitz.

1) *Convergence of $|C_a^N|/N$:* By definition of $N = n + p$, and $C_1^N = [n]$, $C_2^N = [N] \setminus [n]$. Hence, since by assumption, $\kappa(N) = p/n$ converges to some $\kappa \in (0, 1)$, 1) holds.

2) *Agreement of f_ϵ^i with the i th component of g_ϵ holds by construction of y in (S22), f_ϵ^i in (S19), (S20), g_ϵ in (S41) and (S42) along with the partition C_1^N, C_2^N . Convergence of the empirical distribution of $\{\mathbf{y}_i\}_{i \in C_N^a}$ to P_a follows by definition of P_a .*

3) *Initial condition of state evolution:* Follows by assumption v). $\square$

It thus follows from Lemma S2 applied to the smooth approximation ${}_\epsilon \tilde{\mathbf{x}}_i^t$ and Lemma S3 that $(\mathbf{A}, \mathcal{F}_\epsilon, \tilde{x}^0)$ along with iterates ${}_\epsilon \tilde{\mathbf{x}}_i^t$ is a converging AMP instance and Theorem S1 applies.

Theorem S2. Under assumptions (i)-(v), for any pseudo Lipschitz-function of order 2, and for all $t \in [q]$, $a = 1, 2$, almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{|C_a^N|} \sum_{i \in C_a^N} \psi(\epsilon \tilde{\mathbf{x}}_i^t, \mathbf{y}) = E[\psi(\epsilon Z_a^t, Y_a)] ,$$

where $\epsilon Z_a^t \sim N(0, \epsilon \Sigma^t)$, and $\epsilon \Sigma^t$ denotes the covariance matrices from state evolution (S7) and (S8) with function $g_\epsilon(\mathbf{x}, \mathbf{y}, a, t)$

Proof. By Lemma S3 and Theorem S1, for any pseudo Lipschitz function ψ of order 2,

$$\lim_{N \rightarrow \infty} \frac{1}{|C_a^N|} \sum_{i \in C_a^N} \psi(\epsilon \tilde{\mathbf{x}}_i^t, \mathbf{y}) = E[\psi(\epsilon Z_a^t, Y_a)] ,$$

where $\epsilon Z_a^t \sim N(0, \epsilon \Sigma^t)$ as defined in the statement of the Theorem. □

Corollary S2.1. Under assumptions (i)-(v), and initial conditions $\hat{\beta}^0$,

$$\begin{aligned} \epsilon \mu_0 &= \frac{1}{\gamma^2} \lim_{n \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} \\ \epsilon \sigma_0^2 &= \lim_{n \rightarrow \infty} \frac{\|\hat{\beta}^0 - \mu_0 \beta_0\|_2^2}{p} , \end{aligned}$$

for any pseudo Lipschitz-function of order 2, and for all $t \in [q]$, almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\epsilon \nu_j^t, \beta_{0,j}) = E[\psi(\epsilon \sigma_t^2 G, \bar{\beta})] , \quad \lim_{N \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \psi\left(\begin{bmatrix} \eta_{0,j} \\ \epsilon \mathbf{R}_i^t \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon}_j \\ 0 \end{bmatrix}\right) = E\left[\psi\left(\begin{bmatrix} Z_1^t \\ \epsilon Z_2^t \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon} \\ 0 \end{bmatrix}\right)\right] ,$$

where $G \sim N(0, 1)$, $Z_1^t \sim N(0, \gamma^2)$, $\epsilon Z_2^t = \epsilon \mu_t Z_1^t + \kappa^{1/2} \epsilon \sigma_t G$, $G \sim N(0, 1)$, $\bar{\varepsilon} \sim U(0, 1)$ and where $G \perp \bar{\beta}$, $G \perp Z_1^t$ and $(Z_1^t, \epsilon Z_2^t) \perp \bar{\varepsilon}$.

Proof. The proof consists of two parts, which shall be shown in turn: i) For fixed $\epsilon > 0$, extract the iterates $\epsilon \nu^t, \epsilon \mathbf{R}^t$ of (S38) and (S39) from $\epsilon \tilde{\mathbf{x}}_i^t$ and obtain their asymptotic variances according to Theorem S2, and ii) relate these variances to the state evolution of (S35)-(S37).

The first step is similar to the derivations in the proof of Lemma 6 in the supplementary material of Sur and Candès (2019). For $t = 1$, it is easily seen from the AMP instance defined in S4.4.4 and the state evolutions (S7) and (S8) that

$$\begin{bmatrix} \epsilon \Sigma_{1,1}^1 & \epsilon \Sigma_{1,3}^1 \\ \epsilon \Sigma_{3,1}^1 & \epsilon \Sigma_{3,3}^1 \end{bmatrix} = \begin{bmatrix} \lim_{N \rightarrow \infty} \frac{\|\beta_0\|_2^2}{n} & \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} \\ \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \lim_{N \rightarrow \infty} \frac{\|\hat{\beta}^0\|_2^2}{n} \end{bmatrix} , \quad (\text{S43})$$

where $\epsilon \Sigma_{i,j}^1$ denotes the j th entry in the i th row of $\epsilon \Sigma^1$. All other entries of $\epsilon \Sigma^1$ are zero. Now from (S21) it follows that for $i \in [n]$, the first and third entries of $\epsilon \tilde{\mathbf{x}}_i^1$ are $\eta_{0,i}$, $\epsilon \mathbf{R}_i^0 = \mathbf{x}_i^\top \hat{\beta}^0$. Hence, by Theorem S2,

$$\lim_{N \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \psi\left(\begin{bmatrix} \eta_{0,j} \\ \epsilon \mathbf{R}_i^0 \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon}_j \\ 0 \end{bmatrix}\right) = E\left[\psi\left(\mathbf{Z}^1, \begin{bmatrix} \bar{\varepsilon} \\ 0 \end{bmatrix}\right)\right] ,$$

where $\mathbf{Z}^1 = [\mathbf{Z}_1^1, \mathbf{Z}_2^1]^\top \perp \bar{\varepsilon}$ is bivariate normal with mean zero and covariance (S43).

Next, the only nonzero entry of $\epsilon \Sigma^2$ is readily computed as $\kappa^{-2} \mathbb{E} \left[\left\{ \Psi^0(h_\epsilon(\mathbf{Z}_1^1, \bar{\varepsilon}), \mathbf{Z}_2^1) \right\}^2 \right]$. By construction of $\epsilon \tilde{\mathbf{x}}_i^t$ it follows from (S21) and Lemma S3 and Theorem S1 that almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\epsilon \nu_j^1, \beta_{0,j}) = \mathbb{E} [\psi(\epsilon \tau_1^2 G, \bar{\beta})],$$

where $\epsilon \tau_1^2 = \kappa^{-2} \mathbb{E} \left[\left\{ \Psi^0(h_\epsilon(\mathbf{Z}_1^1, \bar{\varepsilon}), \mathbf{Z}_2^1) \right\}^2 \right]$ and $G \sim \mathcal{N}(0, 1) \perp \bar{\beta}$. Next consider $\epsilon \Sigma^3$. By the nested nature of the iterates $\epsilon \mathbf{x}_i^t$, it follows that the first 3×3 principal submatrix is exactly the one given in (S43). Further, it is readily verified that the remaining nonzero entries are given by

$$\epsilon \Sigma_{1,5}^3 = \epsilon \mu_1 \gamma^2, \quad \epsilon \Sigma_{3,5}^3 = \epsilon \mu_1 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n}, \quad \epsilon \Sigma_{5,5}^3 = \epsilon \tau_1^2 \kappa + \epsilon \mu_1^2 \gamma^2,$$

where $\epsilon \mu_1$ comes from recursion (S36) with initial condition μ_0 . Again, Lemma S2 and Theorem S1 yield almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \psi \left(\begin{bmatrix} \eta_{0,j} \\ \epsilon \mathbf{R}_j^0 \\ \epsilon \mathbf{R}_j^1 \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon}_j \\ 0 \\ 0 \end{bmatrix} \right) = \mathbb{E} \left[\psi \left(\mathbf{Z}^3, \begin{bmatrix} \bar{\varepsilon} \\ 0 \\ 0 \end{bmatrix} \right) \right],$$

where $\mathbf{Z}^3 = [\mathbf{Z}_1^3, \mathbf{Z}_2^3, \mathbf{Z}_3^3]^\top \perp \bar{\varepsilon}$ is trivariate normal with mean zero and covariance matrix

$$\begin{bmatrix} \epsilon \Sigma_{1,1}^3 & \epsilon \Sigma_{1,3}^3 & \epsilon \Sigma_{1,5}^3 \\ \epsilon \Sigma_{3,1}^3 & \epsilon \Sigma_{3,3}^3 & \epsilon \Sigma_{3,5}^3 \\ \epsilon \Sigma_{5,1}^3 & \epsilon \Sigma_{5,3}^3 & \epsilon \Sigma_{5,5}^3 \end{bmatrix} = \begin{bmatrix} \lim_{N \rightarrow \infty} \frac{\|\beta_0\|_2^2}{n} & \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \mu_1 \gamma^2 \\ \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \lim_{N \rightarrow \infty} \frac{\|\hat{\beta}^0\|_2^2}{n} & \epsilon \mu_1 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} \\ \epsilon \mu_1 \gamma^2 & \epsilon \mu_1 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \tau_1^2 \kappa + \epsilon \mu_1^2 \gamma^2 \end{bmatrix}.$$

Calculating the nonzero entries of $\epsilon \Sigma^4$ yields that

$$\begin{bmatrix} \epsilon \Sigma_{2,2}^4 & \epsilon \Sigma_{2,4}^4 \\ \epsilon \Sigma_{4,2}^4 & \epsilon \Sigma_{4,4}^4 \end{bmatrix} = \begin{bmatrix} \epsilon \tau_1^2 & \epsilon \rho_{1,2} \\ \epsilon \rho_{1,2} & \epsilon \tau_2^2 \end{bmatrix}, \quad (\text{S44})$$

where

$$\begin{aligned} \epsilon \tau_2^2 &= \kappa^{-2} \mathbb{E} [\left\{ \Psi^1(h_\epsilon(\mathbf{Z}_1^3, \bar{\varepsilon}), \mathbf{Z}_2^3) \right\}^2] \\ \epsilon \rho_{1,2} &= \kappa^{-2} \mathbb{E} [\Psi^0(h_\epsilon(\mathbf{Z}_1^3, \bar{\varepsilon}), \mathbf{Z}_2^3) \Psi^1(h_\epsilon(\mathbf{Z}_1^3, \bar{\varepsilon}), \mathbf{Z}_3^3)]. \end{aligned}$$

Consequently, almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi \left(\begin{bmatrix} \epsilon \nu_j^1 \\ \epsilon \nu_j^2 \end{bmatrix}, \begin{bmatrix} \beta_{0,j} \\ 0 \end{bmatrix} \right) = \mathbb{E} \left[\psi \left(\mathbf{Z}^4, \begin{bmatrix} \bar{\beta} \\ 0 \end{bmatrix} \right) \right],$$

where $\mathbf{Z}^4 = [\mathbf{Z}_1^4, \mathbf{Z}_2^4]^\top$ is bivariate normal with mean zero and covariance matrix (S44). Iterating

further,

$$\begin{bmatrix} \epsilon \Sigma_{1,1}^5 & \epsilon \Sigma_{1,3}^5 & \epsilon \Sigma_{1,5}^5 & \epsilon \Sigma_{1,7}^5 \\ \epsilon \Sigma_{3,1}^5 & \epsilon \Sigma_{3,3}^5 & \epsilon \Sigma_{3,5}^5 & \epsilon \Sigma_{3,7}^5 \\ \epsilon \Sigma_{5,1}^5 & \epsilon \Sigma_{5,3}^5 & \epsilon \Sigma_{5,5}^5 & \epsilon \Sigma_{5,7}^5 \\ \epsilon \Sigma_{7,1}^5 & \epsilon \Sigma_{7,3}^5 & \epsilon \Sigma_{7,5}^5 & \epsilon \Sigma_{7,7}^5 \end{bmatrix} = \begin{bmatrix} \lim_{N \rightarrow \infty} \frac{\|\beta_0\|_2^2}{n} & \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \mu_1 \gamma^2 & \epsilon \mu_2 \gamma^2 \\ \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \lim_{N \rightarrow \infty} \frac{\|\hat{\beta}^0\|_2^2}{n} & \epsilon \mu_1 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \mu_2 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} \\ \epsilon \mu_1 \gamma^2 & \epsilon \mu_1 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \tau_1^2 \kappa + \epsilon \mu_1^2 \gamma^2 & \epsilon \rho_{1,2} \kappa + \epsilon \mu_1 \epsilon \mu_2 \gamma^2 \\ \epsilon \mu_2 \gamma^2 & \epsilon \mu_2 \lim_{N \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} & \epsilon \rho_{1,2} \kappa + \epsilon \mu_1 \epsilon \mu_2 \gamma^2 & \epsilon \tau_2^2 \kappa + \epsilon \mu_2^2 \gamma^2 \end{bmatrix}, \quad (\text{S45})$$

so that almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \psi \left(\begin{bmatrix} \eta_{0,j} \\ \epsilon \mathbf{R}_j^0 \\ \epsilon \mathbf{R}_j^1 \\ \epsilon \mathbf{R}_j^2 \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon}_i \\ 0 \\ 0 \\ 0 \end{bmatrix} \right) = \mathbb{E} \left[\psi \left(\mathbf{Z}^5, \begin{bmatrix} \bar{\varepsilon} \\ 0 \\ 0 \\ 0 \end{bmatrix} \right) \right],$$

where once again $\mathbf{Z}^5 = [\mathbf{Z}_1^5, \dots, \mathbf{Z}_4^5]^\top \perp \bar{\varepsilon}$ is multivariate normal with mean zero and covariance matrix (S45). Hence continuing,

$$\begin{bmatrix} \epsilon \Sigma_{2,2}^6 & \epsilon \Sigma_{2,4}^6 & \epsilon \Sigma_{2,6}^6 \\ \epsilon \Sigma_{4,2}^6 & \epsilon \Sigma_{4,4}^6 & \epsilon \Sigma_{4,6}^6 \\ \epsilon \Sigma_{6,2}^6 & \epsilon \Sigma_{6,4}^6 & \epsilon \Sigma_{6,6}^6 \end{bmatrix} = \begin{bmatrix} \epsilon \tau_1^2 & \epsilon \rho_{1,2} & \epsilon \rho_{1,3} \\ \epsilon \rho_{1,2} & \epsilon \tau_2^2 & \epsilon \rho_{2,3} \\ \epsilon \rho_{1,3} & \epsilon \rho_{2,3} & \epsilon \tau_3^2 \end{bmatrix}, \quad (\text{S46})$$

where

$$\begin{aligned} \epsilon \tau_3^2 &= \kappa^{-2} \mathbb{E} [\{\Psi^2(h_\epsilon(\mathbf{Z}_1^5, \bar{\varepsilon}), \mathbf{Z}_2^5)\}^2] \\ \epsilon \rho_{l,m} &= \kappa^{-2} \mathbb{E} [\Psi^{l-1}(h_\epsilon(\mathbf{Z}_1^5, \bar{\varepsilon}), \mathbf{Z}_{l+1}^5) \Psi^{m-1}(h_\epsilon(\mathbf{Z}_1^5, \bar{\varepsilon}), \mathbf{Z}_{m+1}^5)]. \end{aligned}$$

Therefore almost surely,

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi \left(\begin{bmatrix} \nu_j^1 \\ \nu_j^2 \\ \nu_j^3 \end{bmatrix}, \begin{bmatrix} \beta_{0,j} \\ 0 \\ 0 \end{bmatrix} \right) = \mathbb{E} \left[\psi \left(\mathbf{Z}^6, \begin{bmatrix} \bar{\beta} \\ 0 \\ 0 \end{bmatrix} \right) \right],$$

with $\mathbf{Z}^6 = [\mathbf{Z}_1^6, \mathbf{Z}_2^6, \mathbf{Z}_3^6]^\top$ trivariate normal with mean zero and covariance (S46). Continuing in this vein, one gets that for all $t \in [q]$, by Lemmas S2 and Theorem S1, almost surely

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\nu_j^t, \beta_{0,j}) &= \mathbb{E} [\psi(\tau_t G, \bar{\beta})] \\ \lim_{N \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \psi \left(\begin{bmatrix} \eta_{0,j} \\ \epsilon \mathbf{R}_j^t \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon}_i \\ 0 \end{bmatrix} \right) &= \mathbb{E} \left[\psi \left(\begin{bmatrix} \mathbf{Z}_{t+1}^{2t+1} \\ \mathbf{Z}_{t+2}^{2t+1} \end{bmatrix}, \begin{bmatrix} \bar{\varepsilon} \\ 0 \end{bmatrix} \right) \right], \end{aligned} \quad (\text{S47})$$

where $G \sim \mathcal{N}(0, 1) \perp \bar{\beta}$, and $(\mathbf{Z}_t^{2t+1}, \mathbf{Z}_{t+2}^{2t+1}) \perp \bar{\varepsilon}$ are bivariate normal with mean zero and covariance

$$\begin{bmatrix} \epsilon \Sigma_{1,1}^{t+2} & \epsilon \Sigma_{1,t+2}^{t+2} \\ \epsilon \Sigma_{t+2,1}^{t+2} & \epsilon \Sigma_{t+2,t+2}^{t+2} \end{bmatrix} = \begin{bmatrix} \lim_{N \rightarrow \infty} \frac{\|\beta_0\|_2^2}{n} & \epsilon \mu_t \gamma^2 \\ \epsilon \mu_t \gamma^2 & \epsilon \tau_t^2 \kappa + \epsilon \mu_t^2 \gamma^2 \end{bmatrix}, \quad (\text{S48})$$

and

$$\epsilon \tau_t = \kappa^{-2} \mathbb{E} [\{\Psi^{t-1}(h_\epsilon(\mathbf{Z}_1^{2t-1}, \bar{\varepsilon}), \mathbf{Z}_{t+1}^{2t-1})\}^2], \quad (\text{S49})$$

and where $\mathbf{Z}_1^{2t-1}, \mathbf{Z}_{t+1}^{2t-1} \perp \bar{\varepsilon}$ are bivariate normal with the covariance structure of (S48) with $\epsilon \mu_{t-1}, \epsilon \tau_{t-1}$ in place of $\epsilon \mu_t, \epsilon \tau_t$ respectively.

Next, note that with initialisations ${}_{\epsilon}\mu_0 = \gamma^{-2} \lim_{n \rightarrow \infty} \langle \hat{\beta}^0, \beta_0 \rangle / n$ and ${}_{\epsilon}\sigma_0^2 = \lim_{n \rightarrow \infty} \|\hat{\beta}^0 - \mu_0 \beta_0\|_2^2 / p$, the variances of (S43) correspond to

$$\begin{bmatrix} \gamma^2 & {}_{\epsilon}\mu_0 \gamma^2 \\ {}_{\epsilon}\mu_0 \gamma^2 & {}_{\epsilon}\sigma_0^2 \kappa + {}_{\epsilon}\mu_0^2 \gamma^2 {}_{\epsilon}\mu_0 \gamma^2 \end{bmatrix},$$

Hence by the definition of ${}_{\epsilon}\sigma_t$ in (S37), ${}_{\epsilon}\tau_1 = {}_{\epsilon}\sigma_1$. By induction it follows that ${}_{\epsilon}\tau_t = {}_{\epsilon}\sigma_t$ for all t under consideration. This concludes the proof. $\square$

S4.4.5 Asymptotic equivalence of target and centred AMP

Next it is possible to establish the asymptotic equivalence of the centred recursion (S11)-(S14) with the initial target recursion (S9)-(S10) for appropriately centred iterates. The proof closely follows along the proof of Lemma 4 of the supplementary material of Sur and Candès (2019) which in turn is an adaptation of the proof of Lemma 6.7 of Donoho and Montanari (2016).

Lemma S4. *In the setting of Corollary S2.1, it holds that for initial conditions ${}_{\epsilon}\nu^0 = \hat{\beta}^0 - \mu_0 \beta_0$, ${}_{\epsilon}\mathbf{R}^0 = {}_{\epsilon}\mathbf{S}^0$, $t \in [q]$,*

$$\lim_{N \rightarrow \infty} \frac{1}{p} \left\| {}_{\epsilon}\nu^t - {}_{\epsilon}\hat{\beta}^t - {}_{\epsilon}\mu_t \beta_0 \right\|_2^2 = 0, \quad \lim_{N \rightarrow \infty} \frac{1}{n} \left\| {}_{\epsilon}\mathbf{R}^t - {}_{\epsilon}\mathbf{S}^t \right\|_2^2 = 0,$$

almost surely.

Proof. To facilitate notation, let ${}_{\epsilon}\theta^t = {}_{\epsilon}\hat{\beta}^t - {}_{\epsilon}\mu_t \beta_0$. Then by construction of recursions (S9)-(S10) and (S11)-(S12), and the triangle inequality it holds that

$$\begin{aligned} \left\| {}_{\epsilon}\mathbf{R}^t - {}_{\epsilon}\mathbf{S}^t \right\|_2 &\leq \left\| \mathbf{X} \right\|_2 \left\| {}_{\epsilon}\nu^t - {}_{\epsilon}\theta^t \right\|_2 + \left\| \Psi^{t-1}({}_{\epsilon}\mathbf{y}^*, {}_{\epsilon}\mathbf{R}^{t-1}) - \Psi^{t-1}({}_{\epsilon}\mathbf{y}^*, {}_{\epsilon}\mathbf{S}^{t-1}) \right\| \\ &\leq \left\| \mathbf{X} \right\|_2 \left\| {}_{\epsilon}\nu^t - {}_{\epsilon}\theta^t \right\|_2 + \left\| {}_{\epsilon}\mathbf{R}^{t-1} - {}_{\epsilon}\mathbf{S}^{t-1} \right\|_2, \end{aligned} \quad (\text{S50})$$

where the second inequality follows by Lemma S12 iii) and where ${}_{\epsilon}\mathbf{y}^* = \alpha h_{\epsilon}(\eta_0, \bar{\epsilon}) + (1 - \alpha)/2$.

Similarly,

$$\left\| {}_{\epsilon}\nu^t - {}_{\epsilon}\theta^t \right\|_2 \leq \left\| {}_{\epsilon}\nu^{t-1} - {}_{\epsilon}\theta^{t-1} \right\|_2 + |{}_{\epsilon}q_{t-1} - 1| \left\| {}_{\epsilon}\nu^{t-1} + {}_{\epsilon}\mu_{t-1} \beta_0 \right\|_2 + \left\| {}_{\epsilon}\mu_t - {}_{\epsilon}a_t \right\| \left\| \beta_0 \right\| + \frac{1}{\kappa} \left\| \mathbf{X} \right\|_2 \left\| {}_{\epsilon}\mathbf{R}^{t-1} - {}_{\epsilon}\mathbf{S}^{t-1} \right\|_2.$$

Now since ${}_{\epsilon}\nu^0 = {}_{\epsilon}\theta^0$, an induction argument on the number of iterations t gives that there exists a constant C , which may depend on κ , such that for each $t \in [q]$,

$$\left\| {}_{\epsilon}\nu^t - {}_{\epsilon}\theta^t \right\|_2 \leq (C \left\| \mathbf{X} \right\|_2)^{2t} \left(\sum_{l=1}^{t-1} |{}_{\epsilon}q_l - 1| \left\| {}_{\epsilon}\nu^l + {}_{\epsilon}\mu_l \beta_0 \right\|_2 + \sum_{l=0}^{t-1} |{}_{\epsilon}\mu_l - a_l| \left\| \beta_0 \right\|_2 \right). \quad (\text{S51})$$

Recalling that ${}_{\epsilon}q_t = -\frac{1}{\kappa n} \sum_{j=1}^n \frac{-b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_{\epsilon}(\eta_{0,j}, \bar{\epsilon}_j), \mathbf{R}_j^t))}{1 + b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_{\epsilon}(\eta_{0,j}, \bar{\epsilon}_j), \mathbf{R}_j^t))}$ (cf. (S13) and (S97)), it follows from Lemma S12 iv) and Corollary S2.1 i), that almost surely

$$\lim_{N \rightarrow \infty} {}_{\epsilon}q_t = \frac{1}{\kappa} \mathbb{E} \left[\frac{b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_{\epsilon}(Z_1^t, \bar{\epsilon}), {}_{\epsilon}Z_2^t))}{1 + b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_{\epsilon}(Z_1^t, \bar{\epsilon}), {}_{\epsilon}Z_2^t))} \right] \quad (\text{S52})$$

for $Z_1^t \sim \mathcal{N}(0, \gamma^2)$, ${}_{\epsilon}Z_2^t = {}_{\epsilon}\mu_t Z_1^t + \kappa^{1/2} {}_{\epsilon}\sigma_t G$, $G \sim \mathcal{N}(0, 1)$, $\bar{\epsilon} \sim U(0, 1)$ and where $G \perp \bar{\beta}$, $G \perp Z_1^t$ and

$(Z_1^t, \epsilon Z_2^t) \perp \bar{\epsilon}$. Now the right hand side of (S52) can be rearranged as

$$\frac{1}{\kappa} \mathbb{E} \left[\frac{b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t))}{1 + b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t))} \right] = \frac{1}{\kappa} \mathbb{E} \left[1 - \frac{1}{b_t \zeta''(\text{prox}_{b_t \zeta}(b_t h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t))} \right] = 1,$$

according to state evolution equation (S35).

Again, by Corollary S2.1 i) and $\psi(u, v) = u^2$, it follows that almost surely,

$$\lim_{N \rightarrow \infty} \sum_{j=1} (\epsilon \nu_j^t)^2 = \mathbb{E}[(\epsilon \sigma_t Z)^2] < \infty,$$

for $Z \sim \mathcal{N}(0, 1)$. Further recall that by assumption $\lim_{n \rightarrow \infty} \|\beta_0\|_2^2/n = \gamma^2$ and that by Lemma S13, $\lim_{N \rightarrow \infty} \|\mathbf{X}\|_2 < \infty$ almost surely. Hence it holds almost surely that

$$\lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} (C \|\mathbf{X}\|_2)^{2t} \sum_{l=0}^{t-1} |\epsilon q_l - 1| \|\epsilon \nu_j^l + \epsilon \mu_l \beta_0\|_2 = 0. \quad (\text{S53})$$

Similarly, recalling the definition of ϵa_t in (S40) and invoking Lemma S12 iv) and Corollary S2.1 equation (S47), almost surely

$$\lim_{N \rightarrow \infty} \epsilon a_t = \frac{1}{\kappa} \mathbb{E} \left[\frac{\partial}{\partial Z} \Psi^{t-1}(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_2^t) \Big|_{Z=Z_1^t} \right], \quad (\text{S54})$$

for $Z_1^t \sim \mathcal{N}(0, \gamma^2)$, $\epsilon Z_2^t = \epsilon \mu_t Z_1^t + \kappa^{1/2} \epsilon \sigma_t G$, $G \sim \mathcal{N}(0, 1)$, $\bar{\epsilon} \sim U(0, 1)$ and where $G \perp \bar{\beta}$, $G \perp Z_1^t$ and $(Z_1^t, \epsilon Z_2^t) \perp \bar{\epsilon}$.

To compute the expectation in (S54), Stein's Lemma (Stein, 1981, Lemma 1) shall prove useful. It states that for a normal random variable $z \sim \mathcal{N}(\mu, \sigma^2)$, and a differentiable function $h(z)$, $\mathbb{E}[h(z)(z - \mu)] = \sigma^2 \mathbb{E}[h'(z)]$, where h is derivative of h with respect to z .

To use the Lemma, first note that ϵZ_2^t is a function of Z_1^t . Hence write $\epsilon Z_2^t = t_\epsilon(Z_1^t, G_t) = \epsilon \mu_t Z_1^t + \kappa^{1/2} \epsilon \sigma_t G_t$. Then conditioning on both $\bar{\epsilon}, G_t$, one gets that

$$\mathbb{E} \left[\frac{\partial}{\partial Z} \Psi^{t-1}(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_2^t) \Big|_{Z=Z_1^t} \right] = \mathbb{E}_{G_t, \bar{\epsilon}} \left\{ \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial Z} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=t_\epsilon(Z_1^t, G_t)}} \Big| G_t, \bar{\epsilon} \right] \right\}. \quad (\text{S55})$$

Now since Z_1^t is independent of $G_t, \bar{\epsilon}$, the conditional distribution of Z_1^t given $G_t, \bar{\epsilon}$ is its unconditional distribution, i.e. $Z_1^t | G_t, \bar{\epsilon} \sim \mathcal{N}(0, \gamma^2)$ and thus by Stein's Lemma,

$$\gamma^2 \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial Z_1^t} \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), t_\epsilon(Z_1^t, G_t)) \Big| G_t, \bar{\epsilon} \right] = \mathbb{E}_{Z_1^t} \left[Z_1^t \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), t_\epsilon(Z_1^t, G_t)) \Big| G_t, \bar{\epsilon} \right]. \quad (\text{S56})$$

Now an application of the chain rule to the left hand side of (S56) yields

$$\begin{aligned} \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial Z_1^t} \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), t_\epsilon(Z_1^t, G_t)) \Big| G_t, \bar{\epsilon} \right] &= \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial Z} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=t_\epsilon(Z_1^t, G_t)}} \Big| G_t, \bar{\epsilon} \right] \\ &\quad + \epsilon \mu_{t-1} \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial s} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=t_\epsilon(Z_1^t, G_t)}} \Big| G_t, \bar{\epsilon} \right] \end{aligned} \quad (\text{S57})$$

Combining (S56) and (S57) gives

$$\begin{aligned} \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial Z} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=t_\epsilon(Z_1^t, G_t)}} \Big| G_t, \bar{\epsilon} \right] &= \frac{1}{\gamma^2} \mathbb{E}_{Z_1^t} \left[Z_1^t \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), t_\epsilon(Z_1^t, G_t)) \Big| G_t, \bar{\epsilon} \right] \\ &\quad - \epsilon \mu_{t-1} \mathbb{E}_{Z_1^t} \left[\frac{\partial}{\partial s} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=t_\epsilon(Z_1^t, G_t)}} \Big| G_t, \bar{\epsilon} \right] \end{aligned}$$

Substituting back onto (S55) in conjunction with (S54) gives that almost surely,

$$\lim_{N \rightarrow \infty} \epsilon a_t = \frac{1}{\kappa \gamma^2} \mathbb{E} [Z_1^t \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t)] - \frac{\epsilon \mu_{t-1}}{\kappa} \mathbb{E} \left[\frac{\partial}{\partial s} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=\epsilon Z_2^t}} \right].$$

Now using (S35) and (S97), it follows that

$$\begin{aligned} -\frac{1}{\kappa} \mathbb{E} \left[\frac{\partial}{\partial s} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=\epsilon Z_2^t}} \right] &= \frac{1}{\kappa} \mathbb{E} \left[\frac{b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + \epsilon Z_2^t))}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + \epsilon Z_2^t))} \right] \\ &= \frac{1}{\kappa} \mathbb{E} \left[1 - \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + \epsilon Z_2^t))} \right] \\ &= 1. \end{aligned}$$

Hence, the state evolution of $\epsilon \mu_t$ as specified in (S36) gives that almost surely

$$\begin{aligned} \lim_{N \rightarrow \infty} \epsilon a_t &= \frac{1}{\kappa \gamma^2} \mathbb{E} [Z_1^t \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t)] - \frac{\epsilon \mu_{t-1}}{\kappa} \mathbb{E} \left[\frac{\partial}{\partial s} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) \Big|_{\substack{Z=Z_1^t, \\ s=\epsilon Z_2^t}} \right] \\ &= \frac{1}{\kappa \gamma^2} \mathbb{E} [Z_1^t \Psi(h_\epsilon(Z_1^t, \bar{\epsilon}), \epsilon Z_2^t)] + \epsilon \mu_{t-1} \\ &= \epsilon \mu_t. \end{aligned} \tag{S58}$$

Hence, for any $l \in \mathbb{N}_0$, $\lim_{N \rightarrow \infty} |\epsilon a_t - \epsilon \mu_t| = 0$ almost surely. By almost sure boundedness of $\mathbf{X}$ and since $\|\beta_0\|_2^2/n \rightarrow \gamma^2$ as $n \rightarrow \infty$, it follows that almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \sum_{l=0}^{t-1} |\epsilon a_t - \epsilon \mu_t| \|\beta_0\|_2 = 0. \tag{S59}$$

In conjunction with (S51) and (S53) and (S59), this yields that almost surely,

$$\|\epsilon \nu^t - \epsilon \theta^t\|_2 = 0. \tag{S60}$$

Almost sure convergence of $\|\epsilon \mathbf{R}^t - \epsilon \mathbf{S}^t\|_2$ then follows by an induction argument on t , (S50), (S60) and since $\epsilon \mathbf{S}^0 = \epsilon \mathbf{R}^0$. This concludes the proof. $\square$

S4.4.6 Asymptotic equivalence of AMP and MDYPL

Theorem S3. *In the setting of Lemma S4, assume that there exists a stationary solution $(\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*)$*

to the state evolution (S36)-(S37). Then,

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \left\| \epsilon \hat{\beta}^t - \hat{\beta}^{\text{DY}} \right\|_2^2 = 0$$

almost surely.

Proof. First, a second order Taylor approximation of $\ell(\hat{\beta}^{\text{DY}})$ around $\epsilon \hat{\beta}^t$ gives

$$\ell(\hat{\beta}^{\text{DY}}) = \ell(\epsilon \hat{\beta}^t) + \nabla \ell(\epsilon \hat{\beta}^t)^\top (\hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t) + \frac{1}{2} \mathbf{r}, \quad (\text{S61})$$

where the remainder term $\mathbf{r}$ is, equal to

$$\mathbf{r} = (\hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t)^\top \nabla \nabla^\top \ell(\hat{\beta}^{\text{DY}} - c(\epsilon \hat{\beta}^t - \hat{\beta}^{\text{DY}}))(\hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t),$$

for some constant $c \in [0, 1]$. Now by Sur 2019, Lemma 7, with probability at least $1 - c_1 \exp\{-c_2 n\}$ for absolute constants $c_1, c_2 > 0$, it holds that for all $\beta \in \mathbb{R}^p$

$$\nabla \nabla^\top \ell(\beta) \succ \omega \left(\frac{\|\beta\|_2}{\sqrt{n}} \right) \mathbf{I}_p,$$

where $\mathbf{A} \succ \mathbf{B}$ means that $\mathbf{A} - \mathbf{B}$ is positive definite. Here, $\omega(\cdot) \in (0, 1)$ is a nonincreasing function independent of n . Clearly,

$$\left\| \hat{\beta}^{\text{DY}} - c(\hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t) \right\|_2 \leq \max \left\{ \left\| \hat{\beta}^{\text{DY}} \right\|_2, \left\| \epsilon \hat{\beta}^t \right\|_2 \right\},$$

so that with probability at least $1 - c_1 \exp\{-c_2 n\}$

$$\begin{aligned} \mathbf{r} &\geq \left\| \hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2^2 \omega \left(\frac{\left\| \hat{\beta}^{\text{DY}} - c(\hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t) \right\|_2}{\sqrt{n}} \right) \\ &\geq \left\| \hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2^2 \omega \left(\max \left\{ \frac{\left\| \hat{\beta}^{\text{DY}} \right\|_2}{\sqrt{n}}, \frac{\left\| \epsilon \hat{\beta}^t \right\|_2}{\sqrt{n}} \right\} \right). \end{aligned} \quad (\text{S62})$$

Using (S62) and the Cauchy-Schwarz inequality, (S61) can be rearranged as

$$\left\| \hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2 \leq \frac{2}{\omega \left(\max \left\{ \frac{\left\| \hat{\beta}^{\text{DY}} \right\|_2}{\sqrt{n}}, \frac{\left\| \epsilon \hat{\beta}^t \right\|_2}{\sqrt{n}} \right\} \right)} \left\| \nabla \ell(\epsilon \hat{\beta}^t) \right\|_2,$$

so that it remains to control $\left\| \hat{\beta}^{\text{DY}} \right\|_2, \left\| \epsilon \hat{\beta}^t \right\|_2$ and $\left\| \nabla \ell(\epsilon \hat{\beta}^t) \right\|_2$. Towards bounding $\left\| \nabla \ell(\epsilon \hat{\beta}^t) \right\|_2$, recall that by (S33) and (S34), any by the definition of the function $\Psi^t(\cdot, \cdot)$, and the identity $z - \zeta'(\text{prox}_{b\zeta}(z)) = \text{prox}_{b\zeta}(z)$,

$$-\text{prox}_{\epsilon b_t \zeta}(\epsilon b_t \epsilon \mathbf{y}^* + \epsilon \mathbf{S}^{t-1}) = \epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} + \mathbf{X}_\epsilon \hat{\beta}^t,$$

and thus

$$\epsilon \hat{\beta}^t - \epsilon \hat{\beta}^{t-1} = \frac{\epsilon b_t}{\kappa} \mathbf{X}^\top \left\{ \epsilon \mathbf{y}^* - \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} + \mathbf{X}_\epsilon \hat{\beta}^t \right) \right\}.$$

Therefore,

$$\begin{aligned}\nabla \ell(\epsilon \hat{\beta}^t) &= \mathbf{X}^\top \left\{ \mathbf{y} - \zeta'(\mathbf{X}_\epsilon \hat{\beta}^t) \right\} \\ &= \mathbf{X}^\top \left\{ \mathbf{y} - \epsilon \mathbf{y} \right\} \\ &\quad + \mathbf{X}^\top \left\{ \epsilon \mathbf{y} - \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} - \mathbf{X}_\epsilon \hat{\beta}^t \right) \right\} \\ &\quad + \mathbf{X}^\top \left\{ \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} - \mathbf{X}_\epsilon \hat{\beta}^t \right) - \zeta' \left(\mathbf{X}_\epsilon \hat{\beta}^t \right) \right\}.\end{aligned}$$

almost surely. Further, using that $\zeta'(x)$ is Lipschitz with Lipschitz constant $1/4$,

$$\begin{aligned}\left\| \mathbf{X}^\top \left\{ \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} - \mathbf{X}_\epsilon \hat{\beta}^t \right) - \zeta' \left(\mathbf{X}_\epsilon \hat{\beta}^t \right) \right\} \right\|_2 &\leq \|\mathbf{X}\|_2 \left\| \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} - \mathbf{X}_\epsilon \hat{\beta}^t \right) - \zeta' \left(\mathbf{X}_\epsilon \hat{\beta}^t \right) \right\|_2 \\ &\leq \frac{1}{4} \|\mathbf{X}\|_2 \left\| \epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} + \mathbf{X}_\epsilon \hat{\beta}^t - \mathbf{X}_\epsilon \hat{\beta}^t \right\|_2 \\ &= \frac{1}{4} \|\mathbf{X}\|_2 \left\| \epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} \right\|_2\end{aligned}$$

Using (S33), it immediately follows that

$$\left\| \mathbf{X}^\top \left\{ \epsilon \mathbf{y} - \zeta' \left(\epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} - \mathbf{X}_\epsilon \hat{\beta}^t \right) \right\} \right\|_2 = \frac{\kappa}{b^t} \left\| \epsilon \hat{\beta}^t - \epsilon \hat{\beta}^{t-1} \right\|_2.$$

Therefore, with probability at least $1 - c_1 \exp\{-c_2 n\}$,

$$\left\| \hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2 \leq \frac{2}{\omega \left(\max \left\{ \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}}, \frac{\|\epsilon \hat{\beta}^t\|_2}{\sqrt{n}} \right\} \right)} \left\{ \frac{\kappa}{b^t} \left\| \epsilon \hat{\beta}^t - \epsilon \hat{\beta}^{t-1} \right\|_2 + \frac{1}{4} \|\mathbf{X}\|_2 \left\| \epsilon \mathbf{S}^t - \epsilon \mathbf{S}^{t-1} \right\|_2 + \|\mathbf{X}\|_2 \left\| \mathbf{y} - \epsilon \mathbf{y} \right\|_2 \right\}.$$

To finish the proof initialise the AMP recursion at the stationary solution, i.e. $(\epsilon \mu_0, \epsilon b_0, \epsilon \sigma_0) = (\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*)$. Then it follows immediately from (S47) with $\psi(x, y) = x^2$, in conjunction with Lemma S4 and the stationarity condition that almost surely

$$\lim_{N \rightarrow \infty} \frac{\left\| \epsilon \hat{\beta}^t \right\|_2}{\sqrt{p}} \leq \lim_{N \rightarrow \infty} \frac{\left\| \epsilon \hat{\beta}^t - \epsilon \mu_* \beta_0 \right\|_2}{\sqrt{p}} + \lim_{n \rightarrow \infty} \frac{\|\epsilon \mu_* \beta_0\|_2}{\sqrt{p}} = \epsilon \sigma_* + \frac{\gamma}{\epsilon \mu_* \kappa}. \quad (\text{S63})$$

Further, $\|\hat{\beta}^{\text{DY}}\|_2/\sqrt{n} = \mathcal{O}(1)$ almost surely by Theorem S5. Hence, by the Cauchy property of $\epsilon \nu^t, \epsilon \mathbf{R}^t$, as shown in Lemma S18 along with the equivalence of recursions of Lemma S4, (S63) and Lemma S14 and Lemma S15, it follows that with probability at least $1 - c_1 \exp\{-c_n\}$

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \left\| \epsilon \hat{\beta}^t - \hat{\beta}^{\text{DY}} \right\|_2 = 0.$$

The almost sure claim comes from the first Borel-Cantelli Lemma (e.g. Shiryaev 2016, p. 308). $\square$

It is now possible to finish the proof of Theorem 3.1. First of all, note that by Lemma S16, there exists a unique stationary solution $(\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*)$ to the state evolution (S36)-(S37) that is continuous in ϵ such that $\lim_{\epsilon \rightarrow 0} (\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*) = (\mu_*, b_*, \sigma_*)$, where (μ_*, b_*, σ_*) are a stationary solution to (S15), (S16), (S17).

Now, let ψ be a pseudo Lipschitz function of order 2. Then

$$\begin{aligned}
& \left| \frac{1}{p} \sum_{j=1}^p \psi \left(\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j}, \beta_{0,j} \right) - \frac{1}{p} \sum_{j=1}^p \psi \left(\epsilon \nu^t, \beta_{0,j} \right) \right| \\
& \leq \left| \frac{1}{p} \sum_{j=1}^p \psi \left(\hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \frac{1}{p} \sum_{j=1}^p \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) \right| \\
& \quad + \left| \frac{1}{p} \sum_{j=1}^p \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \frac{1}{p} \sum_{j=1}^p \psi \left(\epsilon \nu^t, \beta_{0,j} \right) \right|. \quad (\text{S64})
\end{aligned}$$

Now

$$\begin{aligned}
& \sum_{j=1}^p \left| \psi \left(\hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) \right| \\
& \leq \sum_{j=1}^p \left| \psi \left(\hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) \right| \\
& \leq C \sum_{j=1}^p \left(1 + \left\| \begin{bmatrix} \hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j} \\ \beta_{0,j} \end{bmatrix} \right\|_2 + \left\| \begin{bmatrix} \epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j} \\ \beta_{0,j} \end{bmatrix} \right\|_2 \right) \left\| \hat{\beta}_j^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2 \\
& \leq C \sqrt{\sum_{j=1}^p \left(1 + \left\| \begin{bmatrix} \hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j} \\ \beta_{0,j} \end{bmatrix} \right\|_2 + \left\| \begin{bmatrix} \epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j} \\ \beta_{0,j} \end{bmatrix} \right\|_2 \right)^2} \left\| \hat{\beta}^{\text{DY}} - \epsilon \hat{\beta}^t \right\|_2, \quad (\text{S65})
\end{aligned}$$

where the first inequality follows from the triangle inequality, the second by virtue of ψ being pseudo Lipschitz of order 2 and the last by the Cauchy-Schwarz inequality.

Further, letting $\mathbf{a}_j = [a_{j1}, a_{j2}]^\top = [\hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j}]^\top$ and $\mathbf{b}_j = [b_{j1}, a_{j2}]^\top = [\hat{\beta}_j^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j}]^\top$,

simple manipulations yield

$$\begin{aligned}
\sum_{j=1}^p (1 + \|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2)^2 &= \sum_{j=1}^p 1 + 2(\|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2) + (\|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2)^2 \\
&= p + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 + 2 \sum_{j=1}^p \|\mathbf{b}_j\|_2 + \sum_{j=1}^p \|\mathbf{a}_j\|_2^2 + \sum_{j=1}^p \|\mathbf{b}_j\|_2^2 + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 \|\mathbf{b}_j\|_2 \\
&= p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 + 2 \sum_{j=1}^p \|\mathbf{b}_j\|_2 + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 \|\mathbf{b}_j\|_2 \\
&\leq p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2 \sum_{j=1}^p |a_{j1}| + 2 \sum_{j=1}^p |b_{j1}| + 4 \sum_{j=1}^p |a_{j2}| + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 \|\mathbf{b}_j\|_2 \\
&= p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2\|\mathbf{a}_{\bullet 1}\|_1 + 2\|\mathbf{b}_{\bullet 1}\|_1 + 4\|\mathbf{a}_{\bullet 2}\|_1 + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 \|\mathbf{b}_j\|_2 \\
&\leq p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2\sqrt{p}\|\mathbf{a}_{\bullet 1}\|_2 + 2\sqrt{p}\|\mathbf{b}_{\bullet 1}\|_2 + 4\sqrt{p}\|\mathbf{a}_{\bullet 2}\|_2 + 2 \sum_{j=1}^p \|\mathbf{a}_j\|_2 \|\mathbf{b}_j\|_2 \\
&\leq p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2\sqrt{p}\|\mathbf{a}_{\bullet 1}\|_2 + 2\sqrt{p}\|\mathbf{b}_{\bullet 1}\|_2 + 4\sqrt{p}\|\mathbf{a}_{\bullet 2}\|_2 \\
&\quad + 2\sqrt{\sum_{j=1}^p a_{j1}^2 + a_{j2}^2} \sqrt{\sum_{j=1}^p b_{j1}^2 + a_{j2}^2} \\
&\leq p + \|\mathbf{a}_{\bullet 1}\|_2^2 + \|\mathbf{b}_{\bullet 1}\|_2^2 + 2\|\mathbf{a}_{\bullet 2}\|_2^2 + 2\sqrt{p}\|\mathbf{a}_{\bullet 1}\|_2 + 2\sqrt{p}\|\mathbf{b}_{\bullet 1}\|_2 + 4\sqrt{p}\|\mathbf{a}_{\bullet 2}\|_2 \\
&\quad + 2(\|\mathbf{a}_{\bullet 1}\|_2 + \|\mathbf{a}_{\bullet 2}\|_2)(\|\mathbf{b}_{\bullet 1}\|_2 + \|\mathbf{a}_{\bullet 2}\|_2), \tag{S66}
\end{aligned}$$

where $\mathbf{a}_{\bullet i} = (a_{1i}, \dots, a_{pi})$, ($i = 1, 2$) and a similar convention applies to $\mathbf{b}_{\bullet 1}$. The fourth line follows since $\sqrt{x^2 + y^2} \leq \sqrt{x^2} + \sqrt{y^2} = |x| + |y|$, the sixth by the Cauchy-Schwarz inequality and since $\|\mathbf{v}\|_1 = |\langle \mathbf{1}, \mathbf{v} \rangle|$, where $\mathbf{1}$ is a p -vector of ones and the absolute value acts elementwise on $\mathbf{v}$. The seventh line follows again by Cauchy-Schwarz, and the last again since $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$. Now, since $\epsilon\mu_* < \infty$, and by assumption and Theorem S5, $\|\beta_0\|_2, \|\hat{\beta}^{\text{DY}}\|_2$ are $\mathcal{O}(\sqrt{n})$, almost surely,

$$\begin{aligned}
\lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\mathbf{a}_{\bullet 1}\|_2 &= \lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\hat{\beta}^{\text{DY}} - \epsilon\mu_*\beta_0\|_2 \leq \lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\hat{\beta}^{\text{DY}}\|_2 + \epsilon\mu_* \frac{1}{\sqrt{p}} \|\beta_0\|_2 < C + \epsilon\mu_*\gamma \\
\lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\mathbf{a}_{\bullet 2}\|_2 &= \lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\beta_0\|_2 = \gamma,
\end{aligned}$$

for some constant $C > 0$, which is independent of t, ϵ . Further, by (S63), almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\mathbf{b}_{\bullet 1}\|_2 = \lim_{N \rightarrow \infty} \frac{1}{\sqrt{p}} \|\hat{\beta}^t - \epsilon\mu_*\beta_0\|_2 = \epsilon\sigma_*^2.$$

Therefore, (S66) gives that almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p (1 + \|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2)^2 \leq 1 + (C + \epsilon\mu_*\gamma) \{2 + C + \epsilon\mu_*\} + \epsilon\sigma_*^2 \{2 + \epsilon\sigma_*^2\} + 4\gamma + 2\{C + \epsilon\mu_*\gamma + \gamma\} \{\epsilon\sigma_*^2 + \gamma\}. \tag{S67}$$

And thus

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p (1 + \|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2)^2 \leq 1 + (C + \mu_*\gamma) \{2 + C + \mu_*\} + \sigma_*^2 \{2 + \sigma_*^2\} + 4\gamma + 2\{C + \mu_*\gamma + \gamma\} \{\sigma_*^2 + \gamma\},$$

where $\lim_{\epsilon \rightarrow 0} (\epsilon \mu_*, \epsilon \sigma_*) = (\mu_*, \sigma_*)$ follows from Lemma S16. In conjunction with Theorem S3 and (S65), this yields that almost surely

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \left| \psi \left(\hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) \right| = 0.$$

Similarly,

$$\begin{aligned} & \left| \frac{1}{p} \sum_{j=1}^p \psi \left(\epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) - \psi \left(\epsilon \nu^t, \beta_{0,j} \right) \right| \\ & \leq C \frac{1}{p} \sqrt{\sum_{j=1}^p \left(1 + \left\| \begin{bmatrix} \epsilon \hat{\beta}^t - \epsilon \mu_* \beta_{0,j} \\ \beta_{0,j} \end{bmatrix} \right\|_2 + \left\| \begin{bmatrix} \epsilon \nu^t \\ \beta_{0,j} \end{bmatrix} \right\|_2 \right)^2} \left\| \epsilon \hat{\beta}^t - \epsilon \mu_* \beta_0 - \epsilon \nu^t \right\|_2, \end{aligned}$$

the right hand side of which is zero almost surely as $N \rightarrow \infty$ for all $t, \epsilon > 0$, by Lemma S4 and (S67). and thus almost surely

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \left| \frac{1}{p} \sum_{j=1}^p \psi \left(\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j}, \beta_{0,j} \right) - \psi \left(\epsilon \hat{\beta}_j^{\text{DY}} - \epsilon \mu_* \beta_{0,j}, \beta_{0,j} \right) \right| = 0.$$

Now recall that from Corollary S2.1, and by the stationarity assumption of the phase transition parameters, $(\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*)$, almost surely

$$\lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\epsilon \nu_j^t, \beta_{0,j}) = \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\epsilon \nu_j^t, \beta_{0,j}) = \mathbb{E} [\psi(\epsilon \sigma_* G, \bar{\beta})],$$

where $G \sim \mathcal{N}(0, 1) \perp \bar{\beta}$. Since ψ is pseudo-Lipschitz of order two,

$$\begin{aligned} |\mathbb{E} [\psi(\epsilon \sigma_* G, \bar{\beta})] - \mathbb{E} [\psi(\sigma_* G, \bar{\beta})]| & \leq \mathbb{E} [|\psi(\epsilon \sigma_* G, \bar{\beta}) - \psi(\sigma_* G, \bar{\beta})|] \\ & \leq C |\epsilon \sigma_* - \sigma_*| \mathbb{E} [(1 + \|\epsilon \sigma_* G, \bar{\beta}\|_2 + \|\sigma_* G, \bar{\beta}\|_2) |G|], \end{aligned}$$

and thus it follows that

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi(\epsilon \nu_j^t, \beta_{0,j}) = \mathbb{E} [\psi(\sigma_* G, \bar{\beta})].$$

In conjunction with (S64), this establishes (S5). Finally, equivalence of stationary solutions (S2)-(S4) with the stationary solutions to (S15)-(S17) is shown in Lemma S19. This concludes the proof of the Theorem.

S5 Proof of Theorem 3.2

Recall that the setup of Theorems 3.2-3.4 is the moderately high-dimensional setting with i.i.d. Gaussian covariates with arbitrary covariance structure. In particular, $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}_p, \Sigma)$ for some positive definite matrix Σ and signal $\beta_0 \in \mathbb{R}^p$ such that $\lim_{n \rightarrow \infty} \text{var}(\mathbf{x}_i^\top \beta_0) = \beta_0^\top \Sigma \beta_0 = \gamma^2$ for $\gamma \in \mathbb{R}_{>0}^p$ and where $\lim_{n \rightarrow \infty} p/n \rightarrow \kappa \in (0, 1)$.

Theorem 3.2. *Let $\hat{\beta}^{\text{DY}}$ be the MDYPL estimator for a logistic regression model with i.i.d. $\mathcal{N}(\mathbf{0}_p, \Sigma)$ covariates and $\limsup_{n \rightarrow \infty} \lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma) < \infty$.*

Assume that (α, κ, γ) are such that (4) admits a solution (μ_, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is non-singular.*

Then for any $t \in \mathbb{R}$,

$$\frac{1}{p} \sum_{j=1}^p \mathbb{1} \left\{ \sqrt{n} \tau_j \frac{\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j}}{\sigma_*} \leq t \right\} \xrightarrow{p} \Phi(t),$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution, $\mathbb{1}(\cdot)$ is the indicator function and $\tau_j^2 = \text{var}(\mathbf{x}_{ij}|\mathbf{x}_{i-j})$ is the conditional variance of the j th predictor given all others. Further, if

$$\frac{1}{p} \sum_{j=1}^p \delta_{\sqrt{n}\tau_j \beta_{0,j}} \xrightarrow{d} \pi_{\bar{\beta}} \quad \text{and} \quad \frac{1}{p} \sum_{j=1}^p n\tau_j^2 \beta_{0,j}^2 \xrightarrow{p} E(\bar{\beta}^2),$$

for some distribution $\pi_{\bar{\beta}}$ with finite second moment and $\bar{\beta} \sim \pi_{\bar{\beta}}$, where $\delta_{(\cdot)}$ is the Dirac delta function, then for any pseudo-Lipschitz function $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}$ of order 2,

$$\frac{1}{p} \sum_{j=1}^p \psi \left(\sqrt{n}\tau_j \left(\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j} \right), \sqrt{n}\tau_j \beta_{0,j} \right) \xrightarrow{p} E[\psi(\sigma_* G, \bar{\beta})], \quad \text{as } n \rightarrow \infty,$$

for $G \sim N(0, 1)$ independent of $\bar{\beta}$.

The proof of Theorem 3.2 is a replication of the arguments in the proof of Theorem 3.2 in Zhao *et al.* (2022) with the substitutions that are outlined in S7. No claim of originality is made.

The following Lemma, which is a generalization of Lemma S8 to arbitrary covariance structures, is the starting point of the analysis. Let $\langle \mathbf{u}, \mathbf{v} \rangle_{\Sigma} = \mathbf{u}^\top \Sigma \mathbf{v}$, and $\|\mathbf{u}\|_{\Sigma}^2 = \langle \mathbf{u}, \mathbf{u} \rangle_{\Sigma}$. Then write $\gamma_n^2 = \|\beta_0\|_{\Sigma}^2 = \text{var}(\mathbf{x}_i^\top \beta_0)$ for $\mathbf{x}_i \sim N(\mathbf{0}_p, \Sigma)$.

Lemma S5. Let $\hat{\beta}^{\text{DY}}$ be the MDYPL estimator from $N(\mathbf{0}_p, \Sigma)$ covariates and signal β_0 . Define

$$\boldsymbol{\xi} = \mathbf{Z} - \left\langle \mathbf{Z}, \frac{\beta_0}{\gamma_n} \right\rangle_{\Sigma} \frac{\beta_0}{\gamma_n},$$

for $\mathbf{Z} \sim N(\mathbf{0}_p, \Sigma^{-1})$ independent of everything else. Further, let

$$\tilde{\mu}_n = \frac{\langle \hat{\beta}^{\text{DY}}, \beta_0 \rangle_{\Sigma}}{\|\beta_0\|_{\Sigma}^2}, \quad \tilde{\sigma}_n^2 = \left\| \hat{\beta}^{\text{DY}} \right\|_{\Sigma}^2 - \frac{\langle \hat{\beta}^{\text{DY}}, \beta_0 \rangle_{\Sigma}^2}{\|\beta_0\|_{\Sigma}^2}.$$

Then

$$\frac{\hat{\beta}^{\text{DY}} - \tilde{\mu}_n \beta_0}{\tilde{\sigma}_n} \stackrel{d}{=} \frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|_{\Sigma}}.$$

Proof. The proof is similar to the proof of Proposition B.1 in the proof of Theorem 3.2 of Zhao *et al.* (2022) and thus omitted. $\square$

Lemma S5 immediately yields the equivalence (S68) below. Define $\mathbf{T}^{\text{approx}}$ as the p -vector with elements

$$\mathbf{T}_j^{\text{approx}} = \tau_j \sqrt{n} \frac{\hat{\beta}_j^{\text{DY}} - \tilde{\mu}_n \beta_{0,j}}{\tilde{\sigma}_n / \sqrt{\kappa}},$$

which ought to approximate

$$\mathbf{T}_j = \tau_j \sqrt{n} \frac{\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j}}{\sigma_*}.$$

Further, let $\mathbf{Z}_j^{\text{scaled}} = \tau_j \mathbf{Z}_j$, so that

$$\mathbf{Z}^{\text{scaled}} \sim N(\mathbf{0}_p, \mathbf{R}), \quad \mathbf{R}_{ij} = \tau_i \tau_j [\Sigma^{-1}]_{ij},$$

and finally let $\tilde{\mathbf{Z}}^{\text{scaled}}$ be the p -vector with entries

$$\tilde{\mathbf{Z}}_j^{\text{scaled}} = \left(\mathbf{Z}_j^{\text{scaled}} - \left\langle \mathbf{Z}_j^{\text{scaled}}, \frac{\beta_0}{\gamma_n} \right\rangle_{\Sigma} \frac{\tau_j \beta_{0,j}}{\gamma_n} \right) \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\Sigma}}.$$

Then

$$\mathbf{T}^{\text{approx}} \stackrel{d}{=} \tilde{\mathbf{Z}}^{\text{scaled}}. \quad (\text{S68})$$

The remainder of the proof is concerned with showing that $\mathbf{T} \approx \mathbf{T}^{\text{approx}}$, $\tilde{\mathbf{Z}}^{\text{scaled}} \approx \mathbf{Z}^{\text{scaled}}$ in an adequate sense and that the empirical CDF of $\mathbf{Z}^{\text{scaled}}$ converges to that of a standard normal.

Lemma S6. *If $\limsup_{n \rightarrow \infty} \lambda_{\max}(\mathbf{\Sigma})/\lambda_{\min}(\mathbf{\Sigma}) < \infty$, then for any univariate Lipschitz function ψ ,*

$$\frac{1}{p} \sum_{j=1}^p \left\{ \psi(\tilde{\mathbf{Z}}_j^{\text{scaled}}) - \psi(\mathbf{Z}_j^{\text{scaled}}) \right\} \xrightarrow{p} 0, \quad \frac{1}{p} \sum_{j=1}^p \left\{ \psi(\mathbf{T}_j^{\text{approx}}) - \psi(\mathbf{T}_j) \right\} \xrightarrow{p} 0.$$

Proof. First consider the pair $\tilde{\mathbf{Z}}^{\text{scaled}}, \mathbf{Z}^{\text{scaled}}$ and let L be the Lipschitz constant of ψ . By the triangle inequality and the definition of $\tilde{\mathbf{Z}}^{\text{scaled}}, \mathbf{Z}^{\text{scaled}}$,

$$\begin{aligned} \left| \frac{1}{p} \sum_{j=1}^p \left\{ \psi(\tilde{\mathbf{Z}}_j^{\text{scaled}}) - \psi(\mathbf{Z}_j^{\text{scaled}}) \right\} \right| &\leq \frac{1}{p} \sum_{j=1}^p \left| \psi(\tilde{\mathbf{Z}}_j^{\text{scaled}}) - \psi(\mathbf{Z}_j^{\text{scaled}}) \right| \\ &\leq L \frac{1}{p} \sum_{j=1}^p \left| \mathbf{Z}_j^{\text{scaled}} \left(1 - \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\mathbf{\Sigma}}} \right) + \left\langle \mathbf{Z}, \frac{\boldsymbol{\beta}_0}{\gamma_n} \right\rangle_{\mathbf{\Sigma}} \frac{\tau_j \boldsymbol{\beta}_{0,j}}{\gamma_n} \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\mathbf{\Sigma}}} \right| \\ &\leq L \left| 1 - \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\mathbf{\Sigma}}} \right| \frac{1}{p} \sum_{j=1}^p |\mathbf{Z}_j^{\text{scaled}}| + \left| \frac{1}{\gamma_n} \left\langle \mathbf{Z}, \frac{\boldsymbol{\beta}_0}{\gamma_n} \right\rangle_{\mathbf{\Sigma}} \right| \frac{L\sqrt{p}}{\|\boldsymbol{\xi}\|_{\mathbf{\Sigma}}} \frac{1}{p} \sum_{j=1}^p |\tau_j \boldsymbol{\beta}_{0,j}|. \end{aligned} \quad (\text{S69})$$

Arguments similar to Lemma S20 establish that $\sqrt{p}/\|\boldsymbol{\xi}\|_{\mathbf{\Sigma}} \rightarrow 1$ almost surely as $n \rightarrow \infty$. Further note that $\langle \mathbf{Z}, \boldsymbol{\beta}_0/\gamma_n \rangle_{\mathbf{\Sigma}}/\gamma_n$ is a univariate Gaussian with $\mathcal{O}(1)$ variance so that it is itself $\mathcal{O}_p(1)$. Hence, it is sufficient to show that

$$\frac{1}{p} \sum_{j=1}^p |\mathbf{Z}_j^{\text{scaled}}| = \mathcal{O}_p(1), \quad \frac{1}{p} \sum_{j=1}^p |\tau_j \boldsymbol{\beta}_{0,j}| = o(1). \quad (\text{S70})$$

For the first note that $\mathbb{E}[|\mathbf{Z}_j^{\text{scaled}}|^2] \leq \mathbb{E}[\mathbf{Z}_j^{\text{scaled}2}] = \tau_j^2 [\mathbf{\Sigma}^{-1}]_{jj} \mathbb{E}[\mathbf{Z}_j^2] = 1$ by definition of τ_j . Now Markov's inequality (e.g. Vershynin (2018, Proposition 1.2.4)) yields that for $\epsilon > 0$,

$$\Pr \left(\frac{1}{p} \sum_{j=1}^p |\mathbf{Z}_j^{\text{scaled}}| \geq \frac{1}{\epsilon} \right) \leq \epsilon \mathbb{E} \left[\frac{1}{p} \sum_{j=1}^p |\mathbf{Z}_j^{\text{scaled}}| \right] \leq \epsilon, \quad (\text{S71})$$

so that $\frac{1}{p} \sum_{j=1}^p |\mathbf{Z}_j^{\text{scaled}}| = \mathcal{O}_p(1)$ indeed. Now, by the Cauchy-Schwarz inequality,

$$\frac{1}{p} \sum_{j=1}^p |\tau_j \boldsymbol{\beta}_{0,j}| = \frac{1}{p} \sum_{j=1}^p |\boldsymbol{\theta}_{0,j}| \leq \frac{1}{\sqrt{p}} \|\boldsymbol{\theta}_0\|_2 = \mathcal{O}(p^{-1/2}), \quad (\text{S72})$$

where $\boldsymbol{\theta}_0 = \mathbf{L}^\top \boldsymbol{\beta}_0$ for $\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^\top$, and by assumption $\|\boldsymbol{\theta}_0\|_2^2 = \text{var}(\mathbf{x}_i^\top \boldsymbol{\beta}_0) \rightarrow \gamma^2$ as $n \rightarrow \infty$, for $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}_p, \mathbf{\Sigma})$. Hence, (S69) is indeed $o(1)$. Next consider $\mathbf{T}, \mathbf{T}^{\text{approx}}$.

$$\begin{aligned}
\left| \frac{1}{p} \sum_{j=1}^p \psi(\mathbf{T}_j^{\text{approx}}) - \psi(\mathbf{T}_j) \right| &\leq \frac{1}{p} \sum_{j=1}^p |\psi(\mathbf{T}_j^{\text{approx}}) - \psi(\mathbf{T}_j)| \\
&\leq \frac{L}{\sigma_*} \sqrt{\frac{n}{p}} \frac{1}{\sqrt{p}} \sum_{j=1}^p \left| \tau_j \hat{\beta}_j^{\text{DY}} \left(1 - \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \right) + \tau_j \beta_{0,j} \left(\frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \tilde{\mu}_n - \mu_* \right) \right| \\
&\leq \frac{L}{\sigma_*} \sqrt{\frac{n}{p}} \left\{ \left| 1 - \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \right| \frac{1}{\sqrt{p}} \sum_{j=1}^p |\tau_j \hat{\beta}_j^{\text{DY}}| + \left| \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \tilde{\mu}_n - \mu_* \right| \frac{1}{\sqrt{p}} \sum_{j=1}^p |\tau_j \beta_{0,j}| \right\}.
\end{aligned} \tag{S73}$$

Now, it follows readily from Lemma S7 and Lemma S9 that $\tilde{\mu}_n \rightarrow \mu_*$ and $\tilde{\sigma}_n \rightarrow \sigma_*$ almost surely as $n \rightarrow \infty$. Indeed, note that by Lemma S7, one can write

$$\tilde{\mu}_n = \frac{(\hat{\beta}^{\text{DY}})^\top \Sigma \beta_0}{\beta_0^\top \Sigma \beta_0} = \frac{(\mathbf{L}^\top \hat{\beta}^{\text{DY}})^\top (\mathbf{L}^\top \beta_0)}{(\mathbf{L}^\top \beta_0)^\top (\mathbf{L}^\top \beta_0)} = \frac{\langle \hat{\theta}^{\text{DY}}, \theta_0 \rangle}{\|\theta_0\|_2^2},$$

for $\Sigma = \mathbf{L} \mathbf{L}^\top$ and where $\hat{\theta}^{\text{DY}}$ is the MDYPL estimator from $N(\mathbf{0}_p, \mathbf{I}_p)$ covariates and signal $\theta_0 = \mathbf{L}^\top \beta_0$. Hence, immediately by Lemma S9, $\tilde{\mu}_n \rightarrow \mu_*$ as $n \rightarrow \infty$ almost surely and the same applies to $\tilde{\sigma}_n$. Secondly, by (S72), $\frac{1}{\sqrt{p}} \sum_{j=1}^p |\tau_j \beta_{0,j}| = o(1)$. Finally, recalling that $\tau_j \hat{\beta}_j^{\text{DY}} = \hat{\theta}_j^{\text{DY}}$, the Cauchy-Schwarz inequality yields

$$\frac{1}{\sqrt{p}} \sum_{j=1}^p |\tau_j \hat{\beta}_j^{\text{DY}}| = \frac{1}{\sqrt{p}} \sum_{j=1}^p |\hat{\theta}_j^{\text{DY}}| \leq \|\hat{\theta}^{\text{DY}}\|_2, \tag{S74}$$

and by Lemma S7 and Theorem S5, $\lim_{n \rightarrow \infty} \|\hat{\theta}^{\text{DY}}\|_2 = \mathcal{O}(1)$ almost surely. Thus, (S73) is $o_p(1)$ as required. $\square$

Taken together, Lemma S6 and (S68) imply that

$$\frac{1}{p} \sum_{j=1}^p \psi(\mathbf{T}_j) - \psi(\mathbf{Z}_j^{\text{scaled}}) = o_p(1),$$

so that it remains to characterise the distribution of $\mathbf{Z}^{\text{scaled}}$, which is fully characterised by the covariance structure Σ . Zhao *et al.* (2022, Section B.2-B.4) show that if $\lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma) = o(\sqrt{p})$, then the empirical CDF of $\mathbf{Z}^{\text{scaled}}$ converges pointwise to that of a standard normal random variable in probability, which establishes the first part of the Theorem.

The proof of the second part of the Theorem follows a similar structure as the first part. In particular, for a bivariate pseudo-Lipschitz function ψ of order two, it is first shown that

$$\frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{T}_j, \tau_j \sqrt{n} \beta_{0,j}) - \frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{T}_j^{\text{approx}}, \tau_j \sqrt{n} \beta_{0,j}) = o_p(1), \tag{S75}$$

and that

$$\frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \tilde{\mathbf{Z}}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) - \frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{Z}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) = o_p(1), \tag{S76}$$

which, in conjunction with (S68) yields

$$\frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{T}_j, \tau_j \sqrt{n} \beta_{0,j}) - \frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{Z}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) = o_p(1),$$

so that it then remains to analyse

$$\frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{Z}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) . \quad (\text{S77})$$

Towards showing (S75), consider

$$\begin{aligned} & \left| \frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{T}_j, \tau_j \sqrt{n} \beta_{0,j}) - \psi(\sigma_* \mathbf{T}_j^{\text{approx}}, \tau_j \sqrt{n} \beta_{0,j}) \right| \\ & \leq \frac{1}{p} \sum_{j=1}^p |\psi(\sigma_* \mathbf{T}_j, \tau_j \sqrt{n} \beta_{0,j}) - \psi(\sigma_* \mathbf{T}_j^{\text{approx}}, \tau_j \sqrt{n} \beta_{0,j})| \\ & \leq \frac{L\sigma_*}{p} \sum_{j=1}^p \left(1 + \|[\sigma_* \mathbf{T}_j, \tau_j \sqrt{n} \beta_{0,j}]^\top\|_2 + \|[\sigma_* \mathbf{T}_j^{\text{approx}}, \tau_j \sqrt{n} \beta_{0,j}]^\top\|_2 \right) \|\mathbf{T}_j - \mathbf{T}_j^{\text{approx}}\|_2 \\ & \leq \frac{L\sigma_*}{p} \left\{ \sum_{j=1}^p (1 + \sigma_* |\mathbf{T}_j| + \sigma_* |\mathbf{T}_j^{\text{approx}}| + 2\sqrt{n} |\tau_j \beta_{0,j}|)^2 \right\}^{1/2} \left\{ \sum_{j=1}^p (\mathbf{T}_j - \mathbf{T}_j^{\text{approx}})^2 \right\}^{1/2}, \end{aligned}$$

where the last line follows by the Cauchy-Schwarz inequality and the monotonicity of ℓ_p norms. Now by construction of $\mathbf{T}_j, \mathbf{T}_j^{\text{approx}}$ and since $(a+b)^2 \leq 2a^2 + 2b^2$,

$$\begin{aligned} \sum_{j=1}^p (\mathbf{T}_j - \mathbf{T}_j^{\text{approx}})^2 &= \frac{n}{\sigma_*^2} \sum_{j=1}^p \left(\tau_j \hat{\beta}_j^{\text{DY}} \left(1 - \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \right) + \tau_j \beta_{0,j} \left(\frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \tilde{\mu}_n - \mu_* \right) \right)^2 \\ &\leq \frac{2n}{\sigma_*^2} \left\{ \left(1 - \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \right)^2 \sum_{j=1}^p (\tau_j \hat{\beta}_j^{\text{DY}})^2 + \left(\frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \tilde{\mu}_n - \mu_* \right)^2 \sum_{j=1}^p (\tau_j \beta_{0,j})^2 \right\} \\ &= \frac{2n}{\sigma_*^2} \left\{ \left(1 - \frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \right)^2 \|\hat{\boldsymbol{\theta}}^{\text{DY}}\|_2^2 + \left(\frac{\sqrt{\kappa} \sigma_*}{\tilde{\sigma}_n} \tilde{\mu}_n - \mu_* \right)^2 \|\boldsymbol{\theta}_0\|_2^2 \right\} \\ &= o(n), \end{aligned}$$

by considerations similar to those in (S72)-(S74). Similarly, one can bound

$$\begin{aligned} \sum_{j=1}^p (1 + \sigma_* |\mathbf{T}_j| + \sigma_* |\mathbf{T}_j^{\text{approx}}| + 2\sqrt{n} |\tau_j \beta_{0,j}|)^2 &\leq 4 \sum_{j=1}^p 1 + \sigma_*^2 \mathbf{T}_j^2 + \sigma_*^2 \mathbf{T}_j^{\text{approx}2} + 4n \tau_j^2 \beta_{0,j}^2 \\ &= 4n + 4\sigma_*^2 \|\mathbf{T}\|_2^2 + 4\sigma_*^2 \|\mathbf{T}^{\text{approx}}\|_2^2 + 16n \|\boldsymbol{\theta}_0\|_2^2. \end{aligned}$$

Now $\mathbf{T}_j = \sqrt{n}(\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_* \boldsymbol{\theta}_{0,j})/\sigma_*$ and by considerations analogous to Lemma S9 and Theorem 3.1, choosing $\psi(t, u) = t^2$, yields that $\|\mathbf{T}\|_2^2 = \mathcal{O}_p(n)$. Similarly, by Lemma S9 and the expansion of (S81), also $\|\mathbf{T}^{\text{approx}}\|_2^2 = \mathcal{O}_p(n)$. Therefore (S75) is indeed $o_p(1)$.

Next, consider the difference (S76). Similarly to above,

$$\begin{aligned} & \left| \frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \tilde{\mathbf{Z}}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) - \psi(\sigma_* \mathbf{Z}_j^{\text{scaled}}, \tau_j \sqrt{n} \beta_{0,j}) \right| \\ & \leq \frac{L\sigma_*}{p} \left\{ \sum_{j=1}^p (1 + |\mathbf{Z}_j^{\text{scaled}}| + |\tilde{\mathbf{Z}}_j^{\text{scaled}}| + 2\sqrt{n} |\tau_j \beta_{0,j}|)^2 \right\}^{1/2} \left\{ \sum_{j=1}^p (\mathbf{Z}_j^{\text{scaled}} - \tilde{\mathbf{Z}}_j^{\text{scaled}})^2 \right\}^{1/2}. \end{aligned}$$

The term

$$\sum_{j=1}^p (1 + |\mathbf{Z}_j^{\text{scaled}}| + |\tilde{\mathbf{Z}}_j^{\text{scaled}}| + 2\sqrt{n} |\tau_j \beta_{0,j}|)^2 = \mathcal{O}_p(n),$$

since $\mathbf{T}^{\text{approx}} \stackrel{d}{=} \tilde{\mathbf{Z}}^{\text{scaled}}$, which was shown to be bounded in the above paragraph and $\mathbf{Z}^{\text{scaled}}$ was appropriately bounded in (S70)-(S71). On the other hand,

$$\begin{aligned} \sum_{j=1}^p \left(\mathbf{Z}_j^{\text{scaled}} - \tilde{\mathbf{Z}}_j^{\text{scaled}} \right)^2 &= \sum_{j=1}^p \left(\mathbf{Z}_j^{\text{scaled}} \left(1 - \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\Sigma}} \right) + \frac{\langle \mathbf{Z}, \boldsymbol{\beta}_0 / \gamma_n \rangle_{\Sigma}}{\gamma_n} \tau_j \boldsymbol{\beta}_{0,j} \right)^2 \\ &\leq 2 \left(1 - \frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\Sigma}} \right)^2 \|\mathbf{Z}^{\text{scaled}}\|_2^2 + \left(\frac{\langle \mathbf{Z}, \boldsymbol{\beta}_0 / \gamma_n \rangle_{\Sigma}}{\gamma_n} \right)^2 \left(\frac{\sqrt{p}}{\|\boldsymbol{\xi}\|_{\Sigma}} \right)^2 \|\boldsymbol{\theta}_0\|_2^2 \\ &= o_p(n), \end{aligned}$$

since arguments similar to Lemma S20 show that $\lim_{n \rightarrow \infty} \sqrt{p} / \|\boldsymbol{\xi}\|_{\Sigma} = 1$ almost surely, $E[\mathbf{Z}_j^{\text{scaled}2}] = 1$ and thus a Markov bound similar to (S70)-(S71) establishes that $\|\mathbf{Z}^{\text{scaled}}\|_2^2 = \mathcal{O}_p(p)$. The second term is bounded since additionally $\lim_{n \rightarrow \infty} \gamma_n^2 = \lim_{n \rightarrow \infty} \|\boldsymbol{\theta}_0\|_2^2 = \gamma^2$, and $\langle \mathbf{Z}, \boldsymbol{\beta}_0 / \gamma_n \rangle_{\Sigma}$ is a normal random variable with $\mathcal{O}(1)$ variance.

Hence, it remains to analyse the behaviour of (S77) which is done in Zhao *et al.* (2022, Lemma C.1). As $\mathbf{Z}^{\text{scaled}}$ solely depends on Σ , this result directly applies to this setting. The conditions of said Lemma are satisfied in the current setting. Zhao *et al.* (2022, Lemma C.1) establishes that

$$\frac{1}{p} \sum_{j=1}^p \psi(\sigma_* \mathbf{Z}_j^{\text{scaled}}, \tau_j \sqrt{n} \boldsymbol{\beta}_{0,j}) \xrightarrow{p} E[\psi(\sigma_* G, \bar{\beta})],$$

for $G \sim N(0, 1) \perp \bar{\beta}$. This concludes the proof the Theorem.

S6 Proof of Theorem 3.3

Theorem 3.3. *Let $\hat{\beta}^{\text{DY}}$ be the MDYPL estimator for a logistic regression model with i.i.d. $N(\mathbf{0}_p, \Sigma)$ covariates.*

Assume that (α, κ, γ) are such that (4) admits a solution (μ_, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is non-singular.*

Then for any regression coefficient such that $\sqrt{n} \tau_j \boldsymbol{\beta}_{0,j} = \mathcal{O}(1)$,

$$\sqrt{n} \tau_j \frac{\hat{\beta}_j^{\text{DY}} - \mu_* \boldsymbol{\beta}_{0,j}}{\sigma_*} \xrightarrow{d} N(0, 1),$$

where $\tau_j^2 = \text{var}(\mathbf{x}_{ij} | \mathbf{x}_{i-j})$.

The proof of Theorem 3.3 is a replication of the arguments in the proof of Theorem 3.1 in Zhao *et al.* (2022) with the substitutions that are outlined in S7 and follows the general outline given in Section 3.3 of the main paper of this text. No claim of originality is made.

Lemma S7. *For any matrix $\mathbf{L} \in \mathbb{R}^{p \times p}$ and $\Sigma = \mathbf{L}\mathbf{L}^{\top}$, let $\hat{\beta}^{\text{DY}}$ be the MDYPL estimator from a logistic regression model with i.i.d. $N(\mathbf{0}_p, \Sigma)$ covariates and signal $\boldsymbol{\beta}_0$. Then*

$$\hat{\boldsymbol{\theta}}^{\text{DY}} = \mathbf{L}^{\top} \hat{\beta}^{\text{DY}}$$

is the MDYPL estimator from a logistic regression with $N(\mathbf{0}_p, \mathbf{I}_p)$ covariates and signal $\boldsymbol{\theta}_0 = \mathbf{L}^{\top} \boldsymbol{\beta}_0$.

Proof. This result follows from the invariance of the MDYPL estimator under linear transformations of the covariates (see for example Zehna (1966)) and the rotational invariance of the normal distribution. Details can be found in Zhao *et al.* 2022, Proposition 2.1. $\square$

Lemma S8. Let $\hat{\boldsymbol{\theta}}^{DY}$ be the MDYPL estimator from a logistic regression model with i.i.d. $N(\mathbf{0}_p, \mathbf{I}_p)$ covariates and signal $\boldsymbol{\theta}_0$. Define the random variables

$$\mu_n = \frac{\langle \hat{\boldsymbol{\theta}}^{DY}, \boldsymbol{\theta}_0 \rangle}{\|\boldsymbol{\theta}_0\|_2^2}, \quad \sigma_n = \left\| \mathbf{P}_{\boldsymbol{\theta}_0^\perp} \hat{\boldsymbol{\theta}}^{DY} \right\|_2, \quad (\text{S78})$$

where $\mathbf{P}_{\boldsymbol{\theta}_0^\perp}$ is the projection onto $\boldsymbol{\theta}_0^\perp$, the orthogonal complement of $\boldsymbol{\theta}_0$. Then

$$\frac{\hat{\boldsymbol{\theta}}^{DY} - \mu_n \boldsymbol{\theta}_0}{\sigma_n}$$

is uniformly distributed on the unit sphere lying in $\boldsymbol{\theta}_0^\perp$.

Proof. The proof is similar to Zhao *et al.* (2022, Lemma 2.1) and therefore omitted. $\square$

Lemma S9 (Zhao *et al.* (2022), Lemma 3.1). Let $\hat{\boldsymbol{\beta}}^{DY}$ be the MDYPL estimator with i.i.d. covariates $\mathbf{x}_i \sim N(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$.

Assume that (α, κ, γ) are such that (4) admits a solution (μ_*, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is non-singular.

Then for μ_n, σ_n as in (S78),

$$\mu_n \xrightarrow{a.s.} \mu_*, \quad \sigma_n \xrightarrow{a.s.} \sqrt{\kappa} \sigma_*,$$

where (μ_*, σ_*) are the solutions corresponding to the system of equations (S2) - (S4).

Proof. The proof is similar to the proof of Lemma 3.1 in Zhao *et al.* (2022) and relies on Lemma S7 and Theorem 3.1.

Given $\boldsymbol{\theta}_0$ such that $\lim_{n \rightarrow \infty} \|\boldsymbol{\theta}_0\|_2 = \gamma$, let $\hat{\boldsymbol{\beta}}^{DY}$ be the MDYPL estimator from $N(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$ covariates and signal $\boldsymbol{\beta}_0 = \sqrt{n} \mathbf{U} \boldsymbol{\theta}_0$ where $\mathbf{U}$ is an orthogonal matrix such that $\mathbf{U} \boldsymbol{\theta}_0 = (\|\boldsymbol{\theta}_0\|_2, \dots, \|\boldsymbol{\theta}_0\|_2)^\top / \sqrt{p}$, which can for example be constructed using a Householder transformation. Then by Lemma S7, $\hat{\boldsymbol{\beta}}^{DY} = \sqrt{n} \mathbf{U} \hat{\boldsymbol{\theta}}^{DY}$ where $\hat{\boldsymbol{\theta}}^{DY}$ is the MDYPL estimator with $N(\mathbf{0}_p, \mathbf{I}_p)$ covariates and signal $\boldsymbol{\theta}_0$. Note that by construction, the empirical distribution of $\boldsymbol{\beta}_0$ converges to a point mass at $\gamma/\sqrt{\kappa}$ and that $\lim_{n \rightarrow \infty} \frac{1}{p} \sum_{j=1}^n \beta_{0,j}^2 = \gamma^2/\kappa$ as required. Hence, Theorem 3.1 gives that for any pseudo-Lipschitz function ψ of order two,

$$\frac{1}{p} \sum_{j=1}^p \psi(\sqrt{n}([U \hat{\boldsymbol{\theta}}^{DY}]_j - \mu_*[U \boldsymbol{\theta}_0]_j), \sqrt{n}[U \boldsymbol{\theta}_0]_j) \xrightarrow{a.s.} E[\psi(\sigma_* G, \bar{\beta})],$$

where $G \sim N(0, 1)$, independent of $\bar{\beta} \sim \delta_{\gamma/\sqrt{\kappa}}$ and μ_*, σ_* solve (S2)-(S4). Then choosing $\psi(x, y) = xy$, gives that

$$\frac{1}{\kappa} \langle U \hat{\boldsymbol{\theta}}^{DY} - \mu_* U \boldsymbol{\theta}_0, U \boldsymbol{\theta}_0 \rangle = \frac{1}{\kappa} \langle \hat{\boldsymbol{\theta}}^{DY} - \mu_* \boldsymbol{\theta}_0, \boldsymbol{\theta}_0 \rangle \xrightarrow{a.s.} 0,$$

from which it follows that

$$\mu_n = \frac{\langle \hat{\boldsymbol{\theta}}^{DY}, \boldsymbol{\theta}_0 \rangle}{\|\boldsymbol{\theta}_0\|_2} \xrightarrow{a.s.} \mu_*. \quad (\text{S79})$$

Similarly, choosing $\psi(x, y) = x^2$, gives

$$\frac{1}{\kappa} \langle U \hat{\boldsymbol{\theta}}^{DY} - \mu_* U \boldsymbol{\theta}_0, U \hat{\boldsymbol{\theta}}^{DY} - \mu_* U \boldsymbol{\theta}_0 \rangle = \frac{1}{\kappa} \langle \hat{\boldsymbol{\theta}}^{DY} - \mu_* \boldsymbol{\theta}_0, \hat{\boldsymbol{\theta}}^{DY} - \mu_* \boldsymbol{\theta}_0 \rangle = \left\| \hat{\boldsymbol{\theta}}^{DY} - \mu_* \boldsymbol{\theta}_0 \right\|_2^2 \xrightarrow{a.s.} \sigma_*^2,$$

which in conjunction with (S79) yields

$$\sigma_n^2 = \left\| \hat{\boldsymbol{\theta}}^{\text{DY}} - \mu_n \boldsymbol{\theta}_0 \right\|_2^2 \xrightarrow{a.s.} \kappa \sigma_*^2$$

as required. $\square$

Towards proving Theorem 3.3, note that Lemma S7 implies that for any fixed coordinate j ,

$$\sqrt{n} \tau_j \frac{\hat{\beta}_j^{\text{DY}} - \mu_* \beta_{0,j}}{\sigma_*} = \sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_* \boldsymbol{\theta}_{0,j}}{\sigma_*}, \quad (\text{S80})$$

where $\hat{\boldsymbol{\theta}}^{\text{DY}} = \mathbf{L}^\top \hat{\boldsymbol{\beta}}^{\text{DY}}$, is the MDYPL estimator from $N(\mathbf{0}_p, \mathbf{I}_p)$ covariates with signal $\boldsymbol{\theta}_0 = \mathbf{L}^\top \boldsymbol{\beta}_0$ for $\mathbf{L}$ being the Cholesky factorization of $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top$ and where $\tau_j^2 = \text{var}(\mathbf{x}_{ij} | \mathbf{x}_{i,-j})$ is the conditional variance of the j th covariate given all other covariates under $\boldsymbol{\Sigma}$ (see Zhao *et al.* (2022, Section 2.1)).

A simple expansion of the RHS of (S80) gives

$$\sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_* \boldsymbol{\theta}_{0,j}}{\sigma_*} = \sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_n \boldsymbol{\theta}_{0,j}}{\sigma_n} \frac{\sigma_n}{\sigma_*} + \sqrt{n} \frac{(\mu_n - \mu_*) \boldsymbol{\theta}_{0,j}}{\sigma_*}. \quad (\text{S81})$$

Now by the assumption of the Theorem, $\sqrt{n} \boldsymbol{\theta}_{0,j} = \sqrt{n} \tau_j \boldsymbol{\beta}_{0,j} = \mathcal{O}(1)$. Hence, by Lemma S9, the second term on the RHS of (S81) is $o(1)$ almost surely. Now by Lemma S8, $(\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_n \boldsymbol{\theta}_{0,j})/\sigma_n = \mathbf{P}_{\boldsymbol{\theta}_0^\perp} \hat{\boldsymbol{\theta}}^{\text{DY}} / \|\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \hat{\boldsymbol{\theta}}^{\text{DY}}\|_2$ is distributed uniformly on the unit sphere lying in $\boldsymbol{\theta}_0^\perp$, where $\mathbf{P}_{\boldsymbol{\theta}_0^\perp}$ is the projection matrix into the orthogonal complement $\boldsymbol{\theta}_0^\perp$ of $\boldsymbol{\theta}_0$. It is a well known property of the Gaussian distribution that for $\mathbf{Z} \sim N(\mathbf{0}_p, \mathbf{I}_p)$, the projection onto the unit sphere is uniformly distributed (see for example Vershynin (2018), Section 3.3.3), and thus by Lemma S8,

$$\frac{\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \hat{\boldsymbol{\theta}}^{\text{DY}}}{\|\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \hat{\boldsymbol{\theta}}^{\text{DY}}\|_2} \stackrel{d}{=} \frac{\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z}}{\|\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z}\|_2},$$

for $\mathbf{Z} \sim N(\mathbf{0}_p, \mathbf{I}_p)$. Expanding $\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z}$ as

$$\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z} = \mathbf{Z} - \left\langle \mathbf{Z}, \frac{\boldsymbol{\theta}_0}{\|\boldsymbol{\theta}_0\|_2} \right\rangle \frac{\boldsymbol{\theta}_0}{\|\boldsymbol{\theta}_0\|_2},$$

whose j th component is thus given by

$$Z_j \left(1 - \frac{\boldsymbol{\theta}_{0,j}^2}{\|\boldsymbol{\theta}_0\|_2^2} \right) + \xi,$$

where Z_j is the j th component of $\mathbf{Z}$ and ξ is a normal random variable independent of Z_j with mean zero and variance

$$\text{var}(\xi) = \frac{\|\boldsymbol{\theta}_{0,-j}\|_2^2}{\|\boldsymbol{\theta}_0\|_2^2} \frac{\boldsymbol{\theta}_{0,j}^2}{\|\boldsymbol{\theta}_0\|_2^2} = \mathcal{O}(n^{-1}),$$

since $\sqrt{n} \boldsymbol{\theta}_j = \mathcal{O}(1)$. Consequently $\xi = \mathcal{O}_p(n^{-1/2})$. Further, by Lemma S20, $\|\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z}\|_2 / \sqrt{p} \rightarrow 1$ almost surely.

Hence, taken together, one gets that

$$\begin{aligned}
\sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_* \boldsymbol{\theta}_{0,j}}{\sigma_*} &= \sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_n \boldsymbol{\theta}_{0,j}}{\sigma_n} \frac{\sigma_n}{\sigma_*} + \sqrt{n} \frac{(\mu_n - \mu_*) \boldsymbol{\theta}_{0,j}}{\sigma_*} \\
&= \sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_n \boldsymbol{\theta}_{0,j}}{\sigma_n} \frac{\sigma_n}{\sigma_*} + o_p(1) \\
&= \sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_n \boldsymbol{\theta}_{0,j}}{\sigma_n} + o_p(1) \\
&\stackrel{d}{=} \sqrt{\frac{n}{p}} \frac{\|\mathbf{P}_{\boldsymbol{\theta}_0^\perp} \mathbf{Z}\|_2}{\sqrt{p}} \left\{ Z_j \left(1 - \frac{\boldsymbol{\theta}_{0,j}^2}{\|\boldsymbol{\theta}_0\|_2^2} \right) + \xi \right\} + o_p(1) \\
&= \frac{1}{\sqrt{\kappa}} \left\{ Z_j \left(1 - \frac{\boldsymbol{\theta}_{0,j}^2}{\|\boldsymbol{\theta}_0\|_2^2} \right) + \xi \right\} + o_p(1) \\
&= \frac{1}{\sqrt{\kappa}} Z_j \sigma_j + o_p(1) \\
&= \frac{1}{\sqrt{\kappa}} Z_j + o_p(1)
\end{aligned}$$

for $\sigma_j = 1 - \boldsymbol{\theta}_{0,j}^2 / \|\boldsymbol{\theta}_0\|_2^2$ which converges to one since $\boldsymbol{\theta}_{0,j} = \mathcal{O}(n^{-1/2})$. By (S80) and (S81),

$$\sqrt{n} \frac{\hat{\boldsymbol{\theta}}_j^{\text{DY}} - \mu_* \boldsymbol{\theta}_{0,j}}{\sigma_*} \stackrel{d}{=} \frac{1}{\sqrt{\kappa}} Z_j + o_p(1), \quad (\text{S82})$$

and the first part of the Theorem follows. The general result follows from (S82) and the rotational invariance of the Gaussian distribution in conjunction with Lemma S7. The argument is completely similar to the one provided in the proof of Theorem 3.1 in Zhao *et al.* (2022) starting below (A.6) and is thus omitted.

S7 Proof of Theorem 3.4

Theorem 3.4. Let $\hat{\boldsymbol{\beta}}^{\text{DY}}$ be the MDYPL estimator for a logistic regression model with i.i.d. $\text{N}(\mathbf{0}_p, \boldsymbol{\Sigma})$ covariates.

Assume that (α, κ, γ) are such that (4) admits a solution (μ_*, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is non-singular.

Consider a fixed subset of indices $I = \{i_1, \dots, i_k\}$, and the DY prior penalized likelihood ratio test statistic

$$\Lambda_I = \max_{\boldsymbol{\beta} \in \mathbb{R}^p} \ell(\boldsymbol{\beta}; \mathbf{y}^*, \mathbf{X}) - \max_{\substack{\boldsymbol{\beta} \in \mathbb{R}^p: \\ \beta_j = 0, j \in I}} \ell(\boldsymbol{\beta}; \mathbf{y}^*, \mathbf{X}).$$

Then, under the null that $\beta_{0,i_1} = \dots = \beta_{0,i_k} = 0$, Λ_I is asymptotically distributed as

$$2\Lambda_I \xrightarrow{d} \frac{\kappa \sigma_*^2}{b_*} \chi_k^2,$$

where χ_k^2 is a Chi-squared random variable with k degrees of freedom.

This follows immediately from Theorem S4 and Lemma S7 as in the discussion of Zhao *et al.* (2022, Section 4).

Theorem S4. Let $\hat{\boldsymbol{\beta}}^{\text{DY}}$ be the MDYPL estimator for a logistic regression model with i.i.d. $\text{N}(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$ covariates and signal $\boldsymbol{\beta}_0$. Assume that (α, κ, γ) are such that (S2)-(S4) admit a solution (μ_*, b_*, σ_*) such that $\mathbf{J}(\mu_*, b_*, \sigma_*)$ is nonsingular. Consider a fixed subset of indices $I = \{i_1, \dots, i_k\}$, and the DY prior

penalized likelihood ratio test statistic

$$\Lambda_I = \max_{\beta \in \mathbb{R}^p} \ell(\beta; \mathbf{y}^*, \mathbf{X}) - \max_{\substack{\beta \in \mathbb{R}^p: \\ \beta_j = 0, j \in I}} \ell(\beta; \mathbf{y}^*, \mathbf{X}).$$

Then, under the null that $\beta_{0,i_1} = \dots = \beta_{0,i_k} = 0$, Λ_I is asymptotically distributed as

$$2\Lambda_I \xrightarrow{d} \frac{\kappa \sigma_*^2}{b_*} \chi_k^2,$$

where χ_k^2 is a Chi-squared random variable with k degrees of freedom.

The proof of Theorem S4 is analogous the proof of Theorem 3 in Sur and Candès (2019) where the MLE is replaced with the MDYPL estimator, the state evolutions in (5) are replaced by (S2)-(S4) and the standard logistic regression log-likelihood $\ell(\beta; \mathbf{y}, \mathbf{X})$ is replaced by the log-likelihood with pseudo responses $\mathbf{y}^*$, i.e. $\ell(\beta; \mathbf{y}^*, \mathbf{X})$ and their Theorem 2 is replaced by Theorem 3.1. A line by line analysis reveals that the arguments in the proof of Sur and Candès (2019) go through with minuscule adjustments. As the proof is lengthy and technical, and there are no noteworthy contributions in the adaptation to the MDYPL estimator, it is omitted.

S8 Boundedness of the MDYPL estimator

Theorem S5. *Let*

$$F(\kappa) = \mathbb{E}[(W - \tau_\kappa)_+], \quad (\text{S83})$$

with $(x)_+ = \max\{0, x\}$, $W = |Z|$, $Z \sim \mathcal{N}(0, 1)$ and where τ_κ is the nonnegative root of

$$\mathbb{E}[\min\{W^2, \tau_\kappa^2\}] = \kappa. \quad (\text{S84})$$

and let $\hat{\beta}^{\text{DY}}$ be the MDYPL estimator for a logistic regression model with i.i.d. $\mathcal{N}(\mathbf{0}_p, n^{-1} \mathbf{I}_p)$ covariates. Then,

- (i) For all $\kappa \in (0, 1)$, $F(\kappa) \in (0, \sqrt{2/\pi})$ and a unique nonnegative solution τ_κ to (S84) exists
- (ii) For any $\kappa \in (0, 1)$, $\alpha \in (0, 1)$, $\gamma > 0$,

$$\limsup_{n \rightarrow \infty} \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} \leq \frac{2 \log(2)}{1 - \alpha} \{F(\kappa)\}^{-1},$$

almost surely.

Proof. The first claim is the content of Lemma S21. Hence, we will now prove the second part of the theorem. By optimality of $\hat{\beta}^{\text{DY}}$, and by (S1),

$$-\ell^*(\hat{\beta}^{\text{DY}}; \mathbf{y}^*, \mathbf{X}) \leq \ell^*(\mathbf{0}_p; \mathbf{y}^*, \mathbf{X}) = \frac{1}{\alpha} n \log(2). \quad (\text{S85})$$

Write $\mathbf{X} = \mathbf{Z}/\sqrt{n}$, where $\mathbf{Z}$ is a $n \times p$ matrix with i.i.d. $N(0, 1)$ entries. By (S85) and Lemma S22,

$$\begin{aligned}
\frac{1}{\alpha} n \log(2) &\geq -\ell^*(\hat{\beta}^{\text{DY}}; \mathbf{y}^*, \mathbf{X}) \\
&> \frac{1-\alpha}{2\alpha} \sum_{i=1}^n |\mathbf{x}_i^\top \hat{\beta}^{\text{DY}}| \\
&= \frac{1-\alpha}{2\alpha} \|\mathbf{X} \hat{\beta}^{\text{DY}}\|_1 \\
&= \frac{1-\alpha}{2\alpha} \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} \left\| \mathbf{Z} \frac{\hat{\beta}^{\text{DY}}}{\|\hat{\beta}^{\text{DY}}\|_2} \right\|_1 \\
&= \frac{1-\alpha}{2\alpha} \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} n \left\| \frac{1}{n} \mathbf{Z} \frac{\hat{\beta}^{\text{DY}}}{\|\hat{\beta}^{\text{DY}}\|_2} \right\|_1 \\
&\geq \frac{1-\alpha}{2\alpha} \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} n \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \left\| \frac{1}{n} \mathbf{Z} \mathbf{u} \right\|_1.
\end{aligned} \tag{S86}$$

Let N_1 be the threshold after which Lemma S10 applies and let N_2 be the smallest $n \in \mathbb{N}$ for which

$$\sqrt{\frac{2 \log(n)}{cn}} < F(\kappa),$$

where c is the universal constant of Lemma S10. Then by Lemma S10, for $n > \max\{N_1, N_2\}$ we have that

$$\Pr \left(\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \|\mathbf{Z} \mathbf{u}\|_1 < F(\kappa) - \sqrt{\frac{2 \log(n)}{cn}} \right) \leq 4n^{-2}.$$

Therefore, using equation (S86), for $n > N$, with probability at least $1 - 4n^{-2}$,

$$\frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} \leq \frac{2 \log(2)}{1-\alpha} \left\{ F(\kappa) - \sqrt{\frac{2 \log(n)}{cn}} \right\}^{-1}.$$

By Borel-Cantelli (see, for example, Shiryaev 2016, p.308), we conclude that

$$\limsup_{n \rightarrow \infty} \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{n}} \leq \limsup_{n \rightarrow \infty} \frac{2 \log(2)}{1-\alpha} \left\{ F(\kappa) - \sqrt{\frac{2 \log(n)}{cn}} \right\}^{-1} = \frac{2 \log(2)}{1-\alpha} \{F(\kappa)\}^{-1}.$$

This concludes the proof. $\square$

Remark 1. We remark, that, in line with the results of Sur and Candès (2019), Candès and Sur (2020), as α approaches one, the upper bound on $\|\hat{\beta}^{\text{DY}}\|_2/\sqrt{n}$ approaches infinity. Similarly, as κ approaches one, our upper bound approaches infinity, in line with the existence of $\hat{\beta}^{\text{DY}}$ which is only guaranteed if $p < n$ (Rigon and Aliverti, 2023).

Lemma S10. *Let $\mathbf{Z}$ be a $n \times p$ matrix with i.i.d. $N(0, 1)$ entries and $p/n \rightarrow \kappa \in (0, 1)$ as $n \rightarrow \infty$. Then*

there exists a $N \in \mathbb{N}$ such that for all $n > N$ and for any $\epsilon > 0$,

$$\Pr \left(\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \|\mathbf{Z}\mathbf{u}\|_1 < F(\kappa) - \epsilon \right) \leq 4 \exp\{-c n \epsilon^2 / 4\},$$

for some universal constant $c > 0$ and $F(\kappa)$ as defined in (S83).

Proof. **Dual norm**

Note that

$$\|\mathbf{u}\|_1 = \sup_{\|\mathbf{v}\|_\infty \leq 1} \mathbf{u}^\top \mathbf{v}.$$

Therefore, let us consider

$$\Psi_n = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{v}^\top \mathbf{Z}\mathbf{u}. \quad (\text{S87})$$

The constraint sets of (S87) are compact, so the Gaussian min max theorem applies (see, for example, Thrampoulidis *et al.* 2015, Theorem 1). In particular, let us define the auxiliary optimisation (AO)

$$\psi_n = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \|\mathbf{u}\|_2 \mathbf{h}^\top \mathbf{v} + \|\mathbf{v}\|_2 \mathbf{g}^\top \mathbf{u} = \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{h}^\top \mathbf{v} + \|\mathbf{v}\|_2 \mathbf{g}^\top \mathbf{u},$$

where $\mathbf{g} \in \mathbb{R}^p, \mathbf{h} \in \mathbb{R}^n$ are independent random vectors with i.i.d. $N(0, 1)$ entries. We therefore have that

$$\Pr(\Psi_n < t) \leq 2 \Pr(\psi_n < t).$$

Analysing the AO

We shall first find the minimum over $\mathbf{u}$ in the AO. To this end, let

$$h(c) = \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{h}^\top \mathbf{v} + c \|\mathbf{v}\|_2,$$

so that

$$\psi_n = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} h(\mathbf{g}^\top \mathbf{u}).$$

Now note that $h(c)$ is nondecreasing in c for any $\mathbf{v}$. Indeed, for $c > c'$, we have that

$$\mathbf{h}^\top \mathbf{v} + c \|\mathbf{v}\|_2 \geq \mathbf{h}^\top \mathbf{v} + c' \|\mathbf{v}\|_2,$$

and the supremum must preserve this inequality. Further note that by Cauchy-Schwarz,

$$\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \mathbf{g}^\top \mathbf{u} = -\|\mathbf{g}\|_2,$$

with minimiser $\mathbf{u} = -\mathbf{g}/\|\mathbf{g}\|_2$. Hence, we have that

$$\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} h(\mathbf{g}^\top \mathbf{u}) = h \left(\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \mathbf{g}^\top \mathbf{u} \right) = h(-\|\mathbf{g}\|_2),$$

and

$$\psi_n = \frac{1}{n} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n; \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{h}^\top \mathbf{v} - \|\mathbf{v}\|_2 \|\mathbf{g}\|_2. \quad (\text{S88})$$

We shall now reparameterise the simplified AO of (S88). To this end, let $v_i = \text{sign}(h_i)a_i$, for $a_i \in [0, 1]$ and let $\mathbf{a} = [a_1, \dots, a_n]^\top$. Further, let $w_i = |h_i|$ and $\lambda_n = \|\mathbf{g}\|_2/\sqrt{n}$. Then (S88) becomes

$$\psi_n = \sup_{\mathbf{a} \in [0, 1]^n} \frac{1}{n} \sum_{i=1}^n w_i a_i - \lambda_n \sqrt{\frac{1}{n} \sum_{i=1}^n a_i^2}. \quad (\text{S89})$$

Let

$$f(\mathbf{a}) = \frac{1}{n} \sum_{i=1}^n \{w_i a_i\} - \lambda_n s(\mathbf{a}), \quad s(\mathbf{a}) = \sqrt{\frac{1}{n} \sum_{i=1}^n a_i^2}.$$

Then maximising $f(\mathbf{a})$ is equivalent to minimising $g(\mathbf{a}) = -f(\mathbf{a})$. Note that g is differentiable on $\mathbb{R}^n \setminus \{0\}$. For $\mathbf{a} = \mathbf{0}_n$, $s(\mathbf{a})$ is not differentiable but has subdifferential

$$\partial s(\mathbf{0}_n) = \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_2 \leq 1/\sqrt{n}\}.$$

The Lagrangian of this minimisation (S89) is given by

$$\mathcal{L}(\mathbf{a}, \boldsymbol{\mu}, \boldsymbol{\nu}) = g(\mathbf{a}) + \sum_{i=1}^n \mu_i (a_i - 1) - \sum_{i=1}^n \nu_i a_i.$$

Since g is convex and Slater holds (take any $\mathbf{a} \in (0, 1)^n$), the KKT conditions are necessary and sufficient and any KKT point is optimal for g , i.e. maximises f (see, for example, Beck 2017, Theorem 3.78). Hence, let us find such a solution. The KKT conditions are:

1. Stationarity: For $\mathbf{a} \neq \mathbf{0}_n$:

$$\frac{\partial \mathcal{L}(\mathbf{a}, \boldsymbol{\mu}, \boldsymbol{\nu})}{\partial a_i} = -\frac{w_i}{n} + \lambda_n \frac{a_i}{n \cdot s(\mathbf{a})} + \mu_i - \nu_i = 0.$$

and for $\mathbf{a} = \mathbf{0}_n$ there must exist a $\mathbf{u} \in \partial s(\mathbf{0}_n)$ such that

$$\lambda_n \mathbf{u}_i - \frac{w_i}{n} + \mu_i - \nu_i = 0.$$

2. Primal feasibility: $a_i \in [0, 1]$ ($i = 1, \dots, n$)
3. Dual Feasibility: $\mu_i, \nu_i \geq 0$ ($i = 1, \dots, n$)
4. Complementary slackness: $\mu_i(a_i - 1) = 0$, $\nu_i a_i = 0$

Consider an optimal $\mathbf{a}$ and let $\tau = \lambda_n/s(\mathbf{a})$. We consider two cases:

1. $\mathbf{a} \in (0, 1)^n$: By complementary slackness, $\mu_i, \nu_i = 0$. Hence, by stationarity, $a_i = w_i/\tau$, which is less than 1 and thus primal feasible whenever $w_i < \tau$.
2. $\mathbf{a} = \mathbf{1}$: By complementary slackness, $\nu_i = 0$ and by stationarity $\mu_i = \frac{w_i - \tau}{n}$ which is weakly greater than zero and thus dual feasible whenever $w_i \geq \tau$.
3. $\mathbf{a} = \mathbf{0}_n$: By complementary slackness, $\mu_i = 0$. By stationarity we need $\mathbf{u}$ and $\boldsymbol{\nu}$ such that

$$\lambda_n u_i = \frac{w_i}{n} + \nu_i, \quad \|\mathbf{u}\|_2 \leq \frac{1}{\sqrt{n}}, \quad \nu_i \geq 0.$$

Let us first focus on the corner case $\mathbf{a} = \mathbf{0}_n$. Consider the candidate solution $\mathbf{v} = \mathbf{0}_n$ and $\mathbf{u} = \mathbf{w}/(n\lambda_n)$. Note that

$$\begin{aligned} \frac{1}{n} \left\| \frac{\mathbf{w}}{\lambda_n} \right\|_2 &= \frac{1}{\sqrt{n}} \frac{\|\mathbf{w}\|_2}{\|\mathbf{g}\|_2} \\ &= \frac{1}{\sqrt{n}} \frac{\|\mathbf{h}\|_2}{\|\mathbf{g}\|_2}, \end{aligned} \quad (\text{S90})$$

so that $\|\mathbf{u}\|_2 \leq 1/\sqrt{n}$ if and only if $\|\mathbf{h}\|_2 \leq \|\mathbf{g}\|_2$.

Now $\mathbf{a} = \mathbf{0}_n$ is optimal if and only if the KKT conditions are satisfied. (S90) shows that if $\|\mathbf{h}\|_2 \leq \|\mathbf{g}\|_2$, then a solution to the KKT conditions exists and $\mathbf{a} = \mathbf{0}_n$ is optimal. On the other hand, we have by Cauchy-Schwarz that

$$\mathbf{h}^\top \mathbf{v} - \|\mathbf{v}\|_2 \|\mathbf{g}\|_2 \leq (\|\mathbf{h}\|_2 - \|\mathbf{g}\|_2) \|\mathbf{v}\|_2,$$

and that this upper bound can be attained for $\mathbf{v} = \mathbf{h}/\|\mathbf{h}\|_2$. Note also that for this $\mathbf{v}$, we have $\|\mathbf{v}\|_\infty \leq 1$ since

$$|v_i| = \frac{|h_i|}{\sqrt{\sum_{i=1}^n h_i^2}} \leq 1 \iff h_i^2 \leq \sum_{i=1}^n h_i^2.$$

Thus there exists a feasible solution with value $\|\mathbf{h}\|_2 - \|\mathbf{g}\|_2$, which is strictly positive if $\|\mathbf{h}\|_2 > \|\mathbf{g}\|_2$. So, $\mathbf{a} = \mathbf{0}_n$ is optimal if and only if $\|\mathbf{g}\|_2 \geq \|\mathbf{h}\|_2$. But note that $\|\mathbf{g}\|_2/\sqrt{p} \rightarrow 1$ and $\|\mathbf{h}\|_2/\sqrt{n} \rightarrow 1$ almost surely (see, for example, Vershynin 2018, Theorem 1.3.1). Hence, for $\kappa \in (0, 1)$, there exists a $N \in \mathbb{N}$ such that for all $n > N$, ψ_n admits an interior solution with probability one.

Thus, we consider such an interior solution next. For fixed τ , we get the thresholding rule

$$a_i(\tau) = \min\{1, w_i/\tau\}. \quad (\text{S91})$$

To make $\mathbf{a}(\tau)$ self-consistent, we note that

$$\|\mathbf{a}(\tau)\|_2^2 = \sum_{i=1}^n \min\{1, w_i^2/\tau^2\}$$

Therefore, the relation $\tau = \lambda_n/s(\mathbf{a})$, yields, for the thresholding rule (S91), that

$$\begin{aligned} \tau &= \frac{\lambda_n}{\sqrt{\frac{1}{n} \|\mathbf{a}(\tau)\|_2^2}} \\ \iff \sqrt{\frac{1}{n} \tau^2 \sum_{i=1}^n \min\{1, w_i^2/\tau^2\}} &= \lambda_n \\ \iff \sqrt{\frac{1}{n} \sum_{i=1}^n \min\{\tau^2, w_i^2\}} &= \lambda_n \\ \iff \frac{1}{n} \sum_{i=1}^n \min\{\tau^2, w_i^2\} &= \lambda_n^2 \end{aligned}$$

Thus define $H_n(\tau) = n^{-1} \sum_{i=1}^n \min\{\tau^2, w_i^2\}$. We have therefore shown that if there exists a τ_n such that

$$H_n(\tau_n) = \lambda_n^2, \quad (\text{S92})$$

then the thresholding function $a_i(\tau_n) = \min\{1, w_i/\tau_n\}$ satisfies the KKT conditions.

Next, we must turn to the existence of a solution τ_n to (S92). First, note that $H_n(\tau)$ is nondecreasing and is strictly increasing on $[0, \max_i w_i]$. It is further continuous in τ with $H_n(0) = 0$ and

$\lim_{\tau \rightarrow \infty} H_n(\tau) = n^{-1} \sum_{i=1}^n w_i^2$. Now, since $w_i = |h_i|$,

$$\frac{1}{n} \sum_{i=1}^n w_i^2 \xrightarrow{\text{a.s.}} 1, \quad \text{as } n \rightarrow \infty.$$

Further,

$$\lambda_n^2 = \frac{\|\mathbf{g}\|_2^2}{n} \xrightarrow{\text{a.s.}} \kappa \in (0, 1), \quad \text{as } n \rightarrow \infty.$$

Therefore, there is a $N \in \mathbb{N}$ such that for all $n > N$, with probability one, there exists a unique solution τ_n . For such a solution τ_n we have the AO value

$$\begin{aligned} \psi_n &= \frac{1}{n} \sum_{i=1}^n w_i \min\{1, w_i/\tau_n\} - \lambda_n \sqrt{\frac{1}{n} \sum_{i=1}^n \min\{1, w_i^2/\tau_n^2\}} \\ &= \frac{1}{n} \sum_{i=1}^n w_i \min\{1, w_i/\tau_n\} - \frac{\lambda_n}{\sqrt{\frac{1}{n} \sum_{i=1}^n \min\{1, w_i^2/\tau_n^2\}}} \frac{1}{n} \sum_{i=1}^n \min\{1, w_i^2/\tau_n^2\} \\ &= \frac{1}{n} \sum_{i=1}^n w_i \min\{1, w_i/\tau_n\} - \tau_n \frac{1}{n} \sum_{i=1}^n \min\{1, w_i^2/\tau_n^2\} \\ &= \frac{1}{n} \sum_{i=1}^n (w_i - \tau_n)_+, \end{aligned}$$

with τ_n solving the equation

$$H_n(\tau) - \lambda_n^2 = 0.$$

Population AO

Consider the population AO

$$\psi = \mathbb{E}[(W - \tau_\kappa)_+],$$

where $W = |Z|$, $Z \sim \mathcal{N}(0, 1)$, and τ_κ is the solution to

$$H(\tau) - \kappa = 0, \quad H(\tau) = \mathbb{E}[\min(\tau^2, W^2)]. \quad (\text{S93})$$

By Lemma S21 $H(\tau)$ is continuous, strictly increasing in τ with $H(0) = 0$ and $\lim_{\tau \rightarrow \infty} H(\tau) = 1$, so that a unique solution τ_κ to (S93) exists for all $\kappa \in (0, 1)$.

Convergence of AO to population AO

Next, we will show that the AO converges almost surely to the population AO. For this we will show the following steps:

1. $\tau_n \xrightarrow{\text{a.s.}} \tau_\kappa$
2. For $T > 0$

$$\sup_{\tau \in [0, T]} |\psi_n(\tau) - \psi(\tau)| \xrightarrow{\text{P}} 0,$$

where

$$\begin{aligned} \psi_n(\tau) &= \frac{1}{n} \sum_{i=1}^n (w_i - \tau)_+ \\ \psi(\tau) &= \mathbb{E}[(W - \tau)_+], \quad W = |Z|, \quad Z \sim \mathcal{N}(0, 1) \end{aligned}$$

$$3. \psi_n(\tau_n) \xrightarrow{P} \psi(\tau_\kappa)$$

$$4. E[\psi_n] \rightarrow \psi$$

To show almost sure convergence of τ_n to τ_κ fix an arbitrary $\epsilon > 0$ and let

$$m_\epsilon = \min\{H(\tau_\kappa + \epsilon) - H(\tau_\kappa), H(\tau_\kappa) - H(\tau_\kappa - \epsilon)\}.$$

Consequently

$$\begin{aligned} H(\tau_\kappa + \epsilon) &\geq \kappa + m_\epsilon \\ H(\tau_\kappa - \epsilon) &\leq \kappa - m_\epsilon, \end{aligned}$$

for n large enough with probability one. Further, since λ_n^2 converges almost surely to κ , we have that for n large enough, $|\lambda_n^2 - \kappa| \leq m_\epsilon/2$ with probability one.

Finally, by a SLLN (see, for example, Vershynin 2018, Theorem 1.3.1), for each fixed τ , $H_n(\tau)$ converges almost surely to $H(\tau)$. Thus, for n large enough:

$$\begin{aligned} |H_n(\tau_\kappa + \epsilon) - H(\tau_\kappa + \epsilon)| &\leq \frac{m_\epsilon}{2} \\ |H_n(\tau_\kappa - \epsilon) - H(\tau_\kappa - \epsilon)| &\leq \frac{m_\epsilon}{2}, \end{aligned}$$

simultaneously with probability one. Thus we have that for n large enough,

$$\begin{aligned} H_n(\tau_\kappa + \epsilon) &= H(\tau_\kappa + \epsilon) + [H(\tau_\kappa + \epsilon) - H_n(\tau_\kappa + \epsilon)] \\ &\geq H(\tau_\kappa + \epsilon) - \frac{m_\epsilon}{2} \\ &\geq \kappa + \frac{m_\epsilon}{2} \\ &\geq \lambda_n^2 \\ &= H_n(\tau_n) \end{aligned},$$

with probability one. Similarly

$$\begin{aligned} H_n(\tau_\kappa - \epsilon) &= H(\tau_\kappa - \epsilon) + [H(\tau_\kappa - \epsilon) - H_n(\tau_\kappa - \epsilon)] \\ &\leq H(\tau_\kappa - \epsilon) + \frac{m_\epsilon}{2} \\ &\leq \kappa - \frac{m_\epsilon}{2} \\ &\leq \lambda_n^2 \\ &= H_n(\tau_n) \end{aligned}.$$

Thus, since $H_n(\tau)$ is strictly increasing, for n large enough, with probability one,

$$\tau_n \in [\tau_\kappa - \epsilon, \tau_\kappa + \epsilon].$$

Since the choice of ϵ was arbitrary, we conclude that $\tau_n \rightarrow \tau_\kappa$ almost surely.

Uniform convergence in probability of ψ_n to ψ for τ in the closed interval $[0, T]$ follows from Newey and McFadden 1994, Lemma 2.4.

Next, since τ_n converges to τ_κ almost surely, fix any $T > \tau_\kappa$ so that $\tau_n \in [0, T]$ eventually with probability one. Further, for any $\epsilon > 0$, we have that for n large enough

$$\begin{aligned} \sup_{\tau \in [0, T]} |\psi_n(\tau) - \psi(\tau)| &\leq \frac{\epsilon}{2} \\ |\psi(\tau_n) - \psi(\tau_\kappa)| &\leq \frac{\epsilon}{2}, \end{aligned}$$

simultaneously with probability approaching one since $\psi_n(\tau)$ converges uniformly in probability to $\psi(\tau)$ on $[0, T]$ and τ_n converges almost surely to τ_κ and $\psi(\tau)$ is continuous in τ . Therefore,

$$\begin{aligned}
|\psi_n(\tau_n) - \psi(\tau_\kappa)| &= |[\psi_n(\tau_n) - \psi(\tau_n)] + [\psi(\tau_n) - \psi(\tau_\kappa)]| \\
&\leq |\psi_n(\tau_n) - \psi(\tau_n)| + |\psi(\tau_n) - \psi(\tau_\kappa)| \\
&\leq \sup_{\tau \in [0, T]} |\psi_n(\tau) - \psi(\tau)| + |\psi(\tau_n) - \psi(\tau_\kappa)| \\
&\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} \\
&= \epsilon
\end{aligned}$$

Since the choice of ϵ was arbitrary, we have established that $\psi_n \rightarrow \psi$ in probability.

Next we will show that ψ_n is uniformly integrable. To that end, note that

$$\psi_n \leq Y_n, \quad Y_n = \frac{1}{n} \sum_{i=1}^n w_i.$$

We shall show that Y_n is uniformly integrable, which in turn establishes that ψ_n is uniformly integrable (see, for example, Gut 2005, Theorem 4.5). To that end, consider

$$\begin{aligned}
\sup_{n \in \mathbb{N}} \mathbb{E}[Y_n^2] &= \sup_{n \in \mathbb{N}} \text{var}[Y_n] + \mathbb{E}[Y_n]^2 \\
&= \sup_{n \in \mathbb{N}} \frac{\text{var}[|Z|]}{n} + \frac{2}{\pi} \\
&= \sup_{n \in \mathbb{N}} \frac{1 - 2/\pi}{n} + \frac{2}{\pi} \\
&< \infty
\end{aligned}$$

for $Z \sim N(0, 1)$. Thus, $\{Y_n\}_{n \in \mathbb{N}}$ is bounded in L^2 and therefore uniformly integrable (see, for example, Gut 2005, Theorem 4.2). Hence ψ_n is uniformly integrable and converges to ψ in expectation (see, for example, Gut 2005, Theorem 5.4).

A tail bound for the AO

Next, we shall establish a tail bound for the AO. Towards this end, note that ψ_n is Lipschitz with Lipschitz constant $\sqrt{2/n}$ in $\mathbf{g}, \mathbf{h}$. Indeed, let

$$\phi_n(\mathbf{g}, \mathbf{h}) = \frac{1}{n} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{h}^\top \mathbf{v} - \|\mathbf{v}\|_2 \|\mathbf{g}\|_2.$$

Since the set $\|\mathbf{v}\|_\infty \leq 1$ is compact, and the map $\mathbf{v} \mapsto \mathbf{h}^\top \mathbf{v} - \|\mathbf{v}\|_2 \|\mathbf{g}\|_2$ is continuous, the maximum is attained. Call $\mathbf{v}^*$ a maximiser for $\mathbf{g}, \mathbf{h}$. Then, for $\mathbf{g}', \mathbf{h}'$, we have

$$\phi_n(\mathbf{g}', \mathbf{h}') \geq \frac{1}{n} \mathbf{h}'^\top \mathbf{v}^* - \|\mathbf{v}^*\|_2 \|\mathbf{g}'\|_2.$$

Consequently,

$$\begin{aligned}
\phi_n(\mathbf{g}, \mathbf{h}) - \phi_n(\mathbf{g}', \mathbf{h}') &\leq \frac{1}{n}(\mathbf{h} - \mathbf{h}')^\top \mathbf{v}^* - \frac{1}{n}\|\mathbf{v}^*\|_2(\|\mathbf{g}\|_2 - \|\mathbf{g}'\|_2) \\
&\leq \frac{1}{n}\|\mathbf{v}^*\|_2\|\mathbf{h} - \mathbf{h}'\|_2 - \frac{1}{n}\|\mathbf{v}^*\|_2(\|\mathbf{g}\|_2 - \|\mathbf{g}'\|_2) \\
&\leq \frac{1}{n}\|\mathbf{v}^*\|_2\|\mathbf{h} - \mathbf{h}'\|_2 + \frac{1}{n}\|\mathbf{v}^*\|_2(\|\mathbf{g} - \mathbf{g}'\|_2) \\
&\leq \frac{1}{\sqrt{n}}\|\mathbf{h} - \mathbf{h}'\|_2 + \frac{1}{\sqrt{n}}(\|\mathbf{g} - \mathbf{g}'\|_2) \\
&\leq \sqrt{\frac{2}{n}}\left\|\begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix} - \begin{bmatrix} \mathbf{g}' \\ \mathbf{h}' \end{bmatrix}\right\|_2
\end{aligned} \tag{S94}$$

The second and third inequality of (S94) follow by Cauchy-Schwarz, the fourth since $\|\mathbf{v}^*\|_\infty < \infty$ so that $\|\mathbf{v}^*\|_2 \leq \sqrt{n}$ and the last since $\sqrt{a} + \sqrt{b} \leq \sqrt{2(a+b)}$. Now swap the roles of $\psi_n(\mathbf{g}, \mathbf{h})$ and $\psi_n(\mathbf{g}', \mathbf{h}')$ to get

$$\phi_n(\mathbf{g}', \mathbf{h}') - \phi_n(\mathbf{g}, \mathbf{h}) \leq \sqrt{\frac{2}{n}}\left\|\begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix} - \begin{bmatrix} \mathbf{g}' \\ \mathbf{h}' \end{bmatrix}\right\|_2. \tag{S95}$$

Multiplying (S95) by -1 yields the lower bound

$$\phi_n(\mathbf{g}, \mathbf{h}) - \phi_n(\mathbf{g}', \mathbf{h}') \geq -\sqrt{\frac{2}{n}}\left\|\begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix} - \begin{bmatrix} \mathbf{g}' \\ \mathbf{h}' \end{bmatrix}\right\|_2,$$

and thus in conclusion

$$|\phi_n(\mathbf{g}, \mathbf{h}) - \phi_n(\mathbf{g}', \mathbf{h}')| \leq \sqrt{\frac{2}{n}}\left\|\begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix} - \begin{bmatrix} \mathbf{g}' \\ \mathbf{h}' \end{bmatrix}\right\|_2.$$

Now Gaussian concentration of Lipschitz functions applies (see, for example, Vershynin 2018, Theorem 5.2.2) and we have that for $t > 0$,

$$\Pr(|\psi_n - \mathbb{E}[\psi_n]| > t) \leq 2 \exp\{-cnt^2/4\},$$

for some constant $c > 0$.

A tail bound for the PO

We are now in a place to get a tail bound for the PO Ψ_n . For this consider

$$\psi_n < \psi - \epsilon \iff (\psi - \mathbb{E}[\psi_n]) + (\mathbb{E}[\psi_n] - \psi_n) > \epsilon.$$

Therefore, the event $\psi_n < \psi - \epsilon$ implies that either $\psi - \mathbb{E}[\psi_n] > \epsilon/2$ or $\mathbb{E}[\psi_n] - \psi_n > \epsilon/2$.

Hence, we have established that there exists a $N \in \mathbb{N}$ such that for all $n > N$,

$$\begin{aligned}
\Pr(\Psi_n < \psi - \epsilon) &\leq 2 \Pr(\psi_n < \psi - \epsilon) \\
&\leq 2 \Pr(\{\psi - \mathbb{E}[\psi_n] > \epsilon/2\} \cup \{\mathbb{E}[\psi_n] - \psi_n > \epsilon/2\}) \\
&\leq 2(\Pr(\psi - \mathbb{E}[\psi_n] > \epsilon/2) + \Pr(\mathbb{E}[\psi_n] - \psi_n > \epsilon/2)) \\
&= 2\mathbb{1}\{\psi - \mathbb{E}[\psi_n] > \epsilon/2\} + 2 \Pr(\mathbb{E}[\psi_n] - \psi_n > \epsilon/2) \\
&\leq 2\mathbb{1}\{\psi - \mathbb{E}[\psi_n] > \epsilon/2\} + 2 \Pr(|\psi_n - \mathbb{E}[\psi_n]| > \epsilon/2) \\
&\leq 2\mathbb{1}\{\psi - \mathbb{E}[\psi_n] > \epsilon/2\} + 4 \exp\{-cn\epsilon^2/4\}.
\end{aligned}$$

Since $E[\psi_n]$ converges to ψ , by Borel-Cantelli (see, for example, Shiryaev 2016, p.308), we have that for n large enough,

$$\Pr \left(\inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \|\mathbf{Z}\mathbf{u}\|_1 < F(\kappa) - \epsilon \right) \leq 4 \exp\{-cn\epsilon^2/4\}.$$

where $F(\kappa)$ is the value ψ of the population AO. This concludes the proof. $\square$

S9 Auxiliary Lemmas

Lemma S11. *Given $\epsilon > 0$, $\alpha \in [0, 1]$, consider the approximation $h_\epsilon(Z, \bar{\epsilon}) = \frac{\alpha}{2} \left\{ 1 - \frac{\bar{\epsilon} - \zeta'(Z)}{|\bar{\epsilon} - \zeta'(Z)| + \epsilon} \right\} + \frac{1-\alpha}{2}$ to $h(Z, \bar{\epsilon}) = \frac{\alpha}{2} \{1 - \text{sign}(\bar{\epsilon} - \zeta'(Z))\} + \frac{1-\alpha}{2}$. Then*

i) $\lim_{\epsilon \rightarrow 0} h_\epsilon(Z, \bar{\epsilon}) = h(Z, \bar{\epsilon})$ for all $Z, \bar{\epsilon}$

ii) $h_\epsilon(Z, \bar{\epsilon})$ is Lipschitz in $[Z, \bar{\epsilon}]^\top$

iii) $\frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon})$ exists for all $Z, \bar{\epsilon} \in \mathbb{R}$, $\epsilon > 0$ and is Lipschitz in $[Z, \bar{\epsilon}]^\top$.

Proof. i) If $\bar{\epsilon} - \zeta'(Z) = 0$, then $h_\epsilon(Z, \bar{\epsilon}) = 0$ for all $\epsilon > 0$. If $\bar{\epsilon} - \zeta'(Z) \neq 0$, then $\lim_{\epsilon \rightarrow 0} h_\epsilon(Z, \bar{\epsilon}) = (\bar{\epsilon} - \zeta'(Z))/|\bar{\epsilon} - \zeta'(Z)| = \text{sign}(\bar{\epsilon} - \zeta'(Z))$. The claim follows.

ii) First it is shown that $f(x) = x/(|x| + \epsilon)$ is Lipschitz for any $\epsilon > 0$ with Lipschitz constant ϵ^{-1} . To that end, note that $\frac{\partial}{\partial x} f(x) = \epsilon/(|x| + \epsilon)^2 \leq \epsilon^{-1}$ and by the mean value theorem (e.g. Rudin 1976, Theorem 5.10),

$$|f(x) - f(y)| \leq \sup_{z \in \mathbb{R}} \left| \frac{\partial}{\partial z} f(z) \right| |x - y| \leq \frac{1}{\epsilon} |x - y|.$$

Further, note that $\zeta''(x)$ is positive and bounded from above by $1/4$. Hence,

$$\begin{aligned} |h_\epsilon(Z, \bar{\epsilon}) - h_\epsilon(Z', \bar{\epsilon}')| &= \frac{\alpha}{2} |f(\bar{\epsilon} - \zeta'(Z)) - f(\bar{\epsilon}' - \zeta'(Z'))| \\ &\leq \frac{\alpha}{2\epsilon} |\bar{\epsilon} - \zeta'(Z) - (\bar{\epsilon}' - \zeta'(Z'))| \\ &\leq \frac{\alpha}{2\epsilon} \{|\bar{\epsilon} - \bar{\epsilon}'| + |\zeta'(Z) - \zeta'(Z')|\} \\ &\leq \frac{\alpha}{2\epsilon} \left\{ |\bar{\epsilon} - \bar{\epsilon}'| + \frac{1}{4} |Z - Z'| \right\} \\ &\leq \frac{\alpha}{8\epsilon} \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'|\} \\ &\leq \frac{\alpha\sqrt{2}}{8\epsilon} \left\| \begin{bmatrix} Z \\ \bar{\epsilon} \end{bmatrix} - \begin{bmatrix} Z' \\ \bar{\epsilon}' \end{bmatrix} \right\|_2, \end{aligned}$$

where the first inequality follows by Lipschitz continuity of $f(\cdot)$, the second by the triangle inequality, the third by the Mean Value Theorem and since $\sup_{x \in \mathbb{R}} |\zeta''(x)| \leq 1/4$, and the last since $(|a| + |b|)^2 \leq 2(|a|^2 + |b|^2)$.

iii) Existence follows by differentiability of $f(x)$ and the chain rule,

$$\frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) = \frac{\alpha}{2} \frac{\epsilon}{(|\bar{\epsilon} - \zeta'(Z)| + \epsilon)^2} \zeta''(Z). \quad (\text{S96})$$

Next, it is shown that $f'(x) = \frac{\partial}{\partial x}x/(|x| + \epsilon) = \epsilon/(|x| + \epsilon)^2$ is Lipschitz in x .

Case 1: $x, y > 0$: For $x > 0$, the function simplifies to

$$f'(x) = \frac{\epsilon}{(x + \epsilon)^2}.$$

Differentiating with respect to x yields

$$f''(x) = \frac{-2\epsilon}{(x + \epsilon)^3},$$

which exists for all $x > 0$, and it is continuous and monotonically increasing in x . Furthermore, as x approaches infinity, $f''(x)$ approaches 0, and as x approaches 0 from the right, $f'(x)$ approaches $-2/\epsilon^2$. Thus, $|f''(x)|$ is bounded by $\frac{2}{\epsilon^2}$ for all $x > 0$.

By the Mean Value Theorem it follows that

$$|f'(x) - f'(y)| \leq \frac{2}{\epsilon^2}|x - y|.$$

Case 2: $x > 0, y < 0$: Note that

$$|f'(x) - f'(y)| = \left| \frac{\epsilon}{(x + \epsilon)^2} - \frac{\epsilon}{(-y + \epsilon)^2} \right| = \left| \frac{\epsilon}{(x + \epsilon)^2} - \frac{\epsilon}{(z + \epsilon)^2} \right| = |f'(x) - f'(z)|,$$

for $z = -y > 0$. Thus analogous to Case 1,

$$|f'(x) - f'(z)| \leq \frac{2}{\epsilon^2}|x - z| < \frac{2}{\epsilon^2}|x + z| = \frac{2}{\epsilon^2}|x - y|,$$

since $z = -y > 0, x > 0$.

Case 3: $x > 0, y = 0$

The difference $|f'(x) - \epsilon^{-1}|$ simplifies to

$$\frac{x^2 + 2\epsilon x}{x^2 + 2\epsilon x + \epsilon^2}.$$

This expression is less than Cx if and only if

$$x^2 + 2\epsilon x \leq Cx^3 + 2C\epsilon x^2 + C\epsilon^2 x.$$

A sufficient condition for this inequality is $C > 2/\epsilon$.

Case 4: $x, y < 0$

By symmetry, the function behaves the same way for $x, y < 0$ as it does for $x, y > 0$. Thus, the Lipschitz constant derived for $x, y > 0$ also applies here.

This establishes that $f'(x)$ is indeed Lipschitz in x for all $x \in \mathbb{R}$. Claim (iii) follows since $\zeta''(x)$ is Lipschitz in x and $\bar{\epsilon} - \zeta'(Z)$ is Lipschitz in both $Z, \bar{\epsilon}$ and by (S96). $\square$

Lemma S12. *The following facts hold for the proximal operator $\text{prox}_{b\zeta}(x)$ and the function $\Psi(y, s) = b(y - \text{prox}_{b\zeta}(by + s))$:*

i) *The proximal operator $\text{prox}_{b\zeta}(x)$ is nonexpansive, i.e. for $x, x' \in \mathbb{R}$:*

$$|\text{prox}_{b\zeta}(x) - \text{prox}_{b\zeta}(x')| \leq |x - x'|.$$

ii) *$\Psi(h_\epsilon(Z, \bar{\epsilon}), s) = b(h_\epsilon(Z, \bar{\epsilon}) - \zeta'(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s)))$ is Lipschitz continuous in $[Z, s, \bar{\epsilon}]^\top$.*

iii) $\Psi(y, s) = b(y - \zeta'(\text{prox}_{b\zeta}(by + s)))$ is Lipschitz continuous in s with Lipschitz constant at most 1.

iv) $\frac{\partial}{\partial Z}\Psi(h_\epsilon(Z, \bar{\epsilon}), s)$ exists for all $Z, \bar{\epsilon}, s \in \mathfrak{R}$, $\epsilon > 0$ and is Lipschitz

Proof. i) Note that $\zeta(x)$ is proper, closed and convex (see e.g. Beck 2017, Chapter 2, for the definition). Indeed, properness of $\zeta(x) : \mathfrak{R} \rightarrow [\infty, \infty]$ follows since it is monotonically increasing in x , $\lim_{x \rightarrow \infty} \zeta(x) = 0$ and $f(0) = 1$. Closedness follows since lower semicontinuity is equivalent to closedness (e.g. Beck 2017, Theorem 2.6) and $\zeta(x)$ is clearly continuous and therefore closed. Hence the claim follows by nonexpansivity of the proximal operator for proper, closed and convex functions (see for example Beck 2017, Theorem 6.4.2b)), for $x, x' \in \mathfrak{R}$,

$$|\text{prox}_{b\zeta}(x) - \text{prox}_{b\zeta}(x')| \leq |x - x'|.$$

ii) It is shown in Lemma S11 ii) that the approximation h_ϵ is Lipschitz.

By nonexpansivity of the proximal operator, which is shown in Lemma S12 i),

$$|\text{prox}_{b\zeta}(x) - \text{prox}_{b\zeta}(x')| \leq |x - x'|.$$

Now letting $x = bh_\epsilon(Z, \bar{\epsilon}) + s$, $x' = bh_\epsilon(Z', \bar{\epsilon}') + s'$ then by Lipschitzianity of h_ϵ , and the triangle inequality and Jensen's inequality, again

$$\begin{aligned} |\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s) - \text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s')| &\leq |b(h_\epsilon(Z, \bar{\epsilon}) - h_\epsilon(Z', \bar{\epsilon}')) + (s - s')| \\ &\leq bL \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'| + |s - s'|\} \\ &\leq \max\{bL, 1\} \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'| + |s - s'|\} \\ &\leq \sqrt{3} \max\{bL, 1\} \|[Z, \bar{\epsilon}, s]^\top - [Z', \bar{\epsilon}', s']^\top\|_2, \end{aligned}$$

for some constant L . Hence, again by boundedness of $\frac{\partial^2}{\partial x^2}\zeta(x)$, it immediately follows that for any $t \in [q]$, $\mathbf{x}, \mathbf{x}'\mathbf{y}, \mathbf{y}' \in \mathfrak{R}^q$, and $b > 0$, note that

$$\begin{aligned} |\Psi(h_\epsilon(Z, \bar{\epsilon}), s) - \Psi(h_\epsilon(Z', \bar{\epsilon}'), s')| &\leq b|h_\epsilon(Z, \bar{\epsilon}) - h_\epsilon(Z', \bar{\epsilon}')| \\ &\quad + b|\zeta'(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s)) - \zeta'(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))| \\ &\leq bL \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'| + |s - s'|\} \\ &\quad + \frac{b}{4} \max\{bL, 1\} \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'| + |s - s'|\} \\ &= \left(bL + \frac{b}{4} \max\{bL, 1\}\right) \{|\bar{\epsilon} - \bar{\epsilon}'| + |Z - Z'| + |s - s'|\} \\ &\leq \sqrt{3} \left(bL + \frac{b}{4} \max\{bL, 1\}\right) \|[Z, s, \bar{\epsilon}]^\top - [Z', s', \bar{\epsilon}']^\top\|_2. \end{aligned}$$

iii) From Donoho and Montanari (2016), Proposition 6.3,

$$\frac{\partial}{\partial s}\Psi(y, s) = -\frac{b\zeta''(\text{prox}_{b\zeta}(by + s))}{1 + b\zeta''(\text{prox}_{b\zeta}(by + s))}, \quad (\text{S97})$$

which is less than one in absolute value and the claim follows by the mean value theorem.

iv) By Proposition 6.3 of Donoho and Montanari (2016), the proximal operator $\text{prox}_{b\zeta}(x)$ is differentiable for $b > 0$, $x \in \mathbb{R}$ with partial derivative

$$\frac{\partial}{\partial x} \text{prox}_{b\zeta}(x) = \frac{1}{1 + b\zeta''(z)} \Big|_{z=\text{prox}_{b\zeta}(x)}.$$

Consequently,

$$\frac{\partial}{\partial x} \zeta'(\text{prox}_{b\zeta}(x)) = \frac{b\zeta''(z)}{1 + b\zeta''(z)} \Big|_{z=\text{prox}_{b\zeta}(x)},$$

and thus using Lemma S11 iii), and the chain rule,

$$\begin{aligned} \frac{\partial}{\partial Z} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) &= \frac{\partial}{\partial y} \Psi(y, s) \Big|_{y=h_\epsilon(Z, \bar{\epsilon})} \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \\ &= b \left(1 - \frac{b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} \right) \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \\ &= \frac{b}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \end{aligned}$$

Thus by simple expansion and the triangle inequality,

$$\begin{aligned} &\left| \frac{\partial}{\partial Z} \Psi(h_\epsilon(Z, \bar{\epsilon}), s) - \frac{\partial}{\partial Z} \Psi(h_\epsilon(Z', \bar{\epsilon}'), s') \right| \\ &= b \left| \left\{ \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} - \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right\} \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right. \\ &\quad \left. + \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \left\{ \frac{\partial}{\partial Z} h_\epsilon(Z', \bar{\epsilon}') - \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right\} \right| \\ &\leq b \left| \frac{b}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} - \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right| \left| \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right| \\ &\quad + b \left| \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right| \left| \frac{\partial}{\partial Z} h_\epsilon(Z', \bar{\epsilon}') - \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right|. \quad (\text{S98}) \end{aligned}$$

Now it is easily seen that the function $f(x) = 1/(1+x)$ is Lipschitz over the nonnegative real line as it is differentiable and its derivative $\frac{\partial}{\partial x} f(x) = -1/(1+x)^2$ is bounded from above by 1 in absolute value for $x \geq 0$. Further, it is easily seen that $\frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon})$ is bounded in absolute value. Hence, using boundedness of $\zeta'''(x)$, Lemma S11 ii) and Lemma S12 i), it follows immediately that

$$\begin{aligned} &b \left| \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} - \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right| \left| \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right| \\ &\leq bC \left| \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s))} - \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right| \\ &\leq b^2 C C' |\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s)) - \zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))| \\ &\leq b^2 C C' C'' |\text{prox}_{b\zeta}(bh_\epsilon(Z, \bar{\epsilon}) + s) - \text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s')| \\ &\leq b^2 C C' C'' C''' \|[Z, s, \bar{\epsilon}]^\top - [Z', s', \bar{\epsilon}']^\top\|, \end{aligned}$$

for some positive constants C, C', C'', C''' . Similarly, by nonnegativity of ζ'' and b , as well as

Lemma S11 iii),

$$\begin{aligned}
b \left| \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(bh_\epsilon(Z', \bar{\epsilon}') + s'))} \right| & \left| \frac{\partial}{\partial Z} h_\epsilon(Z', \bar{\epsilon}') - \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right| \\
& \leq bC \left| \frac{\partial}{\partial Z} h_\epsilon(Z', \bar{\epsilon}') - \frac{\partial}{\partial Z} h_\epsilon(Z, \bar{\epsilon}) \right| \\
& \leq bCC' \|[Z, s, \bar{\epsilon}]^\top - [Z', s', \bar{\epsilon}']^\top\|,
\end{aligned}$$

for some positive constants C, C' . Along with (S98) this concludes the proof. $\square$

Lemma S13. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have i.i.d. entries $x_{ij} \sim N(0, n^{-1})$ and let $\lim_{n \rightarrow \infty} p/n = \kappa \in (0, 1)$. Then almost surely, $\|\mathbf{X}\|_2 < \infty$.

Proof. This follows directly from Rudelson and Vershynin 2010, Theorem 2.1. $\square$

Lemma S14. For $i = 1, \dots, n$, let $\bar{\epsilon}_i$ be i.i.d. $U(0, 1)$ random variables and $\boldsymbol{\eta}_{0,i}$ be i.i.d. $N(0, \gamma^2)$ such that $\bar{\epsilon}_i \perp \boldsymbol{\eta}_{0,j}$ for $i, j = 1, \dots, n$. Further, let

$$y_i^* = \alpha \mathbb{1}\{\bar{\epsilon}_i < \zeta'(\boldsymbol{\eta}_{0,i})\} + \frac{1 - \alpha}{2}, \quad \epsilon y_i^* = \alpha \left\{ 1 - \frac{\bar{\epsilon}_i - \zeta'(\boldsymbol{\eta}_{0,i})}{|\bar{\epsilon}_i - \zeta'(\boldsymbol{\eta}_{0,i})| + \epsilon} \right\} + \frac{1 - \alpha}{2}.$$

Then for $\mathbf{y}^* = (y_1^*, \dots, y_n^*)$, $\epsilon \mathbf{y}^* = (\epsilon y_1^*, \dots, \epsilon y_n^*)$,

$$i) \lim_{n \rightarrow \infty} \frac{1}{p} \|\mathbf{y}^* - \epsilon \mathbf{y}^*\|_2^2 = E[(y_i^* - \epsilon y_i^*)^2] \text{ almost surely,}$$

$$ii) \lim_{\epsilon \rightarrow 0} E[(y_i^* - \epsilon y_i^*)^2] = 0$$

Proof. (i) By definition $\frac{1}{p} \|\mathbf{y}^* - \epsilon \mathbf{y}^*\|_2^2 = \frac{1}{p} \sum_{i=1}^n (y_i^* - \epsilon y_i^*)^2$. By boundedness of $y_i^*, \epsilon y_i^*$, $E[(y_i^* - \epsilon y_i^*)^2]$ exists and thus, by the strong law of large numbers (e.g. Vershynin 2018, Theorem 1.3.1), claim (i) follows. (ii) Note that since $\bar{\epsilon}_i, \boldsymbol{\eta}_{0,i}$ are continuous independent random variables. Hence the event $\{\exists i : \bar{\epsilon}_i = \zeta'(\boldsymbol{\eta}_{0,i})\}$ has measure zero. Hence, $E[(y_i^* - \epsilon y_i^*)^2] = E[(h(\bar{\epsilon}_i, \boldsymbol{\eta}_{0,i}) - \epsilon h(\bar{\epsilon}_i, \boldsymbol{\eta}_{0,i}))^2]$. The claim now follows from boundedness of $E[(y_i^* - \epsilon y_i^*)^2]$, the dominated convergence theorem and Lemma S11 i). $\square$

Lemma S15. The map $f(\boldsymbol{\beta}) = \omega(\|\boldsymbol{\beta}\|_2/\sqrt{n})$ of Lemma 7 in Sur (2019) is such that if $\lim_{\epsilon \rightarrow 0} \|\boldsymbol{\beta}(\epsilon)\|_2 = r$, for some sequence $\boldsymbol{\beta}(\epsilon) \in \mathbb{R}^p$ indexed by ϵ , then $\lim_{\epsilon \rightarrow 0} f(\boldsymbol{\beta}(\epsilon)) = \omega(r/\sqrt{n})$.

Proof. It is readily checked that the function $\omega(\|\boldsymbol{\beta}\|_2/\sqrt{n})$ is of the form $c_1 \inf_{z: |z| \leq c_2 \frac{\|\boldsymbol{\beta}\|_2}{\sqrt{n}}} \zeta''(z)$ for some constants $c_1, c_2 > 0$ that do not depend on $\boldsymbol{\beta}$. Now $\zeta''(z)$ is symmetric and strictly decreasing in $|z|$ so that $\inf_{z: |z| \leq c_2 \frac{\|\boldsymbol{\beta}\|_2}{\sqrt{n}}} \zeta''(z) = \zeta''(c_2 \|\boldsymbol{\beta}\|_2/\sqrt{n})$. The claim follows by taking limits and the continuity of the function ζ'' . $\square$

Lemma S16. Consider the system of equations $F(\mu, b, \sigma, \epsilon) = 0$ given by

$$E \left[\frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right] - 1 + \kappa = 0 \quad (\text{S99})$$

$$\frac{b}{\kappa\gamma^2} E[Z \{ \epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \}] = 0 \quad (\text{S100})$$

$$\frac{b^2}{\kappa^2} E[\{ \epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \}^2] - \sigma^2 = 0, \quad (\text{S101})$$

where $Z \sim N(0, \gamma^2)$, $G \sim N(0, 1)$, and $\bar{\epsilon} \sim U(0, 1)$, all independent of each other, and $\epsilon Y^* = h_\epsilon(Z, \bar{\epsilon})$, $Z_* = \mu Z + \kappa^{1/2} \sigma G$.

Assume there exists a solution $F(\mu_*, b_*, \sigma_*, 0) = 0$ such that the Jacobian with respect to (μ_*, b_*, σ_*) of $F(\mu_*, b_*, \sigma_*, 0)$ is nonsingular.

Then, in a neighbourhood around (μ_*, b_*, σ_*) , $\epsilon = 0$, there exist unique functions $\epsilon\mu_*, \epsilon b_*, \epsilon\sigma_*$ that are continuous in ϵ such that $F(\epsilon\mu_*, \epsilon b_*, \epsilon\sigma_*, \epsilon) = 0$.

Proof. The claim follows by the Implicit Function Theorem (e.g. Rudin 1976, Theorem 9.28), for which it is required that

- (i) There exists a solution $F(\mu_*, b_*, \sigma_*, 0)$,
- (ii) The Jacobian with respect to (μ_*, b_*, σ_*) of $F(\mu_*, b_*, \sigma_*, 0)$ is nonsingular
- (iii) $F(\mu, b, \sigma, \epsilon)$ is continuously differentiable in a neighbourhood around (μ_*, b_*, σ_*) , $\epsilon = 0$

(i) and (ii) are assumed so that it remains to show (iii). Hence, it is sufficient to show that the partial derivatives of (S99)-(S101) exist and are continuous in a neighbourhood around (μ_*, b_*, σ_*) , $\epsilon = 0$. Begin with (S99) and consider

$$\begin{aligned} \frac{\partial}{\partial b} \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} &= - \frac{1}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^2} \frac{\partial}{\partial b} b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \\ &= - \frac{1}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^2} \left\{ \zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \right. \\ &\quad \left. + b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \left[\frac{\partial}{\partial u} \text{prox}_{b\zeta}(u) \Big|_{u=b_\epsilon Y^* + Z_*} \frac{\partial}{\partial b} u + \frac{\partial}{\partial b} \text{prox}_{b\zeta}(u) \Big|_{u=b_\epsilon Y^* + Z_*} \right] \right\} \\ &= - \frac{1}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^2} \left\{ \zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \right. \\ &\quad \left. + b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \left[\frac{\epsilon Y^*}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} - \frac{\zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right] \right\} \\ &= - \frac{1}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^2} \\ &\quad \cdot \left\{ \zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) + \frac{b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \left[\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \right] \right\}, \end{aligned} \quad (\text{S102})$$

where the first and second equality follow by the chain rule and the third uses Proposition A.1 of Donoho and Montanari (2016). Now note that $\zeta'(x) \in (0, 1)$, $\zeta''(x) \in (0, \frac{1}{4}]$, $\zeta'''(x) \in [-\frac{1}{6\sqrt{3}}, \frac{1}{6\sqrt{3}}]$ and $\epsilon Y^* \in [\frac{1-\alpha}{2}, \frac{1+\alpha}{2}]$. Hence (S102) is bounded. Thus, by the Dominated Convergence Theorem

and the Mean Value Theorem,

$$\frac{\partial}{\partial b} \mathbb{E} \left[\frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right] = \mathbb{E} \left[\frac{\partial}{\partial b} \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right]. \quad (\text{S103})$$

And again by dominated convergence, (S103) is continuous if (S102) is. By Lemma S17, $\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)$ is continuous in $(b, {}_\epsilon Y^*, Z_*)$. Since ${}_\epsilon Y^* = \frac{\alpha}{2} \left\{ 1 - \frac{\bar{\epsilon} - \zeta'(Z)}{|\bar{\epsilon} - \zeta'(Z)| + \epsilon} \right\}$ and $Z_* = \mu Z + \kappa^{1/2} \sigma G$, $\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)$ is continuous in $(\mu, b, \sigma, \epsilon)$. The claim follows by continuity of $\zeta'(\cdot), \zeta''(\cdot), \zeta'''(\cdot)$ and since (S102) is the composition of continuous functions. The treatment of the other partial derivatives is similar and thus details are omitted.

Next consider,

$$\frac{\partial}{\partial \mu} \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} = - \frac{b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) Z}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^3}, \quad (\text{S104})$$

which is again dominated by an integrable function and thus $\frac{\partial}{\partial \mu} \mathbb{E} \left[\frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right]$ is continuous in $(\mu, b, \sigma, \epsilon)$.

Similarly,

$$\frac{\partial}{\partial \sigma} \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} = - \frac{b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \kappa^{1/2} G}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^3},$$

and the same conclusion as for (S104) holds for $\frac{\partial}{\partial \sigma} \mathbb{E} \left[\frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \right]$.

Finally,

$$\frac{\partial}{\partial \epsilon} \frac{1}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} = - \frac{b\zeta'''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))}{[1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))]^3} \frac{\alpha}{2} \frac{\bar{\epsilon} - \zeta'(Z)}{(|\bar{\epsilon} - \zeta'(Z)| + \epsilon)^2}, \quad (\text{S105})$$

which is bounded and continuous in $(b, \mu, \sigma, \epsilon)$ for $\bar{\epsilon} \neq \zeta'(Z)$. Since the event $\bar{\epsilon} = \zeta'(Z)$, has measure zero, as $\bar{\epsilon} \sim U(0, 1)$ and $Z \sim N(0, 1)$ $\bar{\epsilon} \perp Z$, dominated convergence still applies and the claim holds.

Next, consider (S100), and differentiate the integrand with respect to b ,

$$\begin{aligned} \frac{\partial}{\partial b} \frac{b}{\kappa\gamma^2} Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \} &= \frac{1}{\kappa\gamma^2} Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \} \\ &\quad - Z \frac{\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))} \{ {}_\epsilon Y - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \}, \end{aligned}$$

of which both terms on the RHS are dominated by an integrable function and which are continuous in $(\mu, b, \sigma, \epsilon)$. Thus $\frac{\partial}{\partial b} \frac{b}{\kappa\gamma^2} \mathbb{E} [Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \}]$ exists and is continuous in $(\mu, b, \sigma, \epsilon)$.

Next, differentiate with respect to μ to get

$$\frac{\partial}{\partial \mu} \frac{b}{\kappa\gamma^2} Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \} = - \frac{b}{\kappa\gamma^2} \frac{Z^2 \zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))}{1 + b\zeta''(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*))},$$

which is again dominated by an integrable function and continuous in $(\mu, b, \sigma, \epsilon)$, so that

$$\frac{\partial}{\partial \mu} \frac{b}{\kappa\gamma^2} \mathbb{E} [Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b_\epsilon Y^* + Z_*)) \}]$$

exists and is continuous in $(\mu, b, \sigma, \epsilon)$.

Differentiating with respect to σ yields

$$\frac{\partial}{\partial \mu} \frac{b}{\kappa \gamma^2} Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} = -\frac{b}{\kappa \gamma^2} \frac{Z \kappa^{1/2} G \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))},$$

and the same conclusion holds for $\frac{\partial}{\partial \sigma} \frac{b}{\kappa \gamma^2} \mathbb{E} [Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}]$.

Finally, differentiating with respect to ϵ ,

$$\frac{\partial}{\partial \mu} \frac{b}{\kappa \gamma^2} Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} = -\frac{b}{\kappa \gamma^2} Z \left\{ 1 - \frac{b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))} \right\} \frac{\alpha}{2} \frac{\bar{\epsilon} - \zeta'(Z)}{(|\bar{\epsilon} - \zeta'(Z)| + \epsilon)^2},$$

and arguing analogous to (S105) yields that $\frac{\partial}{\partial \epsilon} \frac{b}{\kappa \gamma^2} \mathbb{E} [Z \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}]$ exists and is continuous in $(\mu, b, \sigma, \epsilon)$.

Lastly, the same is repeated for (S101).

$$\begin{aligned} \frac{\partial}{\partial b} \frac{b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2 - \sigma^2 &= \frac{2b}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2 - 1 \\ &\quad - \frac{2b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} \frac{\zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))} \\ &\quad \cdot \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}, \end{aligned}$$

$$\frac{\partial}{\partial \mu} \frac{b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2 - \sigma^2 = -\frac{2b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} \frac{Z \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))},$$

$$\begin{aligned} \frac{\partial}{\partial \sigma} \frac{b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2 - \sigma^2 &= -\frac{2b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} \frac{\kappa^{1/2} G \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))} \\ &\quad - 2\sigma, \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \epsilon} \frac{b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2 - \sigma^2 &= -\frac{2b^2}{\kappa^2} \{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \} \\ &\quad \cdot \frac{1}{1 + b \zeta''(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*))} \frac{\alpha}{2} \frac{\bar{\epsilon} - \zeta'(Z)}{(|\bar{\epsilon} - \zeta'(Z)| + \epsilon)^2}, \end{aligned}$$

all of which are dominated by some integrable function and continuous almost everywhere. Thus, again by dominated convergence, all partial derivatives of $\frac{b^2}{\kappa^2} \mathbb{E} [\{ {}_\epsilon Y^* - \zeta'(\text{prox}_{b\zeta}(b {}_\epsilon Y^* + Z_*)) \}^2] - \sigma^2$ exist and are continuous. Thus, $F(\mu, b, \sigma, \epsilon)$ is continuously differentiable in a neighbourhood around $(\mu_*, b_*, \sigma, 0)$ and the Implicit Function Theorem establishes the claim of the Lemma. $\square$

Lemma S17. *The proximal operator $\text{prox}_{b\zeta}(by + z)$ is continuous in (b, y, z) for $b > 0$, $x, y \in \mathfrak{R}$ where $\zeta(x) = \log(1 + e^x)$.*

Proof. Recall that by definition of the proximal operator,

$$\text{prox}_{b\zeta}(by + z) = \arg \min_{u \in \mathfrak{R}} \left\{ b \zeta(u) + \frac{1}{2} (u - [by + z])^2 \right\},$$

and since $\zeta(\cdot)$ is differentiable and strictly convex, it is uniquely determined as the solution to the

first order condition

$$b\zeta'(u) + u - by - z = 0.$$

Now note that $F(u, b, y, z) = b\zeta'(u) + u - by - z$ is continuously differentiable in (u, b, y, z) and that $\frac{\partial}{\partial u}F(u, b, y, z) = b\zeta''(u) + 1 > 0$ for all (u, b, y, z) . Hence, by the Implicit Function Theorem (e.g. Rudin 1976, Theorem 9.28), there exists a unique function $u(b', y', z')$ in a neighbourhood around (b, y, z) that is continuous in its arguments such that $F(u(b', y', z'), b', y', z') = 0$. Hence, the proximal operator is continuous in a neighbourhood around (b, y, z) . Since the choice of (b, y, z) was arbitrary, it is continuous over its entire domain. $\square$

Lemma S18. *In the setting of Corollary S2.1 and assuming that a stationary solution $(\epsilon\mu_*, \epsilon b_*, \epsilon\sigma_*)$ to the state evolution (S36)-(S37) exists, the AMP iterates of (S38) and (S39) satisfy the Cauchy property*

$$\begin{aligned} \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{p} \|\epsilon \boldsymbol{\nu}^{t+1} - \epsilon \boldsymbol{\nu}^t\|_2^2 &= 0 \\ \lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \frac{1}{n} \|\epsilon \mathbf{R}^{t+1} - \epsilon \mathbf{R}^t\|_2^2 &= 0 \end{aligned}$$

almost surely.

Proof. The proof is a direct translation of the Lemmas 6.8-6.9 of Donoho and Montanari (2016) along the lines suggested by Sur and Candès 2019 in the proof of Theorem 7 of the supplementary material. In particular, isolating iterates $\epsilon \boldsymbol{\nu}^{t-1}, \epsilon \boldsymbol{\nu}^t$ in Theorem S2, similar arguments to those in Corollary S2.1 yield that, almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{p} \sum_{j=1}^p \psi \left(\begin{bmatrix} \epsilon \boldsymbol{\nu}_j^{t-1} \\ \epsilon \boldsymbol{\nu}_j^t \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) = \mathbb{E} \left[\psi \left(\begin{bmatrix} \mathbf{Z}_t^{2t} \\ \mathbf{Z}_{t-1}^{2t} \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) \right], \quad \begin{bmatrix} \mathbf{Z}_{t-1}^{2t} \\ \mathbf{Z}_t^{2t} \end{bmatrix} \sim \mathcal{N} \left(\mathbf{0}_2, \begin{bmatrix} \epsilon \tau_{t-1}^2 & \epsilon \rho_{t-1,t} \\ \epsilon \rho_{t-1,t} & \epsilon \tau_t^2 \end{bmatrix} \right)$$

where

$$\begin{aligned} \epsilon \tau_t^2 &= \kappa^{-2} \mathbb{E} [\{\Psi^{t-1}(h_\epsilon(\mathbf{Z}_1^{2t-1}, \bar{\epsilon}), \mathbf{Z}_2^{2t-1})\}^2] \\ \epsilon \rho_{t-1,t} &= \kappa^{-2} \mathbb{E} [\Psi^{t-1}(h_\epsilon(\mathbf{Z}_1^{2t-1}, \bar{\epsilon}), \mathbf{Z}_{t+1}^{2t-1}) \Psi^{t-2}(h_\epsilon(\mathbf{Z}_1^{2t-1}, \bar{\epsilon}), \mathbf{Z}_t^{2t-1})], \end{aligned}$$

which in turn have covariance structure according to the state evolution (S7)-(S8). Now recall that given the covariance structure derived in Corollary S2.1, it was argued that given initial conditions $\hat{\beta}^0$,

$$\begin{aligned} \epsilon \mu_0 &= \frac{1}{\gamma^2} \lim_{n \rightarrow \infty} \frac{\langle \hat{\beta}^0, \beta_0 \rangle}{n} \\ \epsilon \sigma_0^2 &= \lim_{n \rightarrow \infty} \frac{\|\hat{\beta}^0 - \mu_0 \beta_0\|_2^2}{p}, \end{aligned} \tag{S106}$$

for all t , $\epsilon \tau_t = \epsilon \sigma_t$. Now, given a stationary solutions $(\epsilon\mu_*, \epsilon b_*, \epsilon\sigma_*)$ to the state evolutions (S35)-(S37), choose the oracle initialiser $\hat{\beta}^0 = \epsilon\mu_* \beta_0 + \epsilon\sigma_* \boldsymbol{\xi}$, where $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$ which is independent of β_0 . Then the initial conditions of (S106) are

$$\begin{aligned} \epsilon \mu_0 &= \frac{1}{\gamma^2} \lim_{n \rightarrow \infty} \frac{\langle \epsilon\mu_* \beta_0 + \epsilon\sigma_* \boldsymbol{\xi}, \beta_0 \rangle}{n} \stackrel{a.s.}{=} \epsilon\mu_* \\ \epsilon \sigma_0^2 &= \lim_{n \rightarrow \infty} \frac{\|\epsilon\sigma_* \boldsymbol{\xi}\|_2^2}{p} \stackrel{a.s.}{=} \epsilon\sigma_*^2. \end{aligned}$$

Henceforth condition on the event that $\epsilon \mu_0 = \epsilon\mu_*$, $\epsilon \sigma_0^2 = \epsilon\sigma_*^2$. Then $\epsilon \tau_{t-1}^2 = \epsilon \tau_t^2 = \epsilon \sigma_*^2$ and letting

$\psi(\mathbf{u}, \mathbf{v}) = (\mathbf{u}_1 - \mathbf{u}_2)^2$, for $\mathbf{u} = [\mathbf{u}_1, \mathbf{u}_2]^\top$, one gets that almost surely,

$$\lim_{N \rightarrow \infty} \frac{1}{p} \|\epsilon \boldsymbol{\nu}^t - \epsilon \boldsymbol{\nu}^{t-1}\|_2^2 = 2(\epsilon \sigma_*^2 - \epsilon \rho_{t,t-1}).$$

Hence, the goal is to show that $\lim_{t \rightarrow \infty} \epsilon \rho_{t,t-1} = \epsilon \sigma_*^2$.

Towards that end, following the proof of Theorem 4.1 of Donoho and Montanari (2016), define the doubly infinite matrix $\boldsymbol{\Gamma} = \{\Gamma_{s,t}\}_{s,t \in \mathbb{N}_0}$ with entries

$$\Gamma_{s+1,t+1} = \frac{1}{\kappa^2} \mathbb{E} [\Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_s) \Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_t)]$$

where $\Psi^*(y, s) = \epsilon b_* (y - \zeta'(\text{prox}_{\epsilon b_* \zeta}(\epsilon b_* y + s)))$ and

$$\begin{aligned} \epsilon Z_s &= \epsilon \mu_* Z + \kappa^{1/2} \epsilon \tilde{G}_s, & \begin{bmatrix} \epsilon \tilde{G}_s \\ \epsilon \tilde{G}_t \end{bmatrix} &\sim \mathcal{N}\left(\mathbf{0}_2, \begin{bmatrix} \epsilon \sigma_*^2 & \Gamma_{s,t} \\ \Gamma_{s,t} & \epsilon \sigma_*^2 \end{bmatrix}\right), \\ \epsilon Z_t &= \epsilon \mu_* Z + \kappa^{1/2} \epsilon \tilde{G}_t, \end{aligned}$$

where $Z \sim \mathcal{N}(0, \gamma^2)$, $\bar{\epsilon}$, are all pairwise independent and independent of $\tilde{G}_s, \tilde{G}_t$. Further, take the boundary conditions $\Gamma_{0,0} = \epsilon \sigma_*^2$, $\Gamma_{t,0} = \Gamma_{0,t} = 0$ so that in particular $\Gamma_{t,t} = \epsilon \sigma_*^2$.

It is then readily verified that for all t, s , $\Gamma_{t,s} = \epsilon \rho_{t,s}$. Next, let $q_t = \Gamma_{t+1,t} / \epsilon \sigma_*^2$. Then q_t is governed by the recursion

$$\begin{aligned} q_{t+1} &= H(q_t) \\ H(q) &= \frac{1}{\kappa^2 \epsilon \sigma_*^2} \mathbb{E} [\Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_1) \Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_2)], \end{aligned}$$

with

$$\begin{aligned} \epsilon Z_1 &= \epsilon \mu_* Z + \kappa^{1/2} \epsilon \sigma_* G_1, & \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} &\sim \mathcal{N}\left(\mathbf{0}_2, \begin{bmatrix} 1 & q \\ q & 1 \end{bmatrix}\right) \perp Z \sim \mathcal{N}(0, \gamma^2). \\ \epsilon Z_2 &= \epsilon \mu_* Z + \kappa^{1/2} \epsilon \sigma_* G_2, \end{aligned}$$

Next, it is proven that:

- i) $H(1) = 1$
- ii) $H(q)$ is increasing for $q \in [0, 1]$,
- iii) $H(q)$ is strictly convex for $q \in [0, 1]$

Towards i) note that

$$H(1) = \frac{1}{\kappa^2 \epsilon \sigma_*^2} \mathbb{E} [\Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon Z_1)^2],$$

for $\epsilon Z_1 = \epsilon \mu_* Z + \kappa^{1/2} \epsilon \sigma_* G$, $G \sim \mathcal{N}(0, 1) \perp Z$. Thus $H(1) = 1$ since (S37) holds for $(\epsilon \mu_*, \epsilon b_*, \epsilon \sigma_*)$.

Next, given $\bar{\epsilon}, Z$, let $g_{Z, \bar{\epsilon}}(G) = \Psi^*(h_\epsilon(Z, \bar{\epsilon}), \epsilon \mu_* Z + \kappa^{1/2} \epsilon \sigma_* G)$ and let

$$\mathcal{H}(q) = \mathbb{E} \left[g_{Z, \bar{\epsilon}}(G_1) g_{Z, \bar{\epsilon}}(G_2) \middle| \bar{\epsilon}, Z \right], \quad \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}_2, \begin{bmatrix} 1 & q \\ q & 1 \end{bmatrix}\right). \quad (\text{S107})$$

Then, by linearity of expectations and the tower property, it is sufficient to show that $\mathcal{H}(q)$ satisfies ii) and iii).

Following the argument in Donoho and Montanari 2016, Lemma 6.9, let $\{X_t\}_{t \geq 0}$ be the stationary Ornstein-Uhlenbeck process with covariance $\mathbb{E}(X_0 X_t) = e^{-t}$. Then

$$\mathcal{H}(q) = \mathbb{E} [g_{Z, \bar{\epsilon}}(G_1) g_{Z, \bar{\epsilon}}(G_2)] \bigg|_{t=\log(1/q)},$$

where the expectation is taken with respect to the process $\{X_t\}_{t \geq 0}$. Then by the spectral representation for $t = \log(1/q)$,

$$\mathcal{H}(q) = \sum_{l=1}^{\infty} c_l^2 e^{-lt} = \sum_{l=1}^{\infty} c_l^2 q_l, ,$$

and the claim follows since $c_l \neq 0$ at least once for $g_{Z, \bar{\varepsilon}}$ nonlinear.

Hence, it follows that $\lim_{t \rightarrow \infty} q_t = 1$ if and only if $H'(1) \leq 1$.

Further, it holds that

$$H'(1) = \frac{1}{\kappa} \mathbb{E} \left[\left\{ \frac{\partial}{\partial s} \Psi^*(h_\epsilon(\bar{\varepsilon}, Z), \epsilon Z_1) \right\}^2 \right].$$

Indeed, again condition on $\bar{\varepsilon}, Z$ and consider $\frac{\partial}{\partial q} \mathcal{H}(q)$ as defined in (S107). For notational convenience, the conditioning in the expectation of $\mathcal{H}$ is suppressed below and $g_{Z, \bar{\varepsilon}}(G)$ is simply denoted by $g(G)$.

Then, since G_2 can be written as $qG_1 + \sqrt{1-q^2}\tilde{G}$ for $\tilde{G} \sim N(0, 1)$ is independent of all other random variables, it holds that

$$\begin{aligned} \frac{\partial}{\partial q} \mathcal{H}(q) &= \frac{\partial}{\partial q} \mathbb{E} [g(G_1)g(G_2)] \\ &= \mathbb{E} \left[\frac{\partial}{\partial q} g(G_1)g(G_2) \right] \\ &= \mathbb{E} \left[\frac{\partial}{\partial q} g(G_1)g(qG_1 + \sqrt{1-q^2}\tilde{G}) \right] \\ &= \mathbb{E} \left[g(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G})\kappa^{1/2}\epsilon\sigma_* \left\{ G_1 - \frac{q}{\sqrt{1-q^2}}\tilde{G} \right\} \right] \\ &= \mathbb{E} \left[g(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G})\kappa^{1/2}\epsilon\sigma_* G_1 \right] - \frac{q}{\sqrt{1-q^2}} \mathbb{E} \left[g(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G})\kappa^{1/2}\epsilon\sigma_* \tilde{G} \right], \end{aligned} \tag{S108}$$

where the second inequality follows by dominated convergence. Now let $F(G_1, G_2) = g(G_1)g'(G_2)$. Then note that $\frac{\partial}{\partial G_1} F(G_1, G_2) = \kappa^{1/2}\epsilon\sigma_* g'(G_1)g'(G_2)$ and $\frac{\partial}{\partial G_2} F(G_1, G_2) = \kappa^{1/2}\epsilon\sigma_* g(G_1)g''(G_2)$. Hence, using Gaussian integration by parts, which can be derived as a consequence of Stein's Lemma (Stein, 1981, Lemma 1),

$$\begin{aligned} \mathbb{E} \left[g(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G})\kappa^{1/2}\epsilon\sigma_* G_1 \right] &= \mathbb{E} \left[g(G_1)g'(G_2)\kappa^{1/2}\epsilon\sigma_* G_1 \right] \\ &= \kappa^{1/2}\epsilon\sigma_* \mathbb{E} [G_1 F(G_1, G_2)] \\ &= \kappa^{1/2}\epsilon\sigma_* \left\{ \text{cov}(G_1, G_1) \mathbb{E} \left[\frac{\partial}{\partial G_1} F(G_1, G_2) \right] \right. \\ &\quad \left. + \text{cov}(G_1, G_2) \mathbb{E} \left[\frac{\partial}{\partial G_2} F(G_1, G_2) \right] \right\} \\ &= \kappa\epsilon\sigma_*^2 \left\{ \mathbb{E} \left[\frac{\partial}{\partial G_1} F(G_1, G_2) \right] + q \mathbb{E} \left[\frac{\partial}{\partial G_2} F(G_1, G_2) \right] \right\} \\ &= \kappa\epsilon\sigma_*^2 \left\{ \mathbb{E} [g'(G_1)g'(G_2)] + q \mathbb{E} \left[\frac{\partial}{\partial G_2} g(G_1)g''(G_2) \right] \right\}. \end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E} \left[g(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G})\kappa^{1/2}_\epsilon\sigma_*\tilde{G} \right] &= \kappa^{1/2}_\epsilon\sigma_*\mathbb{E} \left[F(G_1, qG_1 + \sqrt{1-q^2}\tilde{G})\tilde{G} \right] \\
&= \kappa^{1/2}_\epsilon\sigma_* \left\{ \text{cov}(G_1, \tilde{G})\mathbb{E} \left[\frac{\partial}{\partial G_1} F(G_1, qG_1 + \sqrt{1-q^2}\tilde{G}) \right] \right. \\
&\quad \left. + \text{cov}(\tilde{G}, \tilde{G})\mathbb{E} \left[\frac{\partial}{\partial \tilde{G}} F(G_1, qG_1 + \sqrt{1-q^2}\tilde{G}) \right] \right\} \\
&= \sqrt{1-q^2}\kappa_\epsilon\sigma_*^2\mathbb{E} \left[g(G_1)g''(qG_1 + \sqrt{1-q^2}\tilde{G}) \right] \\
&= \sqrt{1-q^2}\kappa_\epsilon\sigma_*^2\mathbb{E} [g(G_1)g''(G_2)] ,
\end{aligned} \tag{S109}$$

where it was used that $\text{cov}(G_1, \tilde{G}) = 0$. Thus combining (S108)-(S109), it follows that

$$\frac{\partial}{\partial q}\mathcal{H}(q) = \kappa_\epsilon\sigma_*^2\mathbb{E} [g'(G_1)g'(G_2)] .$$

Marginalizing over $Z, \bar{\varepsilon}$ then gives

$$\begin{aligned}
\frac{\partial}{\partial q}H(q) &= \frac{1}{\kappa^2_\epsilon\sigma_*^2}\frac{\partial}{\partial q}\mathbb{E}_{Z, \bar{\varepsilon}} \left[\mathbb{E} \left[g(G_1)g(G_2) \middle| Z, \bar{\varepsilon} \right] \right] \\
&= \frac{1}{\kappa^2_\epsilon\sigma_*^2}\mathbb{E}_{Z, \bar{\varepsilon}} \left[\mathbb{E} \left[\frac{\partial}{\partial q}g(G_1)g(G_2) \middle| Z, \bar{\varepsilon} \right] \right] \\
&= \frac{1}{\kappa}\mathbb{E}_{Z, \bar{\varepsilon}} \left[\mathbb{E} \left[g'(G_1)g'(G_2) \middle| Z, \bar{\varepsilon} \right] \right] \\
&= \frac{1}{\kappa}\mathbb{E} [g'(G_1)g'(G_2)] \\
&= \frac{1}{\kappa}\mathbb{E} \left[g'(G_1)g'(qG_1 + \sqrt{1-q^2}\tilde{G}) \right] ,
\end{aligned}$$

and it thus follows that $H'(1) = \frac{1}{\kappa}\mathbb{E} \left[\left\{ \frac{\partial}{\partial s}\Psi^*(h_\epsilon(\bar{\varepsilon}, Z), \epsilon Z_1) \right\}^2 \right]$ as required.

Now it is easily checked that $\frac{\partial}{\partial s}\Psi^* \in (0, 1)$ and thus

$$H'(1) = \frac{1}{\kappa}\mathbb{E} \left[\left\{ \frac{\partial}{\partial s}\Psi^*(h_\epsilon(\bar{\varepsilon}, Z), \epsilon Z_1) \right\}^2 \right] \leq \frac{1}{\kappa}\mathbb{E} \left[\frac{\partial}{\partial s}\Psi^*(h_\epsilon(\bar{\varepsilon}, Z), \epsilon Z_1) \right] = 1$$

since (S35) holds. Similar arguments show the Cauchy-property for $\frac{1}{n}\|\epsilon\mathbf{R}^t - \epsilon\mathbf{R}^{t-1}\|_2^2$. $\square$

Lemma S19. *The solution to the system of equations (S2)-(S4) is equivalent to the stationary solution of (S15)-(S17)*

Proof. Begin by imposing stationarity for equations (S15)-(S17), so that they become

$$\begin{aligned}
1 - \kappa &= \mathbb{E} \left[\frac{1}{1 + b_*\zeta''(\text{prox}_{b_*\zeta}(b_*Y^* + Z_*))} \right] \\
0 &= \frac{b_*}{\kappa\gamma^2}\mathbb{E} [Z \{Y^* - \zeta'(\text{prox}_{b_*\zeta}(b_*Y^* + Z_*))\}] \\
\sigma_*^2 &= \frac{b_*^2}{\kappa^2}\mathbb{E} [\{Y^* - \zeta'(\text{prox}_{b_*\zeta}(b_*Y^* + Z_*))\}^2] ,
\end{aligned}$$

for independent $Z \sim N(0, \gamma^2)$, $G \sim N(0, 1)$, and $\bar{\varepsilon} \sim U(0, 1)$, and $Y^* = \alpha \mathbf{1}\{\bar{\varepsilon} \leq \zeta'(Z)\} + \frac{1-\alpha}{2}$, $Z_* = \mu_* Z + \kappa^{1/2} \sigma_* G$. Hence

$$\begin{aligned} \mathbb{E} \left[\frac{1}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* Y^* + Z_*))} \right] &= \mathbb{E} \left[\frac{\zeta'(Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1+\alpha}{2} + Z_*))} \right] + \mathbb{E} \left[\frac{\zeta'(-Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1-\alpha}{2} + Z_*))} \right] \\ &= \mathbb{E} \left[\frac{\zeta'(Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1+\alpha}{2} + Z_*))} \right] + \mathbb{E} \left[\frac{\zeta'(-Z)}{1 + b_* \zeta''(-\text{prox}_{b_* \zeta}(b_* \frac{1-\alpha}{2} + Z_*))} \right] \\ &= \mathbb{E} \left[\frac{\zeta'(Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1+\alpha}{2} + Z_*))} \right] + \mathbb{E} \left[\frac{\zeta'(-Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1+\alpha}{2} - Z_*))} \right] \\ &= 2\mathbb{E} \left[\frac{\zeta'(Z)}{1 + b_* \zeta''(\text{prox}_{b_* \zeta}(b_* \frac{1+\alpha}{2} + Z_*))} \right], \end{aligned}$$

where the first equality follows by conditioning on Y^* , the second follows since $\zeta''(x) = \zeta''(-x)$, the fourth since $-\text{prox}_{b\zeta}(-u) = \text{prox}_{b\zeta}(u + b)$ (cf. p.18 in the supplementary material of Sur and Candès (2019)), which holds since the proximal operator $x^* = \text{prox}_{b\zeta}(-u)$ is the unique solution to $b\zeta'(x^*) + x^* + u = 0$ and similarly $y^* = \text{prox}_{b\zeta}(u + b)$ is the unique solution to $b\zeta'(y^*) + y^* - u - b = 0$. Hence, by substitution and using that $\zeta'(-x) = 1 - \zeta'(x)$, y^* is the unique solution to $b(\zeta'(y) - \zeta'(-x^*)) + y^* - x = 0$ so that $y^* = -x^*$. Therefore, letting $-u = b_*(1 - \alpha)/2 + Z_*$, $u + b_* = b_*(1 + \alpha)/2 - Z_*$ as required. The last equality follows by a simple change of variable and the symmetry of the normal distribution.

Similarly,

$$\begin{aligned} \mathbb{E} [Z \{Y^* - \zeta'(\text{prox}_{b_* \zeta}(b_* Y^* + Z_*))\}] &= \mathbb{E} \left[\zeta'(Z) Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right] \\ &\quad + \mathbb{E} \left[\zeta'(-Z) Z \left\{ \frac{1-\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1-\alpha}{2} + Z_* \right) \right) \right\} \right] \\ &= \mathbb{E} \left[\zeta'(Z) Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right] \\ &\quad + \mathbb{E} \left[\zeta'(-Z) Z \left\{ \frac{1-\alpha}{2} - \left(1 - \zeta' \left(-\text{prox}_{b_* \zeta} \left(b_* \frac{1-\alpha}{2} + Z_* \right) \right) \right) \right\} \right] \\ &= \mathbb{E} \left[\zeta'(Z) Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right] \\ &\quad - \mathbb{E} \left[\zeta'(-Z) Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} - Z_* \right) \right) \right\} \right] \\ &= 2\mathbb{E} \left[\zeta'(Z) Z \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right], \end{aligned}$$

where again, the first equality comes from conditioning on Y^* , the second, from $\zeta'(-x) = 1 - \zeta'(x)$, the third by the symmetry of the proximal operator around b , i.e. $-\text{prox}_{b\zeta}(-u) = \text{prox}_{b\zeta}(u + b)$, and the last by change of variable and symmetry of the normal distribution.

And finally,

$$\begin{aligned}
\mathbb{E} \left[\left\{ Y^* - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* Y^* + Z_* \right) \right) \right\}^2 \right] &= \mathbb{E} \left[\left\{ \zeta'(Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right] \\
&+ \mathbb{E} \left[\left\{ \zeta'(-Z) \left\{ \frac{1-\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1-\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right] \\
&= \mathbb{E} \left[\left\{ \zeta'(Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right] \\
&+ \mathbb{E} \left[\left\{ \zeta'(-Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(-\text{prox}_{b_* \zeta} \left(b_* \frac{1-\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right] \\
&= \mathbb{E} \left[\left\{ \zeta'(Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right] \\
&+ \mathbb{E} \left[\left\{ \zeta'(-Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} - Z_* \right) \right) \right\} \right\}^2 \right] \\
&= 2\mathbb{E} \left[\left\{ \zeta'(Z) \left\{ \frac{1+\alpha}{2} - \zeta' \left(\text{prox}_{b_* \zeta} \left(b_* \frac{1+\alpha}{2} + Z_* \right) \right) \right\} \right\}^2 \right].
\end{aligned}$$

□

Lemma S20. Let $\mathbf{u} \in \mathbb{R}^p$ be a unit vector and $\mathbf{P}_{\mathbf{u}^\perp} = \mathbf{I}_p - \mathbf{u}\mathbf{u}^\top$ be the projection into the subspace $\mathbf{u}^\perp$ that is orthogonal to $\mathbf{u}$. Then for $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$, almost surely,

$$\lim_{p \rightarrow \infty} \frac{\|\mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}\|_2}{\sqrt{p}} = 1,$$

as $p \rightarrow \infty$.

Proof. Consider $\|\mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}\|_2^2 = \mathbf{Z}^\top \mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}$ since $\mathbf{P}_{\mathbf{u}^\perp}$ is idempotent and note that $\text{tr}(\mathbf{P}_{\mathbf{u}^\perp}) = p - 1$. By the Hanson-Wright inequality (e.g. Vershynin (2018, Theorem 6.2.1)), there exists a universal constant $c > 0$ such that for any $\epsilon \in (0, 1)$,

$$\Pr \left(\left| \frac{\mathbf{Z}^\top \mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}}{p-1} - 1 \right| > \epsilon \right) \leq 2 \exp \{ -c\epsilon^2(p-1) \}.$$

Now since for $\delta > 0$, $|z-1| \geq \delta$ implies that $|z^2-1| \geq \max\{\delta, \delta^2\}$ (see Vershynin (2018, Equation 3.2)), one gets that

$$\Pr \left(\left| \frac{\|\mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}\|_2}{\sqrt{p-1}} - 1 \right| > \epsilon \right) \leq 2 \exp \{ -c\epsilon^2(p-1) \}.$$

By the first Borel-Cantelli Lemma, it follows that $\lim_{p \rightarrow \infty} \|\mathbf{P}_{\mathbf{u}^\perp} \mathbf{Z}\|_2 / (p-1) = 1$ almost surely and the claim follows. □

Lemma S21. Let

$$F(\kappa) = \mathbb{E}[(W - \tau_\kappa)_+],$$

with $(x)_+ = \max\{0, x\}$, $W = |Z|$, $Z \sim \mathcal{N}(0, 1)$ and where τ_κ is the nonnegative root of

$$\mathbb{E}[\min\{W^2, \tau_\kappa^2\}] = \kappa.$$

Then, for any $\kappa \in (0, 1)$,

(i) A unique nonnegative solution τ_κ exists

(ii) $F(\kappa)$ is strictly decreasing in κ and $F(\kappa) \in (0, \sqrt{2/\pi})$

Proof. We prove each claim in turn. For (i) we shall consider the mapping $\tau \mapsto \mathbb{E}[\min\{W^2, \tau^2\}]$. First, note that $\mathbb{E}[\min\{W^2, \tau^2\}]$ is strictly increasing in τ on $\mathbb{R}_{\geq 0}$. Indeed, let $\tau > \tau' \geq 0$. Then

1. For $\tau' \geq W$:

$$\min\{W^2, \tau^2\} = \min\{W^2, (\tau')^2\} = W^2.$$

2. For $\tau > W \geq \tau'$:

$$\min\{W^2, \tau^2\} = W^2 > (\tau')^2 = \min\{W^2, (\tau')^2\}.$$

3. For $W \geq \tau$:

$$\min\{W^2, \tau^2\} = \tau^2 > (\tau')^2 = \min\{W^2, (\tau')^2\}.$$

Since $\Pr(W > \tau') = \Pr(|Z| > \tau') = 1 - \Phi(\tau') > 0$ where $\Phi(z)$ is the CDF of a standard normal random variable, we conclude that $\mathbb{E}[\min\{W^2, \tau^2\}] > \mathbb{E}[\min\{W^2, (\tau')^2\}]$ as required. Further note that by dominated convergence

$$\lim_{\tau \rightarrow \infty} \mathbb{E}[\min\{W^2, \tau^2\}] = \mathbb{E}[\lim_{\tau \rightarrow \infty} \min\{W^2, \tau^2\}] = \mathbb{E}[W^2] = 1,$$

since W^2 is a χ^2 -random variable with one degree of freedom. Finally we have that $\mathbb{E}[\min\{W^2, \tau^2\}]$ is continuous in τ on $\mathbb{R}_{\geq 0}$. To see this, note that for any $W \geq 0$, $\min\{W^2, \tau^2\}$ is continuous in τ^2 . Indeed first assume that $|\tau^2 - (\tau')^2| < \epsilon$. WLOG, let $\tau > \tau' \geq 0$ and consider the cases

1. For $\tau' \geq W$:

$$|\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}| = |W^2 - W^2| < \epsilon.$$

2. For $\tau > W \geq \tau'$:

$$|\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}| = |W^2 - (\tau')^2|.$$

Moreover, we have that $\tau^2 - (\tau')^2 > W - (\tau')^2 \geq 0 > (\tau')^2 - \tau^2$. Hence $|W^2 - (\tau')^2| < |\tau^2 - (\tau')^2| < \epsilon$.

3. For $W \geq \tau$:

$$|\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}| = |\tau^2 - (\tau')^2| < \epsilon.$$

Thus, we conclude that $\min\{W^2, \tau^2\}$ is continuous on $\mathbb{R}_{\geq 0}$ for every $W > 0$. Since the map $x \mapsto x^2$ is also continuous on $\mathbb{R}_{\geq 0}$, we conclude that $\tau \mapsto \min\{W^2, \tau^2\}$ is also continuous in τ on $\mathbb{R}_{\geq 0}$. Hence, for any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $W \geq 0$,

$$|\tau - \tau'| < \delta \implies |\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}| < \epsilon.$$

But then,

$$|\mathbb{E}[\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}]| \leq \mathbb{E}[|\min\{W^2, \tau^2\} - \min\{W^2, (\tau')^2\}|] < \mathbb{E}[\epsilon] = \epsilon.$$

So we conclude that $\mathbb{E}[\min\{W^2, \tau^2\}]$. Now for any $\kappa \in (0, 1)$, since $\lim_{\tau \rightarrow \infty} \mathbb{E}[\min\{W^2, \tau^2\}] = 1$, we can find a large enough T such that $\mathbb{E}[\min\{W^2, T^2\}] > \kappa$. Then, finally, by the intermediate value theorem, we conclude that $\mathbb{E}[\min\{W^2, \tau^2\}] - \kappa$ has a root τ_κ in $[0, T]$, which must be unique since the function is strictly increasing.

For (ii), we note that $E[\max\{0, W - \tau\}]$ is strictly decreasing in τ on $\mathfrak{R}_{\geq 0}$. This can be shown with similar arguments as we used to establish that $E[\min\{W^2, \tau^2\}]$ is strictly increasing in τ . Further, τ_κ is strictly increasing in κ . This follows since $E[\min\{W^2, \tau^2\}]$ is strictly increasing in τ on $\mathfrak{R}_{\geq 0}$ and τ_κ solves $E[\min\{W^2, \tau^2\}] - \kappa = 0$. Finally note that

$$\sqrt{2/\pi} = E[W] \geq E[(W - \tau_\kappa)_+] \geq 0.$$

□

Lemma S22. For any $x \in \mathfrak{R}$, $y \in \{0, 1\}$ and $\alpha \in (0, 1)$,

$$\log(1 + \exp\{x\}) - \left(\alpha y + \frac{1 - \alpha}{2}\right)x > \frac{1 - \alpha}{2}|x|.$$

Proof. We consider four cases:

1. $y = 1, x > 0$:

$$\begin{aligned} \log(1 + \exp\{x\}) - \left(\alpha y + \frac{1 - \alpha}{2}\right)x &= \log(1 + \exp\{x\}) - \frac{1 + \alpha}{2}x \\ &> x - \frac{1 + \alpha}{2}x \\ &= \frac{1 - \alpha}{2}x \\ &= \frac{1 - \alpha}{2}|x|. \end{aligned}$$

2. $y = 1, x \leq 0$:

$$\begin{aligned} \log(1 + \exp\{x\}) - \left(\alpha y + \frac{1 - \alpha}{2}\right)x &= \log(1 + \exp\{x\}) - \frac{1 + \alpha}{2}x \\ &> -\frac{1 + \alpha}{2}x \\ &= \frac{1 + \alpha}{2}|x| \\ &> \frac{1 - \alpha}{2}|x|. \end{aligned}$$

3. $y = 0, x > 0$:

$$\begin{aligned} \log(1 + \exp\{x\}) - \left(\alpha y + \frac{1 - \alpha}{2}\right)x &= \log(1 + \exp\{x\}) - \frac{1 - \alpha}{2}x \\ &> x - \frac{1 - \alpha}{2}x \\ &= \frac{1 + \alpha}{2}|x| \\ &> \frac{1 - \alpha}{2}|x|. \end{aligned}$$

4. $y = 0, x \leq 0$:

$$\begin{aligned}\log(1 + \exp\{x\}) - \left(\alpha y + \frac{1 - \alpha}{2}\right)x &= \log(1 + \exp\{x\}) - \frac{1 - \alpha}{2}x \\ &> -\frac{1 - \alpha}{2}x \\ &= \frac{1 - \alpha}{2}|x|.\end{aligned}$$

□

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